Chapter 46. Pneumonia in the Ventilator-Dependent Patient

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Pneumonia in the Ventilator-Dependent Patient: Introduction

Ventilator-associated pneumonia (VAP) is the most frequent intensive care unit (ICU)-acquired infection among patients receiving mechanical ventilation.[1] In contrast to infections of other frequently involved organs (e.g., urinary tract and skin), for which mortality is low, ranging from 1% to 4%, the mortality rate for VAP, defined as pneumonia occurring more than 48 hours after endotracheal intubation and initiation of mechanical ventilation, ranges from 20% to 50% and can even be higher in some specific settings or when lung infection is caused by high-risk pathogens.[1–3] Although the attributable mortality rate for VAP is still debated, it has been shown that these infections prolong both the duration of ventilation and the duration of ICU stay.[1,2] Approximately 50% of all antibiotics prescribed in an ICU are administered for respiratory tract infections.[4] Because several studies have shown that appropriate antimicrobial treatment of patients with VAP significantly improves outcome, more rapid identification of infected patients and accurate selection of antimicrobial agents represent important clinical goals.[2] Consensus, however, on appropriate diagnostic, therapeutic, and preventive strategies for VAP has yet to be reached. In this chapter, we summarize published studies on epidemiology, diagnosis, treatment, and prevention of nosocomial pulmonary infection in critically ill patients mechanically ventilated in the ICU, and present our experience with this infection.

Epidemiology

Accurate data on the epidemiology of VAP are limited by the lack of standardized criteria for its diagnosis. Conceptually, VAP is defined as an inflammation of the lung parenchyma caused by infectious agents not present or incubating at the time mechanical ventilation was started. Despite the clarity of this conception, the past three decades have witnessed the appearance of numerous operational definitions, none of which is universally accepted. Even definitions based on histopathologic findings at autopsy may fail to find consensus or provide certainty. Pneumonia in focal areas of a lobe may be missed, microbiologic studies may be negative despite of presence of inflammation in the lung, and pathologists may disagree on the findings.[5] The absence of a "reference standard" continues to fuel controversy about the adequacy and relevance of many studies in this field.

Incidence of Ventilator-Associated Pneumonia

The exact incidence varies widely depending on the case definition of pneumonia and the population being evaluated.[6–10] All studies, however, have confirmed that nosocomial pneumonia is considerably more frequent in ventilated patients than in other ICU patients, with an incidence increasing by as much as sixfold to 20-fold in this subset of patients.[11,12] VAP occurs in 9% to 27% of all intubated patients and its incidence increases with duration of ventilation.[10,13] The risk of VAP is highest early in the course of hospital stay and is estimated to be 3% per day during the first 5 days of ventilation, 2% per day during days 5 to 10 of ventilation, and 1% per day beginning from day 11.[13] Because most mechanical ventilation is short term, approximately half of all episodes of VAP occur within the first 4 days of mechanical ventilation. In a large epidemiologic study, independent predictors of VAP retained by multivariable analysis were a primary admitting diagnosis of burns, trauma, central nervous system disease, respiratory disease, cardiac disease, mechanical ventilation during the preceding 24 hours, witnessed aspiration, and use of paralytic agents. Exposure to antibiotics conferred protection, but this effect was attenuated over time.[13] According to four studies, the VAP rate was higher in patients with acute respiratory distress syndrome (ARDS) than other ventilated patients, affecting between 34% and more than 70% of patients with ARDS and often leading to the development of sepsis, multiorgan failure, and death.[14–17]

Mortality, Morbidity, and Cost

Mechanically ventilated patients in the ICU with VAP appear to have a twofold to 10-fold higher risk of death as compared with patients without pneumonia. Although these statistics indicate that VAP can be lethal, previous studies have not demonstrated clearly that pneumonia is responsible for the higher mortality rate of these patients.[18] It is often difficult to determine whether ICU patients with severe underlying illness would have survived if VAP had not occurred. VAP, however, has been recognized in several case-controlled studies or studies using multivariate analysis as an important prognostic factor for different groups of critically ill patients.[18–23]

Other factors beyond the simple development of VAP, such as the severity of the disease, the responsible pathogens, or the appropriateness of initial treatment, may be more important determinants of outcome for patients in whom pneumonia develops.[24] Indeed, it may be that VAP increases mortality only in the subset of patients with intermediate severity of illness,[23] when initial treatment is inappropriate,[25–32] and/or in patients with VAP caused by high-risk pathogens, such as *Pseudomonas aeruginosa.*[24,33] Patients with very low severity and early onset pneumonia caused by organisms such as *Haemophilus influenzae* or *Streptococcus pneumoniae* have excellent prognoses with or without VAP, whereas very ill patients with late-onset VAP, occurring while they are in a quasi-terminal state, would be unlikely to survive. Using a multistate progressive disability model that appropriately handled VAP as a time-dependent event in a high-quality database of 2873 mechanically ventilated patients, Nguile-Makao et al showed that VAP attributable mortality was 8.1% overall, varying widely with case mix, severity at admission, time to VAP onset, and severity of organ dysfunction at VAP onset.[24] These results are consistent with the 10.6% value obtained in five German ICUs using also a multistate model progressive disability model.[34]

It is impossible to evaluate precisely the morbidity and excess costs associated with VAP. All studies, however, have shown clearly that patients with VAP have prolonged duration of mechanical ventilation and lengthened ICU and hospital stay as compared with patients who do not have VAP.[1,2,35,36] Summarizing available data, VAP appears to extend the ICU stay by at least 4 to 6 days, with the attributable ICU length of stay being longer for medical than surgical patients and for patients infected with "high-risk" as opposed to "low-risk" organisms.[37] The prolonged hospitalization of patients with VAP underscores the considerable financial burden imposed on the health care system by the development of VAP.[10,35,36,38–41]

Etiologic Agents

Microorganisms responsible for VAP differ according to the population of ICU patients, the durations of hospital and ICU stays, and the specific diagnostic method(s) used to establish the responsible pathogens. A number of studies have shown that gram-negative bacilli cause many of the respiratory infections in this setting.[1,2,42,43] The data from twenty-four studies conducted on ventilated patients, for whom bacteriologic studies were restricted to uncontaminated specimens obtained using a protected specimen brush (PSB) or bronchoalveolar lavage (BAL), confirmed these results: gram-negative bacilli represented 58% of recovered organisms (Table 46-1).[1] The predominant gram-negative bacilli were *P. aeruginosa* and *Acinetobacter* spp., followed by *Proteus* spp., *Escherichia coli, Klebsiella* spp., and *H. influenzae.* A relatively high rate of gram-positive pneumonias was also reported in those studies, with *Staphylococcus aureus* involved in more than 20% of the cases.[42] Many episodes of VAP are caused by multiple pathogens.[1,44]

Table 46-1: Etiology of Ventilator-associated Pneumonia as Documented by Bronchoscopic Techniques in 24 Studies for a Total of 1689 Episodes and 2490 Pathogens

Pathogen	Frequency (%)
Pseudomonas aeruginosa	24.4
Acinetobacter spp.	7.9
Stenotrophomonas maltophilia	1.7
Enterobacteriaceae[a]	14.1
Haemophilus spp.	9.8
Staphylococcus aureus[b]	20.4
Streptococcus spp.	8.0
Streptococcus pneumoniae	4.1
Coagulase-negative staphylococci	1.4
Neisseria spp.	2.6
Anaerobes	0.9
Fungi	0.9
Others (<1% each)[c]	3.8

[a]Distribution when specified: *Klebsiella* spp., 15.6%; *Escherichia coli,* 24.1%; *Proteus* spp., 22.3%; *Enterobacter* spp., 18.8%; *Serratia* spp., 12.1%; *Citrobacter* spp., 5.0%; *Hafnia alvei,* 2.1%.

[b]Distribution when specified: Methicillin-resistant *Staphylococcus aureus* (MRSA), 55.7%; methicillin-susceptible *Staphylococcus aureus* (MSSA), 44.3%.

[c]Including *Corynebacterium* spp.; *Moraxella* spp.; and *Enterococcus* spp.

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Underlying diseases may predispose patients to infection with specific organisms. Patients with chronic obstructive pulmonary disease are at increased risk for *H. influenzae, Moraxella catarrhalis, or S. pneumoniae* infections; cystic fibrosis increases the risk of *P. aeruginosa* and/or *S. aureus* infections, while trauma and neurologic disease increases the risk for *S. aureus* infection. Furthermore, the causative agent for pneumonia differs among ICU surgical populations, with 18% of the nosocomial pneumonias caused by *Haemophilus* or pneumococci, particularly in patients with trauma, but not in patients with malignancy or who underwent transplantation, abdominal, or cardiovascular surgery.[1,2]

Despite somewhat different definitions of early onset pneumonia, varying from onset of less than 3 days to less than 7 days, high rates of *H. influenzae, S. pneumoniae,* methicillin-susceptible *Staphylococcus aureus* (MSSA) or susceptible *Enterobacteriaceae* were constantly found in early onset VAP, whereas *P. aeruginosa, Acinetobacter* spp., methicillin-resistant *Staphylococcus aureus* (MRSA), and multiresistant gram-negative bacilli were significantly more frequent in late-onset VAP.[1] The different pattern of distribution of etiologic agents between early- and late-onset VAP is linked to prior antimicrobial therapy in many patients with late-onset VAP. When multivariate analysis was used to identify risk factors for VAP caused by potentially drug-resistant bacteria such as MRSA, *P. aeruginosa, Acinetobacter baumannii,* and/or *Stenotrophomonas maltophilia* in 135 consecutive episodes of VAP, only three variables remained significant: mechanical ventilation duration of longer than 7 days before onset of VAP, prior antibiotic use, and prior use of broad-spectrum drugs (third-generation cephalosporins, fluoroquinolones, and/or imipenem).[45] Not all studies have confirmed this distribution pattern, and in some studies the most common pathogens associated with early onset VAP were *P. aeruginosa,* MRSA, and *Enterobacter* spp., with similar pathogens associated with late-onset VAP.[46,47] These findings might be explained in part by prior hospitalization and the use of antibiotics before transfer to the ICU.

The incidence of multiresistant pathogens is also closely linked to local factors and varies widely from one institution to another. Consequently, each ICU has to continuously collect meticulous epidemiologic data.[48] Clinicians must clearly be aware of the common microorganisms associated with both early-onset and late-onset VAP in their own hospitals so as to avoid the administration of initial inadequate antimicrobial therapy.

Legionella species, anaerobes, and even *Pneumocystis jiroveci* should be mentioned as potential causative agents, but these microbes are not commonly found when pneumonia is acquired during mechanical ventilation. Herpesviridae, namely herpes simplex virus, can be detected in the lower respiratory tracts of 5% to 64% of ICU patients, depending on the population and the diagnostic method used. In most cases, herpes simplex virus recovery from lower respiratory tract samples of nonimmunocompromised ventilated patients corresponds to viral contamination from the mouth and/or throat. For some patients, however, real herpes simplex virus bronchopneumonitis can develop, and it can evolve into ARDS and/or facilitate the occurrence of bacterial superinfection.[49–51] Cytomegalovirus-induced pneumonia is a rare event in ventilated patients. As for herpes simplex virus bronchopneumonitis, it is impossible to know whether cytomegalovirus detection in the lower respiratory tract is merely a marker of disease severity or signals real disease with its own morbidity and mortality.[52–55]

Isolation of fungi, most frequently *Candida* species, at significant concentrations poses interpretative problems. Invasive disease has been reported in VAP but yeasts are isolated more frequently from respiratory tract specimens in the absence of apparent disease, even when retrieved at high concentrations from bronchoscopic specimens.[56–60] Thus, based on current data, the presence of yeasts in respiratory secretions obtained from nonimmunosuppressed ventilated patients usually indicates colonization rather than infection of the respiratory tract and does not justify by itself a specific antifungal therapy. Evidence of lung tissue invasion is needed for making the diagnosis of *Candida* pneumonia in such a setting. Interactions, however, between *Candida* and bacteria, particularly *Pseudomonas,* have been reported, and colonization of the respiratory tract by yeasts may predispose to bacterial VAP.[61–64]

By examining currently available data, the clinical significance of anaerobes in the pathogenesis and outcome of VAP remains unclear except as etiologic agents in patients with necrotizing pneumonitis, lung abscess, or pleuropulmonary infections. Anaerobic infection and coverage with antibiotics, such as clindamycin or metronidazole, should probably also be considered for patients with respiratory secretions documenting numerous extracellular and intracellular microorganisms after Gram staining in the absence of positive cultures for aerobic pathogens.

Pathogenesis and Predisposing Factors

Pneumonia results from microbial invasion of the normally sterile lower respiratory tract and lung parenchyma caused by either a defect in host defenses, a challenge by a particularly virulent microorganism, or an overwhelming inoculum. The normal human respiratory tract possesses a variety of defense mechanisms that protect the lung from infection. Examples include anatomic barriers, such as the glottis and larynx; cough reflexes; tracheobronchial secretions; mucociliary lining; cell-mediated and humoral immunity; and a dual phagocytic system that involves both alveolar macrophages and neutrophils.[1] When these coordinated components function properly, invading microbes are eliminated and clinical disease is avoided. When these defenses are impaired, however, or if they are overcome by virtue of a high inoculum of organisms or organisms of unusual virulence, pneumonitis may result.

As suggested by the infrequent association of VAP with bacteremia, most of these infections appear to result from aspiration of potential pathogens that have colonized the mucosal surfaces of the oropharyngeal airways. Intubation of the patient not only compromises the natural barrier between the oropharynx and trachea, but may also facilitate the entry of bacteria into the lung by pooling and leakage of contaminated secretions around the endotracheal tube cuff.[65] This phenomenon occurs in most intubated patients, whose supine position may facilitate its occurrence. In previously healthy, newly hospitalized patients, normal mouth flora or pathogens associated with community-acquired pneumonia may predominate. In sicker patients who have been hospitalized more than 5 days, gram-negative bacilli and *S. aureus* frequently colonize the upper airway.[66]

Uncommonly, VAP may arise in other ways. Observed "macroaspirations" of gastric material initiate the process in some patients. Allowing condensates in ventilator tubing to drain into the patient's airway may have the same effect. Fiber-optic bronchoscopy, tracheal suctioning, or manual ventilation with contaminated equipment may also bring pathogens to the lower respiratory tract. More recently, concerns have focused on the potential role of contaminated inline medication nebulizers, but these devices are infrequently associated with VAP.

Risk factors for tracheobronchial colonization with gram-negative bacilli appear to be the same as those that favor pneumonia and include more severe illness, longer hospitalization, prior or concomitant use of antibiotics, malnutrition, intubation, azotemia, and underlying pulmonary disease.[67] Experimental investigations have linked some of these risk factors to changes in adherence of gram-negative bacilli to respiratory epithelial cells. Although formerly attributed to losses of cell-surface fibronectin, these changes in adherence more likely reflect alterations of cell-surface carbohydrates. Bacterial adhesins and prior antimicrobial therapy appear to facilitate the process. Interestingly, *Enterobacteriaceae* usually appear in the oropharynx first, whereas *P. aeruginosa* more often appears first in the trachea.[68]

Other sources of pathogens causing VAP include the paranasal sinuses, dental plaque, and the subglottic area between the true vocal cords and the endotracheal tube cuff. Not all authors agree that the gastropulmonary route of infection is truly operative in ICU patients.[69] Progression of colonization from the stomach to the upper respiratory tract with subsequent episodes of VAP could not be demonstrated in several studies and efforts to eliminate the gastric reservoir with antimicrobial therapy without decontaminating the oropharyngeal cavity have generally failed to prevent VAP.[69–71] In fact, there is more than one potential pathway for colonization of the oropharynx and trachea in such a setting, including fecal–oral cross-infection on the hands of health care personnel, and contaminated respiratory therapy equipment. Patient-care activities, such as bathing, oral care, tracheal suctioning, enteral feeding, and the tube manipulations, provide ample opportunities for transmission of pathogens when infection-control practices are substandard.[72]

Risk Factors

Risk factors provide information on the probability of lung infection developing in individuals and populations. Thus, they may contribute to the elaboration of effective preventive strategies by indicating which patients might be most likely to benefit from prophylaxis against pneumonia. Table 46-2 summarizes the independent factors for VAP that were identified by multivariate analyses in selected studies.[13,25,38,73–80]

Table 46-2: Independent Factors for Ventilator-Associated Pneumonia Identified by Multivariate Analysis

Host Factors	Intervention Factors
Serum albumin <2.2 g/dL Age ≥ 60 years ARDS COPD, pulmonary disease Coma or impaired consciousness Burns, trauma Organ failure Severity of illness Large-volume gastric aspiration Gastric colonization and pH Upper respiratory tract colonization Sinusitis	H_2 blockers ± antacids Paralytic agents, continuous intravenous Sedation >4 units of blood products Intracranial pressure monitoring MV >2 days Positive end-expiratory pressure Frequent ventilator-circuit changes Reintubation Nasogastric tube Supine head position Transport out of the ICU Prior antibiotic or no antibiotic therapy[a] *Other factor:* Season: fall, winter

Abbreviations: ARDS, acute respiratory distress syndrome; *COPD,* chronic obstructive pulmonary disease; ICU, intensive care unit; *MV,* mechanical ventilation.

[a]See text for explanations.

Surgery

Postsurgical patients are at increased risk for VAP. In a 1981 report, the pneumonia rate during the postoperative period was 17%.[81] Those authors stated that the development of pneumonia was closely associated with preoperative markers of severity of the underlying disease, such as low serum albumin concentration and a high score on the American Society of Anesthesiologists preanesthesia physical status classification. A history of smoking, longer preoperative stays, longer surgical procedures and thoracic or upper abdominal surgery were also significant risk factors for postsurgical pneumonia. Another study comparing adult ICU populations demonstrated that postoperative patients had consistently higher rates of nosocomial pneumonia than did medical ICU patients, with a risk ratio of 2.2.[79] Multiple regression analysis was performed to identify independent predictors of nosocomial pneumonia in the two groups; for surgical ICU patients, mechanical ventilation (>2 days) and Acute Physiology and Chronic Health Evaluation (APACHE) score were retained by the model; for the medical ICU population, only mechanical ventilation (>2 days) remained significant. It has been suggested that different surgical ICU patient populations may have different risks for nosocomial pneumonia: Cardiothoracic surgery and trauma (particularly head trauma) patients were more likely to develop VAP than medical or other types of surgical patients.[13]

Antimicrobial Agents

The use of antibiotics in the hospital setting is associated with an increased risk of nosocomial pneumonia and selection of resistant pathogens.[22,25,45,66,82–85] In a cohort study of 320 patients, prior antibiotic administration was identified by logistic regression analysis to be one of the four variables independently associated with VAP along with organ failure, age greater than 60 years, and the patient's head positioning (i.e., flat on the patient's back or supine vs. head and thorax raised 30 degrees to 40 degrees or semirecumbent).[22] Other investigators, however, found that antibiotic administration during the first 8 days was associated with a lower risk of early onset VAP.[86] For example, Sirvent et al showed that a single dose of a first-generation cephalosporin given prophylactically was associated with a lower rate of early-onset VAP in patients with structural coma.[87] Moreover, multiple logistic regression analysis of risk factors for VAP in 358 medical ICU patients identified the absence of antimicrobial therapy as one of the factors independently associated with VAP onset.[88] Finally, the results of the multicenter Canadian study on the incidence of and risk factors for VAP indicated that antibiotic treatment conferred protection against VAP.[13] This apparent protective effect of antibiotics disappears after 2 to 3 weeks, suggesting that a higher risk of VAP cannot be excluded beyond this point. Thus, risk factors for VAP change over time, thereby explaining why they differ from one series to another.

Stress-Ulcer Prophylaxis

In theory, patients receiving stress-ulcer prophylaxis that does not change gastric acidity, such as sucralfate, should have lower rates of gastric bacterial colonization and, consequently, a lower risk for nosocomial pneumonia, than those receiving antacids or H_2 blockers.[89–90]

According to meta-analyses of the efficacy of stress-ulcer prophylaxis in ICU patients, respiratory tract infections were significantly less frequent in patients treated with sucralfate than those receiving antacids or H_2 blockers.[91,92] This conclusion, however, was not fully confirmed in a very large, multicenter, randomized, blinded, placebo-controlled trial that compared sucralfate suspension (1 g every 6 hours) with the H_2-receptor antagonist ranitidine (50 mg every 8 hours) for the prevention of upper gastrointestinal bleeding in 1200 ventilated patients.[93] Clinically relevant gastrointestinal bleeding developed in ten of the 596 (1.7%) patients receiving ranitidine, as compared with twenty-three of the 604 (3.8%) receiving sucralfate (relative risk [RR], 0.44; 95% confidence interval [CI], 0.21 to 0.92; $p = 0.02$). In the ranitidine group, 114 of 596 (19.1%) patients had VAP, as diagnosed by an adjudication committee using a modified version of the Centers for Disease Control and Prevention criteria, versus ninety-eight of 604 (16.2%) in the sucralfate group (RR, 1.18; 95% CI, 0.92 to 1.51; $p = 0.19$). VAP, however, occurred significantly less frequently in patients receiving sucralfate when the diagnosis of pneumonia was based on Memphis VAP Consensus Conference criteria (if there was radiographic evidence of abscess and a positive needle aspirate, or histologic proof of pneumonia at biopsy or autopsy) ($p = 0.03$).[93]

Sucralfate appears to have a small protective effect against VAP because stress-ulcer prophylactic medications that raise the gastric pH might themselves increase the incidence of pneumonia.[94,95] This contention is supported by direct comparisons of trials of H_2-receptor antagonists versus no prophylaxis, which showed a trend toward higher pneumonia rates among the patients receiving H_2-receptor antagonists (odds ratio [OR], 1.25; 95% CI, 0.78 to 2.00).[91] Furthermore, the comparative effects of sucralfate and no prophylaxis are unclear. Among 226 patients enrolled in two randomized trials, those receiving sucralfate tended to develop pneumonia more frequently than those given no prophylaxis (OR, 2.11; 95% CI, 0.82 to 5.44).[96,97]

Endotracheal Tube, Reintubation, and Tracheotomy

The presence of an endotracheal tube by itself circumvents host defenses, causes local trauma and inflammation, and increases the probability of aspiration of nosocomial pathogens from the oropharynx around the cuff. Scanning electron microscopy of 25 endotracheal tubes revealed that 96% had partial bacterial colonization and 84% were completely coated with bacteria in a biofilm or glycocalyx.[98] The authors hypothesized that bacterial aggregates in biofilm dislodged during suctioning might not be killed by antibiotics or effectively cleared by host immune defenses. Clearly, the type of endotracheal tube may also influence the likelihood of aspiration. Use of low-volume, high-pressure endotracheal cuffs reduced the rate to 56% and the advent of high-volume, low-pressure cuffs further lowered it to 20%.[99] Leakage around the cuff allows secretions pooled above the cuff to enter the trachea; this mechanism, recently confirmed, underlines the importance of maintaining adequate intracuff pressure for preventing VAP.[100]

In addition to the presence of endotracheal tubes, reintubation is, per se, a risk factor for VAP.[101] This finding probably reflects an increased risk of aspiration of colonized oropharyngeal secretions into the lower airways by patients with subglottic dysfunction or impaired consciousness after several days of intubation. Another explanation is direct aspiration of gastric contents into the lower airways, particularly when a nasogastric tube is kept in place after extubation.

Some investigators postulated that early tracheotomy could lower VAP rate because it can permit easier oral hygiene and bronchopulmonary toilet or less time spent deeply sedated.[102] Such benefit, however, was not confirmed in other studies, including two large recent randomized trials having systematically evaluated this issue.[103–106]

Nasogastric Tube, Enteral Feeding, and Position of the Patient

Almost all ventilated patients have a nasogastric tube inserted to evacuate gastric and enteral secretions, prevent gastric distension, and/or provide nutritional support. The nasogastric tube is not generally considered to be a potential risk factor for VAP, but it may increase oropharyngeal colonization, cause stagnation of oropharyngeal secretions, and increase reflux and the risk of aspiration. A multivariate analysis retained the presence of a nasogastric tube as one of the three independent risk factors for nosocomial pneumonia based on a series of 203 patients admitted to the ICU for 72 hours or more.[77]

Early initiation of enteral feeding is generally regarded as beneficial in critically ill patients, but it may increase the risk of gastric colonization, gastroesophageal reflux, aspiration, and pneumonia.[107,108] The aspiration rate generally varies as a function of differences in the patient population, neurologic function, type of feeding tube, location of the feeding port, and method of evaluating aspiration. Clinical impressions and preliminary data suggest that postpyloric or jejunal feeding entails less risk of aspiration and may therefore be associated with fewer infectious complications than gastric feeding, although this point remains controversial.[109,110] Nonetheless, aspiration can easily occur should the feeding tube be inadvertently dislodged. A retrospective study of non–critically ill adult patients showed a 40% rate of accidental feeding tube dislodgment, but all the patients whose tube was dislodged were confused or disoriented or had altered awareness, as is frequently observed in ICU patients.[111]

Maintaining ventilated patients with a nasogastric tube in place in a supine position is also a risk factor for aspiration of gastric contents into the lower airways. When radioactive material was injected through a nasogastric tube directly into the stomach of nineteen ventilated patients, the mean radioactive counts in endobronchial secretions were higher in a time-dependent fashion in samples obtained from patients in a supine position than in those obtained from patients in a semirecumbent position.[112] The same microorganisms were isolated from the stomach, pharynx, and endobronchial samples of 32% of the specimens taken while patients were lying supine. The same investigators conducted a randomized trial comparing semirecumbent and supine positions.[113] The trial, which included eighty-six intubated and ventilated patients, was stopped after the planned interim analysis because the frequency and the risk of VAP were significantly lower for the semirecumbent group. These findings were indirectly confirmed by the demonstration that the head position of the supine patient during the first 24 hours of mechanical ventilation was an independent risk factor for acquiring VAP.[22] To what degree of elevation, however, the head of bed should be targeted remains controversial.[114–117]

Respiratory Equipment

Ventilators with humidifying cascades often have high levels of tubing colonization and condensate formation that may also be risk factors for pneumonia. The rate of condensate formation in the ventilator circuit is linked to the temperature difference between the inspiratory-phase gas and the ambient temperature, and may be as high as 20 to 40 mL/hour.[118,119] Examination of condensate colonization in twenty circuits detected a median level of 2.0×10^5 organisms/mL, and 73% of the fifty-two gram-negative isolates present in the patients' sputum samples were subsequently isolated from condensates.[119] Because most of the tubing colonization was derived from the patients' secretions, the highest bacterial counts were present near the endotracheal tube. Simple procedures, such as turning the patient or raising the bed rail, may accidentally spill contaminated condensate directly into the patient's tracheobronchial tree.[120] Inoculation of large amounts of fluid with high bacterial concentrations is an excellent way to overwhelm pulmonary defense mechanisms and cause pneumonia. Heating ventilator tubing markedly lowers the rate of condensate formation, but heated circuits are often nondisposable and are expensive. Inline devices with one-way valves to collect the condensate are probably the easiest way to handle this problem; they must be correctly positioned into disposable circuits and emptied regularly.

To decrease condensation and moisture accumulation in ventilator circuits, several studies have investigated the use of heat–moisture exchangers in place of conventional heated-water humidification systems. Slightly lower VAP rates were observed in four studies and a significant difference in a fifth study, suggesting that heat–moisture exchangers are at least comparable to heated humidifiers and may be associated with lower VAP rates than heated humidifiers.[121–125] Changing the heat–moisture exchangers every 48 hours did not affect ventilator circuit colonization and the authors concluded that the cost of mechanical ventilation might be substantially reduced without any detriment to the patient by prolonging the time between heat–moisture exchangers changes from 24 to 48 hours.[126] Furthermore, using heat–moisture exchangers may decrease the nurses' workload (no need to refill cascades, to void water traps on circuits, and so on), decrease the number of septic procedures (it was clearly shown that respiratory tubing condensates must be handled as an infectious waste), and reduce the cost of mechanical ventilation, especially when used for prolonged periods without change. Because some observational studies, however, have documented an increased resistive load and a larger dead space associated with exchangers,[127,128] their use should be discouraged in patients with ARDS ventilated with a low tidal volume and in patients with chronic obstructive pulmonary disease during the weaning period, if pressure support, and not T-piece trials, are used.

There is no apparent advantage to changing ventilator circuits frequently for VAP prevention. This holds true whether circuits are changed every 2 days or every 7 days compared with no change at all, and whether they are changed weekly as opposed to three times per week.[129–131] A policy of no circuit changes or infrequent circuit changes is simple to implement and the costs are likely lower than those generated by regular, frequent circuit changes; thus, such a policy is strongly recommended by the 1997 Centers for Disease Control and Prevention guidelines and other guidelines.[132–134]

Sinusitis

While many studies have compared the risk of nosocomial sinusitis as a function of the intubation method used and the associated risk of VAP, only a few were adequately powered to give a clear answer. In one study of 300 patients who required mechanical ventilation for at least 7 days and were randomly assigned to undergo nasotracheal or orotracheal intubation, computed tomographic evidence of sinusitis was observed slightly more frequently in the nasal than oral endotracheal group ($p = 0.08$), but this difference disappeared when only bacteriologically confirmed sinusitis was considered.[135]

The rate of infectious maxillary sinusitis and its clinical relevance were also prospectively studied in 162 consecutive critically ill patients, who had been intubated and ventilated for 1 hour to 12 days before enrollment.[136] All had a paranasal computed tomography scan within 48 hours of admission, which was used to divide them into three groups (no, moderate, or severe sinusitis), according to the radiologic appearance of the maxillary sinuses. Patients who had no sinusitis at admission ($n = 40$) were randomized to receive endotracheal and gastric tubes via the nasal or oral route and, based on radiologic images, respective sinusitis rates were 96% and 23% ($p < 0.03$); yet, no differences in the rates of infectious sinusitis were documented according to the intubation route. VAP, however, was more common in patients with infectious sinusitis, with 67% of them developing lung infection in the days following the diagnosis of sinusitis.[136] Therefore, although it seems clear that infectious sinusitis is a risk factor for VAP, no studies have yet been able to definitively demonstrate that orotracheal intubation decreases the infectious sinusitis rate compared to nasotracheal intubation. Thus, no firm recommendations on the best route of intubation to prevent VAP can be advanced.

Intrahospital Patient Transport

A prospective cohort study conducted in 531 ventilated patients evaluated the impact of transporting the patient out of the ICU to other sites within the hospital.[137] Results showed that 52% of the patients had to be moved at least once, for a total of 993 transports, and that 24% of the transported patients developed VAP compared with 4% of the patients confined to the ICU ($p < 0.001$). Multiple logistic regression analysis confirmed that transport out of the ICU was independently associated with VAP (OR = 3.8; $p < 0.001$).

Diagnosis

VAP is typically suspected when a patient has new or progressive radiographic infiltrates and clinical findings suggesting infection, such as the new onset of fever, purulent sputum, leukocytosis, increased minute ventilation, and/or a decline in arterial oxygenation. Because interpretation of chest radiographs is difficult, particularly in patients with prior abnormalities, such as ARDS, it is also mandatory to consider the diagnosis of VAP in ventilated patients who clinically deteriorate, and/or in whom vasopressors should be increased to maintain blood pressure, even in the absence of a clear-cut progression of the radiographic abnormalities.

The systemic signs of infection, however, such as fever, tachycardia, and leukocytosis, are nonspecific findings that can be caused by any condition that releases cytokines. In trauma and other surgical patients, fever and leukocytosis should prompt the physician to suspect infection, but during the early posttraumatic or postoperative period (i.e., during the first 72 hours), these findings usually are not conclusive. Later, fever and leukocytosis are more likely to be caused by pulmonary or nonpulmonary (vascular catheter infection, gastrointestinal infection, urinary tract infection, sinusitis, or wound infection) infections, but even then, other events associated with an inflammatory response (e.g., devascularized tissue, open wounds, pulmonary edema, and/or infarction) can be responsible for these findings. Although the plain (usually portable) chest roentgenogram remains an important component in the evaluation of ventilated patients with suspected pneumonia, it is most helpful when it is normal and rules out pneumonia. When infiltrates are evident, the particular pattern is of limited value for differentiating among cardiogenic pulmonary edema, noncardiogenic pulmonary edema, pulmonary contusion, atelectasis (or collapse), and pneumonia, even when using computed tomographic scanning.[8,17,138–142] Because the tracheobronchial tree of mechanically ventilated patients is frequently rapidly colonized by potential pathogens, the presence of bacteria at that level is not a sufficient argument to diagnose true lung infection, which constitutes another major obstacle for the diagnosis of VAP.[8,143]

In 1991, a composite clinical score, the clinical pulmonary infection score (CPIS) was proposed, based on seven variables (temperature, blood leukocyte count, volume and purulence of tracheal secretions, oxygenation, pulmonary radiography, and semiquantitative culture of tracheal aspirate) accorded 0, 1, or 2 points.[144] This scoring system, however, is quite tedious to calculate and difficult to use in clinical practice, because several variables, such as progression of pulmonary infiltrates and results of semiquantitative cultures of tracheal secretions, can lead to different calculations depending on the observer.[145] Furthermore, its value was not validated in several subsequent prospective studies, especially in patients with bilateral pulmonary infiltrates.[146–154]

Thus, as soon as a ventilated patient is suspected of developing pneumonia, a more complete diagnostic workup should be undertaken, targeting two objectives. The first objective is the immediate recognition of a true VAP or of an extrapulmonary bacterial infection, so as to start effective antibiotics against the microorganisms responsible for infection as soon as possible.[1,2] Numerous studies indicate that failure to initiate prompt appropriate antimicrobial treatment in this setting is a major risk factor for an increased morbidity and mortality.[155–163] The second objective is to avoid overusing antibiotics in patients with only proximal airways colonization and no ongoing bacterial infection. Epidemiologic investigations have clearly demonstrated that indiscriminate use of antimicrobial agents in ICU patients may have immediate and long-term consequences, which contribute to emergence of multiresistant pathogens and increase the risk of serious superinfections.[164–169] This risk is not limited to one patient. Instead, the risk of colonization or infection by multidrug-resistant strains is increased in patients throughout the ICU and even the entire hospital. Virtually all reports emphasize that better antibiotic control programs to limit bacterial resistance are urgently needed in ICUs, and that patients without true infection should not receive antimicrobial treatment.[164]

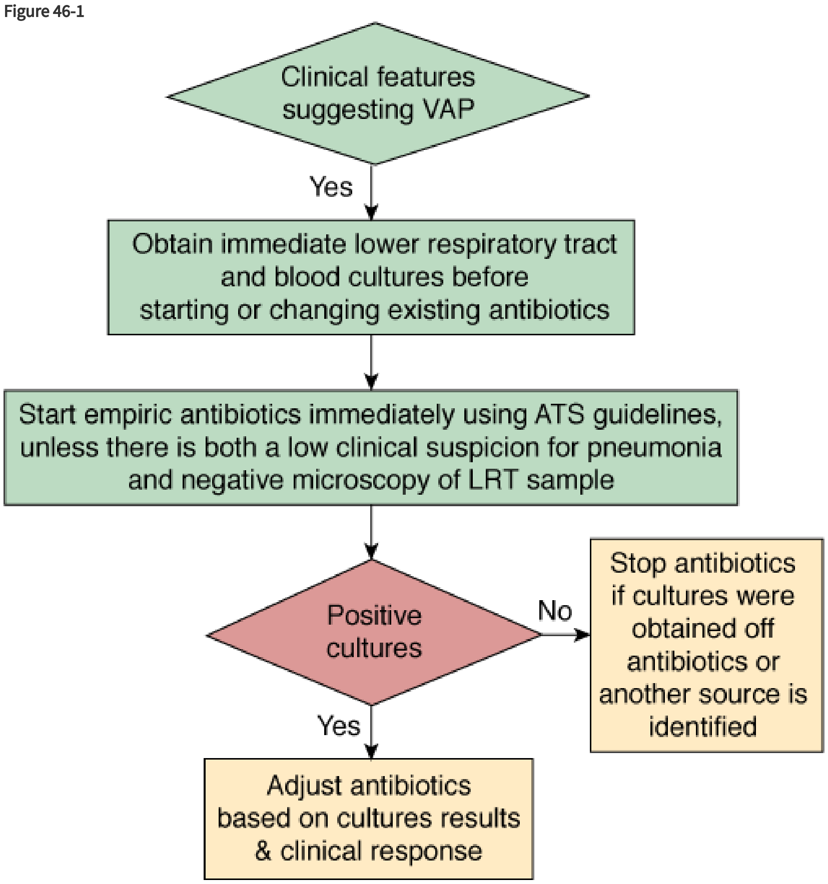
To reach these objectives, all diagnostic strategies should follow three consecutive steps: (a) obtaining a respiratory tract sample (from proximal or distal airways) for microscopy and culture (qualitative, semiquantitative, or quantitative) before introduction of new antibiotics; (b) immediately starting empiric antimicrobial treatment, unless there are both a negative microscopy and no signs of severe sepsis; and (c) reevaluating treatment on day 2 or 3, based on microbiologic cultures results and clinical outcome.[1,2]

Qualitative Cultures of Endotracheal Aspirates

The first option is to use a clinical strategy and to treat every patient clinically suspected of having a pulmonary infection with new antibiotics (even when the likelihood of infection is low), arguing that several studies showed that immediate initiation of appropriate antibiotics was associated with reduced mortality.[28,32,157,170–175] Using this strategy, all patients suspected of having VAP are treated with new antibiotics after having obtaining an endotracheal aspirate for microscopy and qualitative culture. The selection of appropriate empirical therapy is based on risk factors and local microbiologic and resistance patterns, and involves qualitative testing to identify possible pathogens. The initial antimicrobial therapy is adjusted according to culture results and clinical response (Fig. 46-

1). Antimicrobial treatment is discontinued if and only if the following three criteria are fulfilled on day 3: (a) clinical diagnosis of VAP is unlikely (there are no definite infiltrates found on chest radiography at follow-up and no more than one of the three following findings are present: temperature higher than 38.3°C [100.9°F], leukocytosis or leukopenia, and purulent tracheobronchial secretions) or an alternative noninfectious diagnosis is confirmed; (b) tracheobronchial aspirate culture results are nonsignificant; and (c) severe sepsis or shock are not present.[176]

Figure 46-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Diagnostic and therapeutic strategy applied to patients managed with the "clinical" strategy. *ATS,* American Thoracic Society; *LRT,* lower respiratory tract; *VAP,* ventilator-associated pneumonia.

This clinical approach has two undisputable advantages: first, no specialized microbiologic techniques are required, and, second, the risk of missing a patient who needs antimicrobial treatment is minimal when all suspected patients are treated with new antibiotics. Because, however, tracheobronchial aspirate culture results are rarely negative secondary to the high rate of proximal airways colonization observed in patients receiving mechanical ventilation, discontinuation of antibiotics on day 3 is difficult to perform, leading to antibiotic overuse in many ICU patients. Qualitative endotracheal aspirate cultures contribute indisputably to the diagnosis of VAP only when they are completely negative for a patient with no modification of prior antimicrobial treatment. In such a case, the negative-predictive value is very high and the probability of the patient having pneumonia is close to zero.[5] This is why some investigators have proposed to replace qualitative cultures of endotracheal aspirates by semiquantitative or quantitative cultures of the same specimens.[177]

Quantitative Cultures of Endotracheal Aspirates

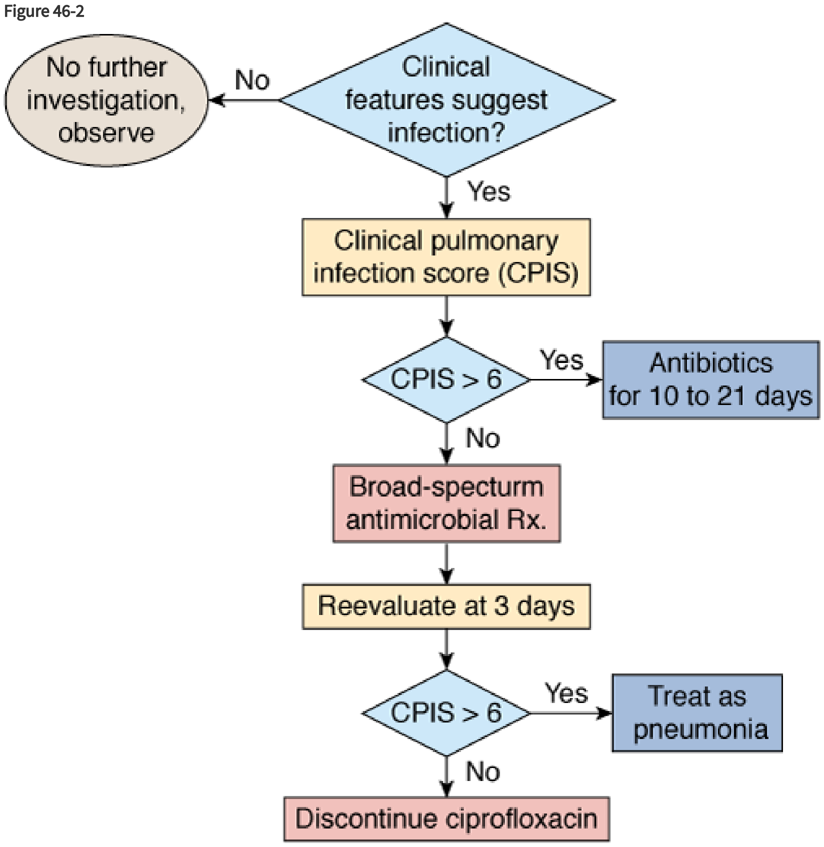
Several studies using quantitative culture techniques suggest that endotracheal aspirate cultures may have an acceptable overall diagnostic accuracy, similar to that of several other more invasive techniques.[177] Not all studies, however, have confirmed this conclusion. To assess the reliability of that method, bronchoscopy with PSB and BAL was used to study fifty-seven episodes of suspected lung infection in thirty-nine ventilator-dependent patients with no recent changes of antimicrobial therapy.[178] The operating characteristics of endotracheal aspirate cultures were calculated over a range of cutoff values (from 10^3 to 10^7 colony-forming units [CFU]/mL); the threshold of 10^6 CFU/mL appeared to be the most accurate, with a sensitivity of 68% and a specificity of 84%. When this threshold was applied to the study population, however, almost one-third of the patients with pneumonia were not identified. Furthermore, only 40% of microorganisms cultured in endotracheal aspirate samples coincided with those obtained from PSB specimens. Other authors have emphasized that, although quantitative endotracheal aspirate cultures can correctly identify patients with pneumonia, microbiologic results cannot be used to infer which microorganisms present in the trachea are really present in the lungs. In a study comparing quantitative endotracheal aspirate culture results to postmortem quantitative lung biopsy cultures, only 53% of the microorganisms isolated from the former samples at concentrations greater than 10^7 CFU/mL were also found in the latter cultures.[179]

The inherent advantage of quantitative cultures of endotracheal aspirates is that they are more specific, permitting the discontinuation of antibiotics in more patients than when using only qualitative cultures. But it must be kept in mind that this technique has several potential pitfalls. First, many patients may not be identified using the cutoff value of 10^6 CFU/mL. Second, as soon as a lower threshold is used, specificity declines sharply and overtreatment becomes a problem. Finally, selecting antimicrobial therapy solely on the basis of endotracheal aspirate culture results can lead to either unnecessary antibiotic therapy or overtreatment with broad-spectrum antimicrobial agents.

The "Singh" Strategy

Another option when using the clinical approach is to follow the strategy described by Singh et al, in which decisions regarding initial antibiotic therapy are based on CPIS, a clinical score constructed from seven variables.[180] Patients with CPIS greater than 6 are treated as having VAP with antibiotics for 10 to 21 days, and antibiotics are discontinued after 3 days if the CPIS is 6 or less (Fig. 46-2). Such an approach avoids prolonged treatment of patients who have a low likelihood of infection, while allowing immediate treatment of patients who are more likely to have VAP.

Figure 46-2



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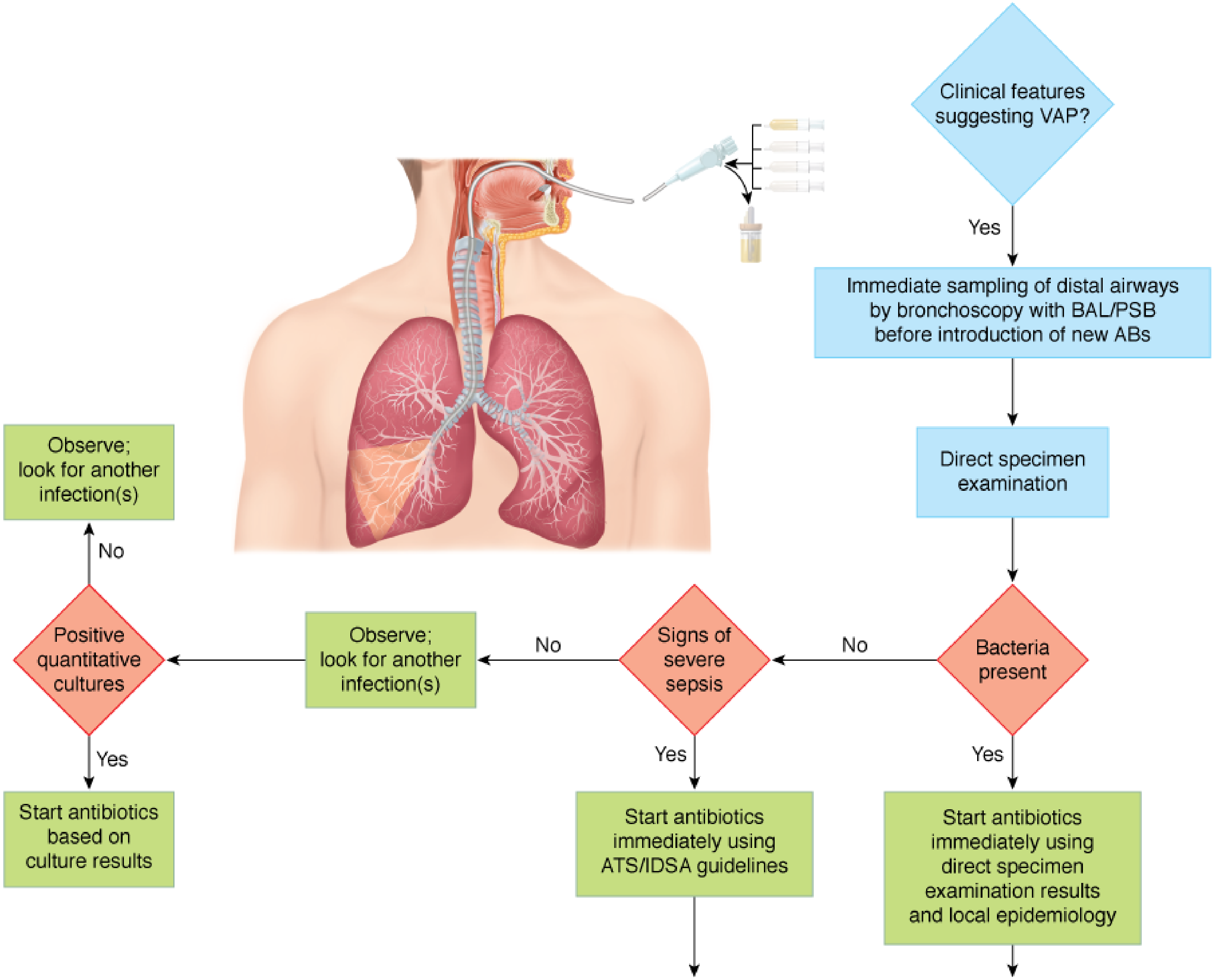
Diagnostic and therapeutic strategy applied to patients managed with the strategy proposed by Singh et al.[180]

Two conditions must be fulfilled when using this strategy. First, the selection of initial antimicrobial therapy should be based on the most common microbes responsible for VAP at each institution. For example, ciprofloxacin would not be the right choice in hospitals with a high prevalence of MRSA infections. Second, physicians should reevaluate antimicrobial treatment on day 3, when susceptibility patterns of the microorganism(s) recovered from pulmonary secretions are available, so as to select treatment with a narrower spectrum antibiotic.

Quantitative Cultures of Distal Specimens Obtained by Bronchoscopy

This strategy uses quantitative cultures of lower respiratory secretions (BAL or PSB collected with a bronchoscope) to define both the presence of pneumonia and the etiologic pathogen(s). Pathogens are present in inflammatory secretions of the lower respiratory tract at concentrations of at least 10^5 to 10^6 CFU/mL, whereas contaminants are generally present at less than 10^4 CFU/mL.[181] The diagnostic thresholds proposed for PSB and BAL are based on this concept. Because PSB collects between 0.001 and 0.01 mL of secretions, the presence of greater than 10^3 bacteria in the originally diluted sample (1 mL) actually represents 10^5 to 10^6 CFU/mL of pulmonary secretions. Similarly, 10^4 CFU/mL for BAL, which collects 1 mL of secretions in 10 to 100 mL of effluent, represents 10^5 to 10^6 CFU/mL.[182–184] Using this strategy, therapeutic decisions are tightly protocolized, using the results of direct examination of distal pulmonary samples and results of quantitative cultures in deciding whether to start antibiotic therapy, which pathogens are responsible for infection, which antimicrobial agents to use, and whether to continue therapy (Fig. 46-3).

Figure 46-3

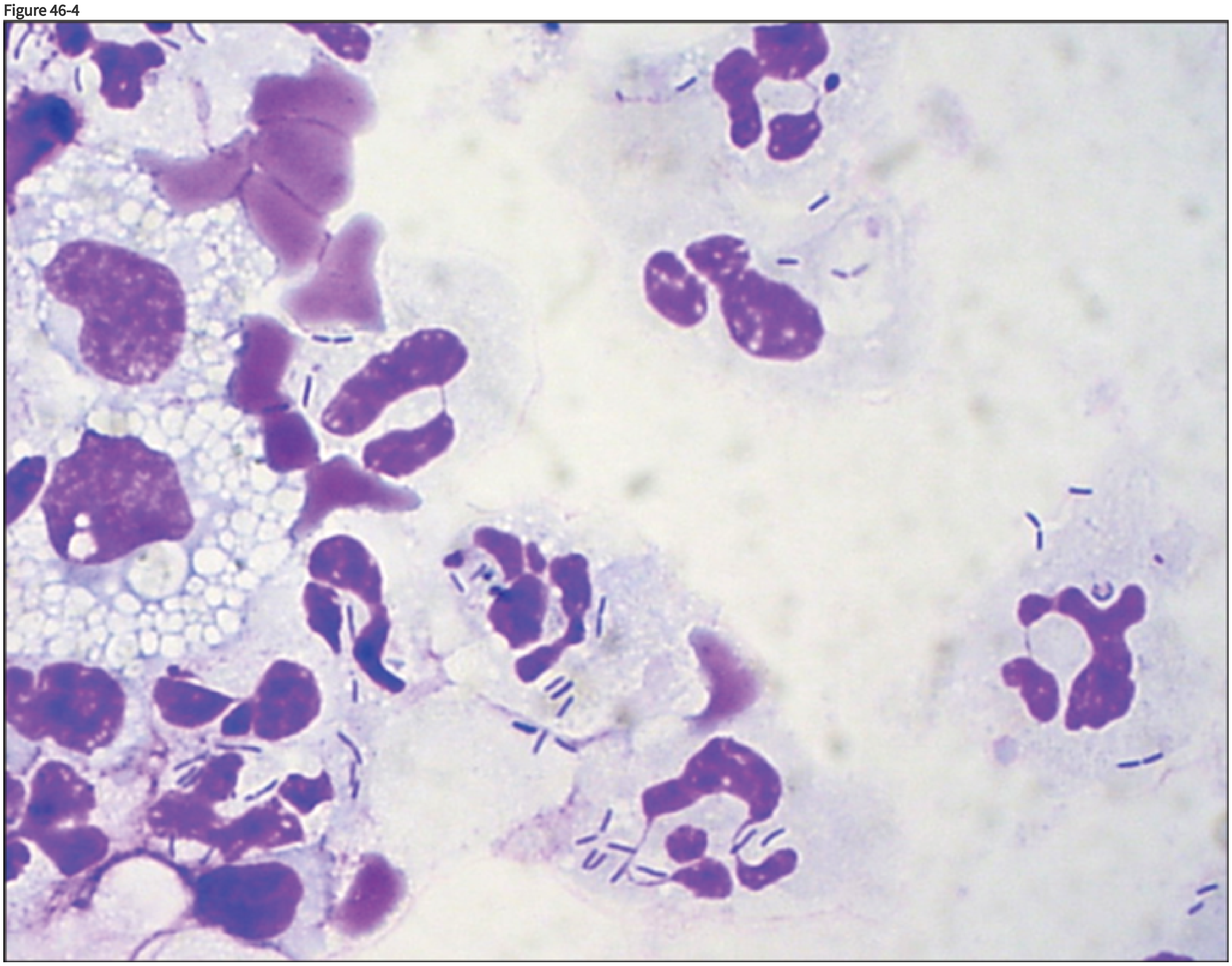


Diagnostic and therapeutic strategy applied to patients managed with the "invasive" strategy. *AB,* antibiotic; *ATS,* American Thoracic Society; *BAL,* bronchoalveolar lavage; *IDSA,* Infectious Disease Society of America; *PSB,* protector specimen brush; *VAP,* ventilator-associated pneumonia.

One major technical problem with all bronchoscopic techniques is proper selection of the sampling area in the tracheobronchial tree. Almost all intubated patients have purulent-looking secretions and the secretions first seen may represent those aspirated from another site into gravity-dependent airways or from upper-airway secretions aspirated around the endotracheal tube. Usually, the sampling area is selected based on the location of infiltrate on chest radiograph or the segment visualized during bronchoscopy as having purulent secretions.[185] Collection of secretions in the lower trachea or mainstem bronchi, which may represent recently aspirated secretions around the endotracheal tube cuff, should be avoided. In patients with diffuse pulmonary infiltrates or minimal changes in a previously abnormal chest radiograph, determining the correct airway to sample may be difficult. In these cases, sampling should be directed to the area where endobronchial abnormalities are maximal.[186] In case of doubt, and because autopsy studies indicate that VAP frequently involves the posterior portion of the right lower lobe, this area should probably be sampled as a priority.[187] Although bilateral sampling in the immunosuppressed host with diffuse infiltrates has been advocated, there is no convincing evidence that multiple specimens are more accurate than single specimens for diagnosing nosocomial bacterial pneumonia in ventilated patients.

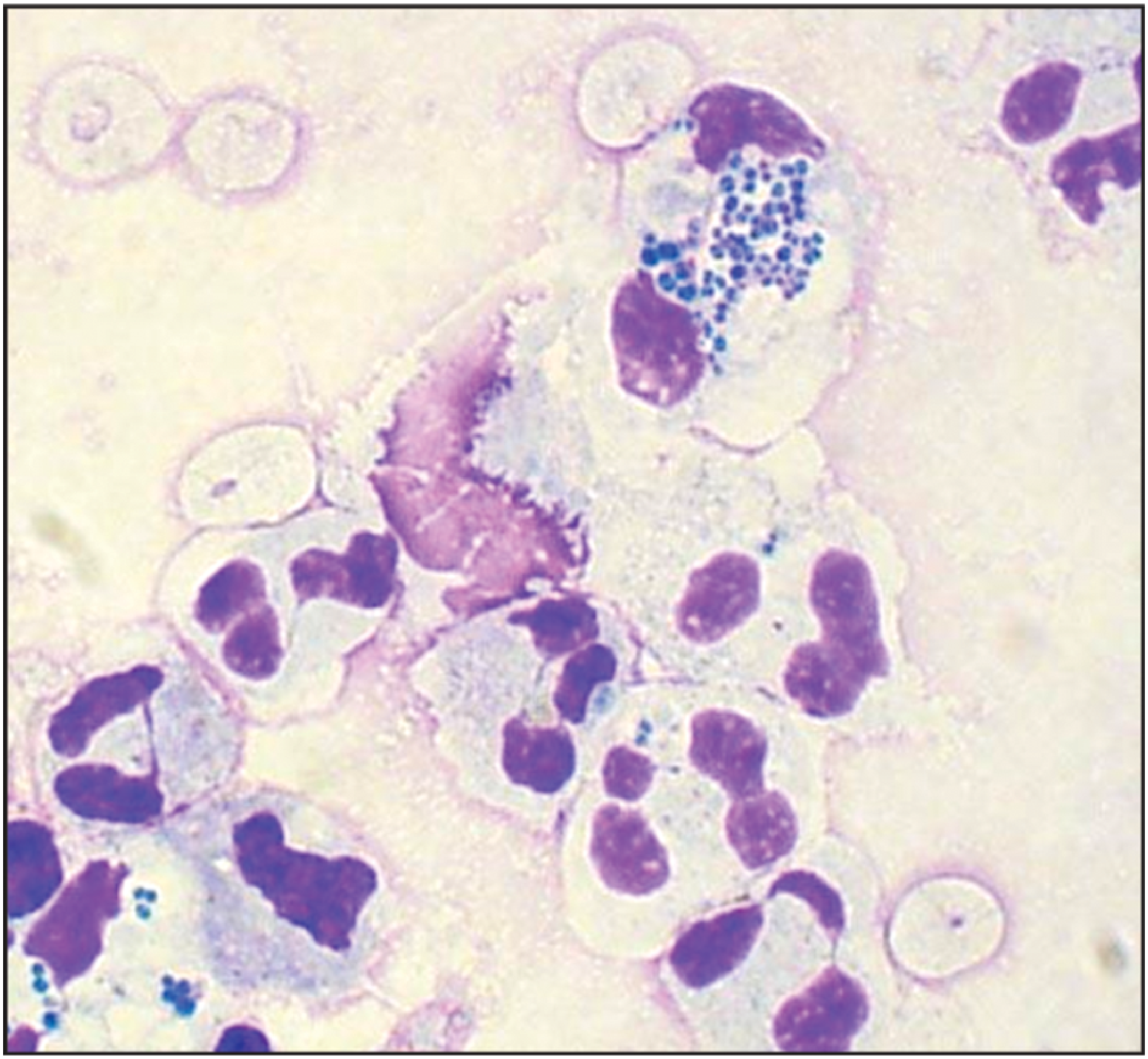
Because BAL harvests of cells and secretions from a large area of the lung and specimens can be microscopically examined immediately after the procedure to detect the presence or absence of intracellular or extracellular bacteria in the lower respiratory tract, it is particularly well suited to provide rapid identification of patients with pneumonia (Figs. 46-4 and 46-5).[183,188–193] Assessment of the degree of qualitative agreement between Gram stains of BAL fluid and PSB quantitative cultures for a series of fifty-one patients with VAP, however, showed correspondence to be complete for 51%, partial for 39%, and nonexistent for 10% of the cases.[190]

Figure 46-4



Light micrograph of cells recovered by BAL from a patient with *Pseudomonas aeruginosa* pneumonia (Diff-Quick stain).

Figure 46-5



Light micrograph of cells recovered by BAL from a patient with *Staphylococcus aureus* pneumonia (Diff-Quick stain).

Many groups have investigated the value of quantitative BAL culture for the diagnosis of pneumonia in ICU patients.[183,194,195] When the results of the eleven studies evaluating BAL fluids from a total of 435 ICU patients with nosocomial pneumonia were pooled, overall accuracy was very close to that of PSB: The Q value was 0.84 (Q represents the intersection between the summary receiver operating characteristics curve and a diagonal from the upper-left corner to the lower-right corner of the receiver operating characteristics space).[194] Similar conclusions were drawn in another meta-analysis, which pooled the results of twenty-three studies: Sensitivity and specificity of BAL were 73% ± 18% and 82% ± 19%, respectively.[195] When analysis in these studies was restricted to patients without prior antibiotics or when only lung tissue cultures were used as the reference standard, results of bronchoscopic techniques pneumonia were much better: Sensitivity was always greater than 80%.

Other studies confirm the accuracy of bronchoscopic techniques for diagnosing nosocomial pneumonia. In a study evaluating spontaneous lung infections occurring in ventilated baboons with permeability pulmonary edema, Johanson et al found excellent correlation between the bacterial content of lung tissue and results of quantitative culture of lavage fluid.[196] BAL recovered 74% of all species present in lung tissue, including 100% of species present at a concentration equal to or greater than 10^4 CFU/g of tissue. Similarly, in twenty ventilated patients who had not developed pneumonia before the terminal phase of disease and who had no recent changes in antimicrobial therapy, Chastre et al found that bronchoscopic BAL specimens obtained just after death identified 90% of all species present in the lung, with a strong correlation between the results of quantitative cultures of both specimens.[183] These findings confirm that bronchoscopic BAL samples very reliably identify, both qualitatively and quantitatively, microorganisms present in lung segments, even when the pneumonia develops as a superinfection in a patient already receiving antimicrobial treatment for several days.

Values within 1 $\log_{10}$ of the cutoff must, however, be interpreted cautiously, and bronchoscopy should be repeated in symptomatic patients with a negative (<10^4 CFU/mL) result.[197] Many technical factors, including medium and adequacy of incubation, and antibiotic or other toxic components, may influence results. Reproducibility of PSB sampling has been recently evaluated by three groups.[198–200] Although in vitro repeatability was excellent and in vivo qualitative recovery 100%, quantitative results were more variable. In 14% to 17% of patients, results of replicate samples fell on both sides of the 10^3 CFU/mL threshold, and results varied by more than 1 $\log_{10}$ in 59% to 67% of samples.[198–200] This variability is presumably related to both the irregular distribution of organisms in secretions and the very small volume actually sampled by PSB. As with all diagnostic tests, borderline PSB and/or BAL quantitative culture results should be interpreted cautiously, and the clinical circumstances should be considered before reaching any therapeutic decision.

The most compelling argument for invasive techniques coupled with quantitative cultures of PSB or BAL specimens is that they can reduce excessive antibiotic use. There is little disagreement that the clinical diagnosis of nosocomial pneumonia is overly sensitive and leads to the unnecessary use of broad-spectrum antibiotics. Because bronchoscopic techniques may be more specific, their use would reduce antibiotic pressure in the ICU, thereby limiting the emergence of drug-resistant strains and the attendant increased risks of superinfection.[22,201] When culture results are available, BAL and/or PSB techniques facilitate precise identification of the offending organisms and their susceptibility patterns. Such data are invaluable for optimal antibiotic selection in patients with a true VAP. They also increase the confidence and comfort level of health care workers in managing patients with suspected nosocomial pneumonia.[202] The more targeted use of antibiotics also could reduce overall costs, despite the expense of bronchoscopy and quantitative cultures, and minimize antibiotic-related toxicity. This is particularly true in patients who have late-onset VAP, in whom expensive combination therapy is commonly recommended. A conservative cost analysis in a trauma ICU suggested that the discontinuation of antibiotics upon the return of negative bronchoscopic quantitative culture results could lead to a savings of more than $1700 per patient suspected of VAP.[203]

Finally, a major benefit of a negative bronchoscopy is to direct attention away from the lungs as the source of fever. Many hospitalized patients with negative bronchoscopic cultures have other potential sites of infection that can be identified via a simple diagnostic protocol. In fifty patients with suspected VAP who underwent a systematic diagnostic protocol designed to identify all potential causes of fever and pulmonary densities, Meduri et al confirmed that lung infection was present in only 42% of cases; the frequent occurrence of multiple infectious and noninfectious processes justifies a systematic search for the source of fever in this setting.[141] Delay in diagnosis or definitive treatment of the true site of infection may lead to prolonged antibiotic therapy, more antibiotic-associated complications, and induction of further organ dysfunction.[204]

Quantitative Cultures of Distal Specimens Obtained Without Bronchoscopy

At least fifteen studies have described a variety of nonbronchoscopic techniques using various types of endobronchial catheters for sampling distal lower respiratory tract secretions; globally, results have been similar to those obtained with bronchoscopy.[205] Compared to conventional PSB and/or BAL, nonbronchoscopic techniques are less invasive, can be performed by clinicians not qualified to perform bronchoscopy, have lower initial costs than bronchoscopy, avoid potential contamination by the bronchoscopic channel, are associated with less compromise of gas exchange during the procedure, and can be performed even in patients intubated with small endotracheal tubes. Disadvantages include the potential sampling errors inherent in a blind technique and the lack of airway visualization. Although autopsy studies indicate that pneumonia in ventilator-dependent patients has often spread into every pulmonary lobe and predominantly involves the posterior portion of the lower lobes, several clinical studies on ventilated patients with pneumonia contradict those findings, as some patients had sterile cultures of PSB specimens from the noninvolved lung.[17,206]

Furthermore, although the authors of most studies concluded that the sensitivities of nonbronchoscopic and bronchoscopic techniques were comparable, the overall concordance was only approximately 80%, emphasizing that, in some patients, the diagnosis could be missed by a blind technique, especially in the case of pneumonia involving the left lung.[17]

Patients Already Receiving Antimicrobial Therapy

Performing microbiologic cultures of pulmonary secretions for diagnostic purposes after initiation of new antibiotic therapy in patients suspected of having developed VAP leads to a high rate of false-negative results, regardless of the method of obtaining the secretions. In fact, all microbiologic techniques are of limited value in patients with a recent infiltrate who have received new antibiotics, even if they have received the new antibiotics for less than 24 hours. A negative finding could indicate that the patient has been successfully treated for pneumonia and the bacteria are eradicated, or that the patient had no lung infection to begin with. Using both PSB and BAL, Souweine et al prospectively investigated sixty-three episodes of suspected VAP.[207] If patients had been treated with antibiotics but did not have a recent change in antibiotic class, sensitivity of PSB and BAL culture (83% and 77%, respectively) were similar to the sensitivities achieved in patients not being treated with antibiotics. In other words, prior therapy did not reduce the yield of diagnostic testing among patients receiving current antibiotics given to treat a prior infection. Conversely, if therapy was recent, sensitivity of invasive diagnostic methods, using traditional thresholds, was only 38% with BAL and 40% with PSB.[207] These two clinical situations should be clearly distinguished before interpreting the results of pulmonary secretion cultures, irrespective of how they were obtained. In the second situation, when the patient receives new antibiotics after the appearance of signs suggesting VAP, no conclusion concerning the presence or absence of pneumonia can be drawn if culture results are negative.[207–209] Pulmonary secretions therefore need to be obtained before starting new antibiotics, as is the case for all types of microbiologic samples.

Use of Procalcitonin and Other Biologic Markers

Procalcitonin (PCT), a 116-amino-acid peptide that is one of the precursors of the hormone calcitonin, has been described as a good diagnostic marker of bacterial infection in patients with community-acquired infections, especially in patients with lower respiratory tract infection.[210–213] Moreover, several interventional trials have shown that PCT could be used to start or to postpone antibiotic treatment in community-acquired lower respiratory tract infections.[214–218] In patients with nosocomial infections, and particularly in patients with VAP, its usefulness as a diagnostic marker is more doubtful.[219–223] There are several reasons to explain why PCT is not a good diagnostic marker in patients with suspected VAP. First, pneumonia may be a localized infection; thus, as for other localized infections, PCT can be synthesized locally without systemic release, explaining its low serum level or apparent decrease in patients with true pulmonary infections. Second, ICU patients may suffer from previous severe sepsis or septic shock, multiorgan failure, or may have developed a systemic inflammatory response syndrome after surgery or trauma, conditions known to increase blood level of biomarkers including PCT in the absence of infection.[222] Thus, a high level of PCT the day VAP is suspected is not useful, because it is not possible to distinguish an elevation caused by a previous noninfectious condition from an elevation caused by an active infection. Third, it is known that a time lag of 24 to 48 hours can exist between bacterial infection onset and peak PCT release, and that may also explain the apparent low level of PCT on the day of VAP onset. Incorporating PCT values in clinical score (such as CPIS) did not improve its diagnostic value.[221,222]

The soluble triggering receptor expressed on myeloid cells-1 (sTREM-1) molecule is known to be specifically released during several infectious processes.[224] Although it was apparently a reliable marker of pneumonia, especially VAP, more recent studies obtained contradictory findings, thereby raising doubt as to its usefulness for VAP diagnosis.[220,225–227] Pending additional studies, and because this marker is not routinely available, sTREM-1 is not recommended as an indicator to guide antibiotic use in such situations.

Gram-negative bacilli cause more than 80% of VAP episodes and are associated with high mortality. Because gram-negative bacilli pneumonia might be diagnosed more rapidly by endotoxin measurement in BAL fluid, several investigators tested this hypothesis.[228–231] Applying a threshold of greater than 5 endotoxin units (EU/mL) in BAL fluid yielded the best operating characteristics for gram-negative bacilli pneumonia diagnosis (100% sensitivity; 75% specificity; area under the receiver operating characteristics curve: 0.88) in a series of sixty-three hospitalized adults suspected of having lung infection.[229] Three other studies confirmed the potential contribution of this tool.[228,230,231] These findings suggest that endotoxin determination in BAL fluid might become an acceptable adjunct for the rapid diagnosis of gram-negative bacilli pneumonia in the near future, when it will be available at bedside.

Summary of the Evidence

Aside from decision-analysis studies[232,233] and a single retrospective study,[202] five trials have used a randomized scheme to assess the effect of a diagnostic strategy on antibiotic use and outcome in patients suspected of having VAP.[26,27,234–236] In three randomized studies conducted in Spain, no differences in mortality and morbidity were found when either invasive (PSB and/or BAL) or noninvasive (quantitative endotracheal aspirate cultures) techniques were used to diagnose VAP.[26,27,234] These studies were relatively small, ranging from fifty-one to eighty-eight patients. Antibiotics were continued in all patients despite negative cultures, thereby offsetting the potential advantage of the specific diagnostic test in patients with suspected VAP. Several prospective studies have concluded that antibiotics can be stopped in patients with negative quantitative cultures, without adversely affecting the recurrence of pneumonia and mortality.[188,237,238]

In a French study in which 413 patients were randomized, those receiving bacteriologic management using BAL and/or PSB had a lower mortality rate on day 14, lower sepsis-related organ failure assessment scores on days 3 and 7, and less antibiotic use.[235] Pertinently, twenty-two nonpulmonary infections were diagnosed in the bacteriologic strategy group and only five in the clinical strategy group, suggesting that overdiagnosis of VAP can lead to errors in identifying nonpulmonary infections. A randomized trial conducted by the Canadian Critical Care Trials Group investigated the effect of different diagnostic approaches on outcomes of 740 patients suspected of having VAP.[236] There was no difference in the 28-day mortality rate in patients in whom BAL was used versus those in whom endotracheal aspiration was used as the diagnostic strategy. The BAL group and the endotracheal aspiration group also had similar rates of targeted antibiotic therapy on day 6, days alive without antibiotics, and maximum organ-dysfunction scores. Unfortunately, information about how the decision algorithms were followed in the two diagnostic arms once cultures were available was not provided, raising uncertainties about how de-escalation of antibiotic therapy was pursued in patients with negative BAL cultures. Obviously, the potential benefit of using a diagnostic tool such as BAL for safely restricting unnecessary antimicrobial therapy in such a setting can only be obtained when decisions regarding antibiotics are closely linked to bacteriologic results, including both direct examination and cultures of respiratory specimens.

Our personal bias is that use of bronchoscopic techniques to obtain PSB and BAL specimens from an affected area of the lung in ventilated patients with signs suggestive of pneumonia enables the formulation of a therapeutic strategy superior to that based exclusively on clinical evaluation. Bronchoscopic techniques, when performed before the introduction of new antibiotics, enable physicians to identify most patients who need immediate treatment, and help select optimal therapy in a safe and well-tolerated manner. These techniques also avoid resorting to broad-spectrum coverage of all patients who develop a clinical suspicion of infection.[239] The full impact of this decision tree on patient outcome remains controversial.[235,236] Yet, being able to withhold antimicrobial treatment from some patients without infection may constitute a distinct advantage in the long-term: It minimizes the emergence of resistant microorganisms in the ICU and redirects the search for another (the true) infection site.

In patients with clinical evidence of severe sepsis and rapid worsening organ dysfunction, hypoperfusion, or hypotension, or patients with a very high pretest probability of disease, the initiation of antibiotic therapy should not be delayed while awaiting bronchoscopy. Patients should be given immediate antibiotics. In this situation, simple nonbronchoscopic procedures find their best justification, allowing distal pulmonary secretions to be obtained on a 24-hour basis, just before starting new antimicrobial therapy.

Despite broad experience with PSB and BAL, it remains unclear which should be used. Most investigators prefer BAL over PSB to diagnose bacterial pneumonia, because BAL (a) has a slightly higher sensitivity to identify VAP-causative microorganisms, (b) enables better selection of an empiric antimicrobial treatment before culture results are available, based on microscopically examined cytocentrifuged preparations, (c) is less dangerous for many critically ill patients, (d) is less costly, and (e) may provide useful clues for the diagnosis of other types of infections. Nevertheless, a very small return on BAL may contain only diluted material from the bronchial rather than alveolar level, and thus give rise to false-negative results, particularly in patients with very severe chronic obstructive pulmonary disease. In these patients, the value of BAL is greatly diminished and PSB is preferred.[184]

When bronchoscopy is not available, we recommend replacing bronchoscopy in the algorithm in Figure 46-3 by one of the simplified nonbronchoscopic diagnostic techniques, or following the strategy described by Singh et al (see Figure 46-3). Such an approach avoids prolonged treatment of patients with a low likelihood of infection, while allowing immediate treatment of patients with VAP.

Treatment

Antimicrobial therapy of patients with VAP is a two-stage process. The first stage involves administering broad-spectrum antibiotics to avoid inappropriate treatment in patients with true bacterial pneumonia.[2] The second stage focuses on trying to achieve this objective without overusing and abusing antibiotics. In general, the first goal can be accomplished by identifying patients with pneumonia in a rapid fashion and starting therapy with an empirical regimen that is likely to be accurate. This requires that the choice be driven by anticipation of the likely etiologic pathogens, modified by knowledge of local patterns of antimicrobial resistance and local microbiology. The second goal involves combining a number of different steps, including stopping therapy in patients with a low probability of the disease, commitment to focus and narrow treatment once the etiologic agent is known, switching to monotherapy after day 3, and shortening duration of therapy to 7 to 8 days in most patients, as dictated by the patient's clinical response and information about the bacteriology.

Initial Therapy

Failure to initiate prompt appropriate and adequate therapy (the etiologic organism is sensitive to the therapeutic agent, the dose is optimal, and the correct route of administration is used) has been a consistent factor associated with increased mortality.[32,155–157] Because pathogens associated with inappropriate initial empiric antimicrobial therapy mostly include antibiotic-resistant microorganisms, such as *P. aeruginosa, Acinetobacter* spp., *Klebsiella pneumoniae, Enterobacter* spp., and MRSA, patients at risk for infection with these organisms should initially receive a combination of agents that can provide a very broad spectrum of coverage (Table 46-3).[2,173] Several observational studies have now confirmed that the use of a regimen that combines initially a broad-spectrum β-lactam with an aminoglycoside increases the proportion of patients appropriately treated as compared to monotherapy or to a regimen combining a β-lactam with a fluoroquinolone.[162,240–242] Only patients with early onset infection, mild or moderate disease severity, and no specific risk factors for multiresistant strains, such as prolonged duration of hospitalization (≥5 days), admission from a health care–related facility, recent prolonged antibiotic therapy, and specific local epidemiologic data, can be treated with a narrow-spectrum drug, such as a nonpseudomonal third-generation cephalosporin.[1,2,243]

Table 46-3: Initial Empiric Therapy for Ventilator-Associated Pneumonia in Patients with Late-Onset Disease or Risk Factors for Multidrug-Resistant Pathogens

Potential Pathogens	Combination Antibiotic Therapy
MDR Pathogens ◦ *Pseudomonas aeruginosa* ◦ *Klebsiella pneumoniae* (ESBL) ◦ *Enterobacter* species ◦ *Serratia marcescens* ◦ *Acinetobacter* species Methicillin-resistant *Staphylococcus aureus* (MRSA)	Antipseudomonal cephalosporin (cefepime, ceftazidime) *or* Antipseudomonal carbapenem (imipenem or meropenem or doripenem) *or* β-lactam, β-lactamase inhibitor (piperacillin-tazobactam) *plus* Antipseudomonal fluoroquinolone (ciprofloxacin or high-dose levofloxacin) *or* Aminoglycoside (amikacin, gentamicin, or tobramycin) *plus* Linezolid or vancomycin (if MRSA risk factors are present or there is a high incidence locally)

Abbreviations: ESBL, extended-spectrum β-lactamase; *MDR,* multidrug resistant; *MRSA,* methicillin-resistant *Staphylococcus aureus.*

Source: Modified from Guidelines for the management of adults with hospital-acquired, ventilator-associated, and healthcare-associated pneumonia. *Am J Respir Crit Care Med*. 2005;171:388–416, with permission.

When risk factors for multiresistant pathogens are present, the choice of agents should be based on local patterns of antimicrobial susceptibility and anticipated side effects. Having a current and frequently updated knowledge of local bacteriologic patterns can increase the likelihood that appropriate initial antibiotic treatment will be prescribed.[48] The choice should also take into account which therapies patients recently received (within the prior 2 weeks), striving not to repeat the same antimicrobial class, if possible.[244–246]

Use of endotracheal aspirate surveillance cultures two or three times weekly may also make it possible to increase the proportion of patients receiving initially appropriate antimicrobial therapy.[247–251] This strategy rests upon the observation that VAP caused by potentially multiresistant pathogens is typically preceded by colonization of the oropharynx and the proximal airways by the same strains. To be of clinical use in directing initial antibiotic therapy, surveillance cultures must be able to detect this colonization rapidly and with high sensitivity, as false-negative results would place the patient at risk for inappropriate therapy. Moreover, a focused antibiotic choice, with limitation of unnecessary broad-spectrum drugs, requires a low number of false-positive surveillance results. Patients with a prolonged hospital stay and numerous previous antibiotics will benefit the most. Thus, such a strategy can only be recommended when the local prevalence of multiresistant microorganisms is high, when current empirical therapy is suboptimal and cannot be easily increased through adaptation of a decision tree, and when the resources for the microbiologic workup are available.[248]

Stopping Therapy When the Diagnosis of Infection Becomes Unlikely

Because clinical signs of infection are nonspecific and can be caused by any condition associated with an inflammatory response, many more patients than necessary are initially treated with antibiotics. Thus, it is important to use serial clinical evaluations and microbiologic data to reevaluate therapy after 48 to 72 hours.[2,252]

The decision tree should contain an explicit statement that patients with a low probability of infection will be identified and therapy stopped when infection appears unlikely. The algorithm cannot be exactly the same for a “clinical” or an “invasive” strategy, depending on the general principles and microbiologic techniques on which the diagnostic strategy is constructed (see “Diagnosis” above). Using a "clinical strategy" in which all patients with clinically suspected pulmonary infection are treated with new antibiotics, even when the likelihood of infection is low, the decision whether to continue antibiotics or not on day 3 will be based essentially on a combination of clinical signs.[176] Briefly, antibiotics are discontinued if and only if the clinical diagnosis of VAP is unlikely (there are no definite infiltrates found on chest radiography at follow-up), tracheobronchial aspirate culture results are nonsignificant, and there is no severe sepsis or shock. Other decision trees for stopping antibiotics at 3 days can be constructed, for example, by incorporating results of serial levels of procalcitonin in the blood or sTREM-1 in bronchial secretions, but still require validation.[220,223,253]

The decision algorithm for withholding or withdrawing antibiotics using the "invasive strategy" is based on results of direct examination of distal pulmonary samples obtained by bronchoscopic or nonbronchoscopic BAL and results of quantitative cultures (see Fig. 46-3). Briefly, antibiotics are withheld in patients with no bacteria on gram-stained cytocentrifuged preparations and no signs of severe sepsis or septic shock; and discontinued when quantitative culture results are below the cutoff defining a positive result, except in patients with proven extrapulmonary infection and/or severe sepsis.[235] As demonstrated by several studies, patients managed with such a bacteriologic strategy receive fewer antibiotics, and more patients have all their antibiotics discontinued compared to the clinical strategy group, thereby confirming that the two strategies actually differed.[193,202,235,237,239] Future studies should, however, compare bronchoscopy against and in addition to a clinical strategy incorporating an explicit statement for stopping antibiotics in patients with a low probability of infection, for example using the algorithm described above or the CPIS score, as proposed by Singh et al.[146,176,180,254] Formal economic analysis is also required because prevention of resistance and better antibiotic control may result in cost savings. Whatever the diagnostic strategy used, each ICU team should monitor the adherence of their physicians to it and implement corrective measures, as needed.

Focusing Therapy Once the Agent of Infection Is Identified

Once the results of respiratory tract and blood cultures become available, therapy can often be focused or narrowed, based on the identity of specific pathogens and their susceptibility to specific antibiotics, so as to avoid prolonged use of a broader spectrum of antibiotic therapy than is justified by the available information.[2,42,161,249,255–258] For many patients, including those with late-onset infection, therapy can be narrowed because an anticipated organism (such as *P. aeruginosa, Acinetobacter* spp., or MRSA) was not recovered or because the organism isolated is sensitive to a narrower-spectrum antibiotic than used in the initial regimen. For example, vancomycin and linezolid should be stopped if no MRSA is identified, unless the patient is allergic to β-lactams and has developed an infection caused by a gram-positive microorganism.[249] Very broad-spectrum agents, such as carbapenems, piperacillin-tazobactam, and/or cefepime should also be restricted to patients with infection caused by pathogens only susceptible to these agents. In case of infection caused by a piperacillin-susceptible *P. aeruginosa* strain, antimicrobial treatment should be streamlined to this specific drug. Similarly, in the absence of an infection caused by a nonfermenting gram-negative bacilli or an extended-spectrum β-lactamase-producing *Enterobacteriaceae,* the β-lactam should be changed to a non-antipseudomonal antibiotic, such as ceftriaxone or cefotaxime. Clinicians must be aware, however, that emergence of stable, derepressed, resistant mutants may lead to treatment failure when third-generation cephalosporins are chosen in the case of infections caused by *Enterobacter, Citrobacter, Morganella morganii,* indole-positive *Proteus,* and *Serratia* spp., even if the isolate appears susceptible on initial testing.

Unfortunately, several studies have shown that although de-escalation was not associated with any adverse outcomes, it was not consistently performed in many ICUs.[259–263]

Optimizing Antimicrobial Therapy

Several published reports have demonstrated a relationship among serum concentrations of β-lactams or other antibiotics, the minimal inhibitory concentration (MIC) of the infecting organism, and the rate of bacterial eradication from respiratory secretions in patients with lung infection.[264–270] Consequently, clinical and bacteriologic outcomes can be improved by optimizing a therapeutic regimen according to pharmacokinetic–pharmacodynamic properties of the agent(s) selected for treatment.[161,265,271–278] Most investigators distinguish between antimicrobial agents that kill by a concentration-dependent mechanism (e.g., aminoglycosides and fluoroquinolones) from those that kill by a time-dependent mechanism (e.g., β-lactams and vancomycin). Multivariate analyses based on seventy-four acutely ill, mostly VAP patients, who were treated with intravenous ciprofloxacin (200 mg BID to 400 mg TID), demonstrated that the most important independent factor for probability of cure was a pharmacodynamic variable, that is, the 24-hour area under the concentration–time curve divided by the MIC (AUIC).[271] For AUIC less than 125, the probabilities of clinical and microbiologic cures were 42% and 26%, respectively; with AUIC greater than 125, the probabilities were 80% and 82%, respectively. Routine dosages of fourth-generation cephalosporins, carbapenems, and fluoroquinolones may not achieve the target AUICs for resistant gram-negative bacteria, such as *P. aeruginosa* and *Acinetobacter* spp. Higher dosing regimens and/or prolonged duration of infusion are frequently needed in that case.[264,265,279]

Pharmacokinetic-pharmacodynamic models have also been used to optimize aminoglycoside therapy for VAP caused by gram-negative bacilli, using the first measured maximum concentration of drug in serum (C_{max}).[272] Seventy-eight patients with VAP were analyzed, and the investigators reported an 89% success rate for temperature normalization by day 7 of therapy for C_{max}/MIC greater than 4.7, and an 86% success rate for leukocyte count normalization by day 7 of therapy for C_{max}/MIC greater than 4.5. Logistic regression analysis predicted a 90% probability of temperature and leukocyte count normalizations by day 7 if a C_{max}/MIC greater than 10 was achieved within the first 48 hours of aminoglycoside administration. Aggressive aminoglycoside doses immediately followed by pharmacokinetic monitoring for each patient would ensure that C_{max}/MIC target ratios are achieved early during therapy.

These findings confirm the need to adjust the target dose of antimicrobial agents (used in treating severe pulmonary infection) to an individual patient's pharmacokinetics and putative bacterial pathogens' susceptibilities. Altered pharmacokinetic secondary to increase in volume of distribution in critically ill patients can result in insufficient serum β-lactam concentrations when standard dosages are administered, emphasizing the need to carefully monitor peak and trough levels of antibiotics when treating resistant pathogens, such as gram-negative bacilli.[280–282] Development of a priori dosing algorithms based on MIC, patient creatinine clearance and weight, and the clinician-specified AUIC target might be a valid way to improve treatment of these patients, leading to a more precise approach than current guidelines for use of antimicrobial agents.[273–277]

Switching to Monotherapy at Days 3 to 5

The two most commonly cited reasons to use combination therapy for the duration of antibiotic treatment are to achieve synergy and to prevent the emergence of resistant strains. Synergy, however, has only been clearly documented to be valuable in vitro in the therapy of *P. aeruginosa* or other difficult-to-treat gram-negative bacilli and in patients with neutropenia[283] or bacteremic infection,[284,285] which is uncommon in hospital-acquired pneumonia or VAP. When combination therapy was evaluated in randomized controlled studies, its benefit was inconsistent or null, even when the results were pooled in a meta-analysis or the analysis was restricted to patients infected by *P. aeruginosa.*[286–293] Combination therapy did not prevent the emergence of resistance during therapy, but did lead to a significantly higher rate of nephrotoxicity. In a retrospective analysis of 115 episodes of *P. aeruginosa* bacteremia, the use of adequate combination antimicrobial therapy as empirical treatment until receipt of the sensitivity was associated with a better rate of survival at 30 days than the use of monotherapy.[294] Adequate combination antimicrobial therapy given as definitive treatment for *P. aeruginosa* bacteremia, however, did not improve the rate of survival compared to adequate definitive monotherapy.

Based on these data, therapy could be switched to monotherapy in most patients after 3 to 5 days, provided that initial therapy was appropriate, clinical course appears favorable, and that microbiologic data do not suggest a very difficult-to-treat microorganism, with a very high in vitro minimal inhibitory concentration, as it can be observed with some nonfermenting gram-negative bacilli.

Shortening Duration of Therapy

Efforts to reduce the duration of therapy for VAP are justified by studies of the natural history of the response to therapy. Dennesen et al demonstrated that when VAP was adequately treated, significant improvements were observed for all clinical parameters, generally within the first 6 days of antibiotics.[295] The consequence of prolonged therapy to 14 days or more was newly acquired colonization, especially with *P. aeruginosa* and *Enterobacteriaceae,* generally during the second week of therapy. These data support the premise that most patients with VAP, who receive appropriate antimicrobial therapy, have a good clinical response within the first 6 days.[295–297] Prolonged therapy simply leads to colonization with antibiotic-resistant bacteria, which may precede a recurrent episode of VAP.

Reducing duration of therapy in patients with VAP has led to good outcomes with less antibiotic use with a variety of different strategies. Singh et al used a modification of the CPIS scoring system to identify low-risk patients (CPIS ≤6) with suspected VAP who could be treated with 3 days of antibiotics as opposed to the conventional practice of 10 to 21 days of antibiotic therapy.[180] Patients receiving the shorter course of antibiotic therapy had better clinical outcomes than the patients receiving longer therapy, with fewer subsequent superinfections attributed to antibiotic-resistant pathogens, although many of these patients may not have had pneumonia. A multicenter, randomized, controlled trial demonstrated, in a large series of 413 patients with microbiologically proven VAP, that patients who received appropriate initial empiric therapy for 8 days had outcomes similar to patients who received therapy for 15 days.[298] A trend toward greater rates of relapse for short-duration therapy was seen if the etiologic agent was *P. aeruginosa* or *Acinetobacter* spp., but clinical outcomes were exactly the same. These results were recently confirmed by two other studies, including a prospective, randomized trial of 290 patients evaluating an antibiotic discontinuation policy.[254,299]

Based on these data, an 8-day regimen can probably be standard for patients with VAP. Possible exceptions to this recommendation include immunosuppressed patients, those whose initial antimicrobial treatment was not appropriate for the causative microorganism(s), and patients whose infection was caused by very difficult-to-treat microorganisms and had no improvement in clinical signs of infection.

Many clinicians, however, remain hesitant about prescribing fewer fixed days of antibiotics for patients with VAP, and would prefer to customize antibiotic duration based on the clinical course of the disease and/or using serial determinations of a biologic marker of infection, such as PCT. The rationale for using a biomarker to tailor antibiotic-treatment duration relies on the fact that the inflammatory response is most often proportional to infection severity.[212] When that response is absent or low, it might be logical to discontinue antibiotics earlier. Moreover, it is well known that PCT levels reflect the inflammatory response intensity and are related to outcome.[223,300,301] Thus, adapting antimicrobial treatment duration to PCT kinetics seems reasonable, and has been demonstrated as useful in several randomized trials targeting patients with acute respiratory infection, including five trials conducted in the ICU.[211,214–218,222,302–308]

Stolz et al randomized 101 patients with VAP to be managed with an antibiotic discontinuation strategy according to guidelines (control group) or to serum PCT concentrations.[304] In patients randomized in the PCT group, attending physicians were recommended to stop antibiotics when the PCT concentration was below 0.5 ng/m, or had decreased by more than 80% from the baseline value. Using this algorithm, the number of antibiotic-free days alive within 28 days after VAP diagnosis was significantly higher in the PCT group than in the control group (median [interquartile range], 13 days [2 to 21] vs. 9.5 days [1.5 to 17]). This reduction in antibiotic administration was not associated with a worse outcome, because ventilator-free days, ICU-free days, length of hospital stay, 28-day mortality, and hospital mortality rates were similar in the two groups.[304] Similar results were observed by Bouadma et al in the PRORATA (PROcalcitonin to Reduce Antibiotic Treatments in Acutely ill patients) trial, a randomized trial that included 621 ICU patients, of whom 141 had microbiologically proven VAP.[305] Using predefined algorithms to initiate or discontinue antibiotics according to serum PCT levels, the duration of antibiotic treatment for the first episode was significantly decreased in the seventy-five patients randomized in the PCT-guided group compared to the control group managed using international and local guidelines (7.3 ± 5.3 days vs. 9.4 ± 5.7 days, $p = 0.02$, respectively), with no detrimental effects on patients' outcome.[305]

Aerosolized Therapy

Because insufficient dosing of antibiotics at the site of infection in patients with VAP may lead to clinical and microbiologic failures, efforts to optimize pulmonary penetration of antimicrobial agents are warranted. Directly delivering the drug to the site of infection via aerosolization may represent a valid option, providing that this technique actually allows improved lung-tissue concentrations at the infected site. This mode of administration, by achieving high pulmonary antibiotic concentrations, could increase the antibacterial activity of concentration-dependent antibiotics, such as aminoglycosides, or restore the bactericidal activity of antibiotics in the case of infections caused by pathogens of impaired sensitivity. Furthermore, by limiting systemic exposure, it could also allow the administration of antibiotics characterized by a high systemic toxicity, such as aminoglycosides and polymyxins.

Pooling the results of the five randomized controlled trials that examined the potential benefit of inhaled or endotracheally instilled antibiotics for the treatment of patients with hospital-acquired pneumonia or VAP, a statistically higher success rate was demonstrated in patients receiving antimicrobial agents via the respiratory tract.[309] No difference in mortality, however, could be documented and the meta-analysis was based on a very limited number of patients. It should also be noted that the bronchial deposition of aerosolized antibiotics might have rendered cultures of endotracheal samples falsely "negative," which could have artificially increased the rate of success in patients randomized to aerosolized or endotracheally instilled antibiotics, casting some doubt on the validity of the results.

Several recent studies, based on a new generation of nebulizers with improved technology, have renewed the interest in aerosolized antibiotic therapy for patients with VAP.[310–313] In anesthetized piglets on prolonged mechanical ventilation for a severe experimental *E. coli* bronchopneumonia, amikacin lung-tissue concentrations were markedly higher following aerosolization as compared to intravenous administration.[314] Seventy-one percent of lung segments were found sterile after two nebulizations and 25 hours of treatment, whereas cultures of lung segments were comparable in nontreated and intravenously treated animals. In a recent study using a new device with a vibrating plate and multiple apertures to produce an aerosol of amikacin conducted in sixty-nine patients with gram-negative bacilli VAP, the authors found that the nebulized drug was well distributed in the lung parenchyma, with high tracheal and alveolar levels but low serum concentration, below the renal toxicity threshold.[315] Moreover, aerosolized amikacin was well tolerated, without any severe adverse event, and patients who received amikacin twice daily required significantly less antibiotics than patients given placebo.[316]

Data on the impact of aerosolized antibiotics active against gram-positive bacteria are scarce. In a placebo-controlled trial, Palmer et al randomized forty-three patients with purulent tracheobronchitis and gram-stain-identified microorganisms to receive aerosolized antibiotics ($n = 19$) or placebo ($n = 24$).[311] The antibiotic was chosen according to tracheal aspirate gram-staining results (vancomycin for gram-positive, gentamicin for gram-negative). Antibiotic aerosolization led to faster resolution of clinical signs of pneumonia than placebo, fewer subsequent VAP episodes, less bacterial resistance and use of systemic antibiotics, and perhaps accelerated weaning from mechanical ventilation.[311]

Aerosolized polymyxin is also being used increasingly for treating patients with infections caused by multidrug-resistant gram-negative bacilli, mainly *A. baumannii* and *P. aeruginosa,* with mixed results.[312,313] In a randomized trial that included 100 patients with VAP caused by gram-negative bacilli (predominantly multidrug-resistant *A. baumannii* and/or *P. aeruginosa*), patients who were treated with a combination of systemic antibiotics and nebulized colistin had a higher rate of favorable microbiologic outcome than did patients who were treated with systemic antibiotics alone (microbiologic eradication or presumed eradication 61% vs. 38%), but there was no differences in clinical outcome (51% vs. 53%).[313] In a retrospective case-control study that included eighty-six patients with VAP caused by multidrug-resistant gram-negative bacilli (predominantly *A. baumannii*) treated with a combination of intravenous and aerosolized colistin compared with intravenous colistin alone, there was only a trend toward improved rates of clinical cure, pathogen eradication, and mortality in the patients who received aerosolized and intravenous colistin.[312]

Thus, although the results of recent investigations emphasize the potential contribution of aerosolized antibiotics to treat VAP as an efficient adjunctive therapy to intravenous antibiotics, the clinical impact of such a strategy has not yet been definitively established. At present, aerosolized antibiotics can only be recommended to treat patients with multidrug-resistant VAP, for which no effective intravenous antibiotics are available. Indisputably, large prospective trials are needed to evaluate the potential usefulness of this therapeutic modality.

Prevention

Because VAP is associated with increased morbidity, longer hospital stay, increased health care costs, and higher mortality rates, prevention is a major challenge for intensive care medicine.[2,317,318] A number of recommendations for the prevention of VAP are empiric rather than based on controlled observations, which make evaluation of the impact of such interventions difficult in this setting for several reasons: (a) the difficulty in obtaining an accurate diagnosis of VAP, that is, to distinguish patients with true infection from patients with tracheal colonization and/or other pathologic processes as only patients who develop true VAP are likely to benefit from preventive measures; (b) the difficulty of precisely determining the impact of prophylactic measure on the overall mortality of a general ICU population, that is, to identify preventable deaths, directly attributable to VAP, among all deaths occurring in a population of ventilated ICU patients; and (c) the difficulty of evaluating the consequences of a preventive measure on a potentially pathogenic mechanism, for example, to evaluate the exact role played by prevention or reduction of tracheal colonization in modifying the development of VAP.

Conventional Infection-Control Approaches

These measures should be the first step taken in any prevention program.[319] The design of the ICU has a direct effect on the potential for nosocomial infections. Adequate space and lighting, proper functioning of ventilation systems, and facilities for handwashing lead to lower infection rates.[320] It should, however, be kept in mind that physical upgrading of the environment does not per se reduce the infection rate unless personnel attitude and practices are improved. In any ICU, one of the most important factors is the health care staff, including the number, quality, and motivation of medical, nursing, and ancillary members.[321] The team should include a sufficient number of nurses to minimize them moving from one patient to another and to avoid having them working under constant pressure.[322–325] The importance of personal cleanliness and attention to aseptic procedures must be emphasized at every possible opportunity. At the same time, unnecessarily rigid restrictions should be avoided.[326] The importance of personal cleanliness and attention to aseptic procedures must be emphasized at every opportunity. It is clear that careful monitoring, decontamination, and compliance with guidelines for the use of respiratory equipment all reduce the incidence of nosocomial pneumonia.[324] In particular, handwashing and handrubbing with alcohol-based solutions remain the most important components of effective infection control practices in the ICU.[132,324,327]

A bacterial monitoring policy facilitates the early recognition of colonization and infection and is associated with significant reductions in nosocomial infection rates.[328] The focal point for infection control activities in the ICU is a surveillance system designed to establish and maintain a database that describes endemic rates of nosocomial infection. Awareness of the endemic rates enables the recognition of the onset of an epidemic when infection rates rise above a calculated threshold.

Adoption of an antibiotic policy restricting the prescription of broad-spectrum agents and useless antibiotics is of major importance.[22,164,235,329] Better use of antibiotics in the ICU can be achieved by implementing strict guidelines, avoiding the treatment of patients who do not have bacterial infections, using narrow-spectrum antibiotics whenever possible, and reducing the duration of treatment. Similarly, transfusion of red blood cells and other allogenic blood products should follow a strict policy, because several studies have identified exposure to allogenic blood products as a risk factor for postoperative infection and pneumonia.[330–335] Some very simple, safe, inexpensive, and logical measures may have major effects on the frequency of VAP in ventilated patients. These include avoiding nasal insertion of endotracheal and gastric tubes, maintaining the endotracheal tube cuff pressure above 20 cm H_2O to prevent leakage of bacteria around the cuff into the lower respiratory tract, prompt reintubation of patients who are likely to fail extubation, removing tubing condensate, and providing adequate oral hygiene with tooth brushing.[101,119,136,336,337]

Specific Prophylaxis Against Ventilator-Associated Pneumonia

Specific strategies aimed at reducing the duration of mechanical ventilation (a major risk factor for VAP), such as improved methods of sedation, use of protocols to facilitate and accelerate weaning, using low tidal volume and adequate levels of positive end-expiratory pressure, and use of intensive insulin therapy to control blood glucose should be considered as integral parts of any infection-control program.[338–343] All are based on the application of strict protocols. Noninvasive ventilation is an alternative approach to using artificial airways to avoid infectious complications and injury of the trachea in patients with acute respiratory failure. Many observational studies and seven randomized trials suggest that patients who tolerate noninvasive ventilation have a lower incidence of pneumonia than those tracheally intubated.[338,344–352]

Apart from protocols aimed at reducing the duration of mechanical ventilation, eight prophylactic approaches have been studied: semirecumbent positioning, oscillating and rotating beds, continuous or intermittent aspiration of subglottic secretions, ventilator circuits management, methods of enteral feeding, stress-ulcer prophylaxis, oral decontamination with antiseptics, and selective digestive decontamination.

Semirecumbent Positioning

Supine positioning is independently associated with the development of VAP.[22] Placing ventilated patients in a semirecumbent position to minimize reflux and aspiration of gastric contents is a simple measure, although some practical problems can occur in unstable patients. Only a few trials have evaluated the efficacy of semirecumbent positioning.[112–114,117,353–355] In a randomized trial based on a small number of patients, Drakulovic et al observed lower rates of both clinically suspected and bacteriologically confirmed VAP, and identified supine positioning as an independent risk factor for VAP with enteral nutrition, ventilation for 7 days or longer and a Glasgow coma score of less than 9 points. The feasibility and efficacy of this intervention in a larger patient population, however, remain unknown, all the more as its efficacy was not confirmed in a subsequent trial that included 221 ventilated patients or in a recent meta-analysis.[117,353] Raising the head of bed to 30 degrees or higher may also have some detrimental skin effects and may increase the incidence of pressure ulcer formation.[115] Pending additional studies, most experts currently recommend maintaining the head of the bed elevated to at least 20 degrees to 30 degrees in all ventilated patients who are hemodynamically stable, particularly when they are receiving enteral nutrition.[134,317,356–361]

Oscillating and Rotating Bed

Immobility in critically ill patients treated with mechanical ventilation results in atelectasis, impaired secretions drainage, and potentially predisposes to pulmonary complications including VAP. Oscillating and rotating beds may help in preventing pneumonia.[317] Six randomized trials, which included mostly surgical and trauma patients, ventilated or not, and summarized in a meta-analysis by Choi and Nelson,[362] compared continuous lateral rotational therapy with standard beds. The meta-analysis found a significant reduction in the risk for pneumonia, principally concerning early onset (<5 days) pneumonia and a decreased duration of ICU stay. Notably, the only randomized, controlled trial—not included in the metaanalysis—conducted on a general ICU population did not show any differences in pneumonia rates but showed a significantly shorter length of ICU stay.[363] Some adverse events have been described with these beds including disconnection of catheters or pressure ulceration; in addition, nursing care is potentially complicated with oscillating beds. Finally, despite the cost of such beds, cost-to-benefit analyses suggested favorable results, mainly caused by the reduction of ICU length of stay.

Oral Decontamination with Antiseptics

Topical application of chlorhexidine or other antiseptics to the oral mucosa may decrease respiratory pathogen colonization and secondary lung infection in ventilated patients. Randomized, controlled trials, however, have reported mixed results: Some showed little effect whereas others found a reduction in the incidence of VAP.[364–376] Combining the results of the seven randomized controlled trials that evaluated the potential efficacy of chlorhexidine, a 30% relative reduction in the risk of VAP was observed, but no effect of chlorhexidine on reduction of mortality or duration of mechanical ventilation could be demonstated.[377] The varying concentration of the chlorhexidine solutions used in these studies may have affected the results. In the study by Koeman et al,[369] a 2% solution of chlorhexidine was used, a much higher concentration than in the other published studies, most of which used a 0.12% or 0.2% solution; this may partially explain the benefit of chlorhexidine for reducing VAP in this study. Reported adverse effects of oral use of chlorhexidine include staining of the teeth, which is reversible with professional cleaning, and a transient abnormality of taste.[377] The optimal concentration, frequency of application, effect on promoting resistance among oropharyngeal flora, and cost-effectiveness of chlorhexidine should be addressed in future studies.

Aspiration of Subglottic Secretions and Use of Specialized Endotracheal Tubes

Repeated micro inhalations of colonized oropharyngeal (subglottic) secretions are the major mechanism of VAP. Continuous or intermittent suctioning of oropharyngeal secretions has been proposed as a means to avoid chronic aspiration of secretions through the tracheal cuff of intubated patients. Aspiration of subglottic secretions requires the use of specially designed endotracheal tube with a separate lumen that opens into the subglottic region. Thirteen randomized controlled trials have studied aspiration of subglottic secretions for the prevention of VAP for a total of 2442 randomized patients.[378–386] Of the thirteen studies, twelve reported a reduction in VAP rates in the subglottic secretion drainage arm. When the results were combined in a meta-analysis, the overall risk ratio for VAP was 0.55 (95% CI, 0.46 to 0.66; p <.00001) with no heterogeneity, and the use of subglottic secretion drainage was associated with reduced ICU length of stay, decreased duration of mechanical ventilation, and increased time to first episode of VAP.[386] No effect, however, on hospital or ICU mortality could be demonstrated.[386] Some experimental data in sheep and ICU patients suggest the possibility of tracheal damage with the use of this type of tube.[382,387,388]

Bacterial aggregates in biofilm dislodged during suctioning might not be eradicated by antibiotics or effectively cleared by host immune defenses, thereby constituting dangerous inoculums for the lung. Preliminary data obtained in animal models and from small randomized human studies support the hypothesis that an endotracheal tube coated externally and internally with a potent antiseptic product such as silver could have a sustained antimicrobial effect within the proximal airways and block biofilm formation at its surface.[389–394] Such a device was evaluated in a large, randomized, multicenter, single-blind trial by Kollef et al.[395] The authors conclude that the new device was able to lower the VAP frequency from 7.5% for the control group to 4.8% for the group receiving the silver-coated endotracheal tube. The silver-coated tube, however, did not reduce mortality rates, the duration of intubation, hospital length of stay, or the frequency or severity of adverse effects.

Ventilator Circuit Management

Decreased frequency of ventilator circuit change, replacement of heated humidifiers by heat and moisture exchangers, decreased frequency of heat and moisture exchanger change, and closed suctioning systems have been tested for preventing VAP.[2,317,318,396] Four randomized trials of decreased frequency of ventilator circuit changes have been published. Changes every 2 days, 7 days, and no scheduled change did not find significant difference in the rate of VAP as summarized in a recent meta-analysis.[397] One meta-analysis summarized the results of five randomized, controlled trials which compared the effects of heated humidifiers and heat and moisture exchangers on the risk of VAP.[318] Only one of these five studies found a significant reduction of VAP rate with the use of heat and moisture exchangers.[125] Efficacy of both humidification strategies seems comparable. Two studies, however, reported increased rates of endotracheal tube occlusion with the use of heat and moisture exchangers; the increased resistive load can cause difficulties in ventilation and weaning in patients with severe acute respiratory distress syndrome—related to larger dead space. No other adverse effects were observed. No effect on mortality was reported. Finally, one study has evaluated the impact of less frequent changes (daily vs. every 5 days) in heat and moisture exchangers on the development of VAP.[398] No difference in the VAP rates was observed.

To avoid hypoxia, hypotension, and contamination of suction catheters entering the tracheal tube, investigators have examined closed suctioning systems.[396,399–400] They either found a nonsignificantly lower prevalence of VAP for patients managed with the closed system compared to the open system, without any adverse effect,[400] or they found that its use was associated with an increased frequency of endotracheal colonization.[399] Closed-suction systems also failed to reduce cross-transmission and acquisition rates of the most relevant gram-negative bacteria in ICU patients in a prospective crossover study in which 1110 patients were enrolled.[396]

Methods of Enteral Feeding

Nearly all ventilated patients have a nasogastric tube inserted to manage gastric and enteral secretions, prevent gastric distension, or provide nutritional support. A nasogastric tube may increase the risk for gastroesophageal reflux, aspiration, and VAP.[77] Four randomized, controlled trials have evaluated methods of enteral feeding aimed at preventing VAP: postpyloric or jejunal feeding (vs. gastric feeding), the use of motility agents (metoclopramide vs. placebo), acidification of feeding (with addition of hydrochloric acid), and intermittent (vs. continuous) feeding.[108,401,402] These studies did not find differences in incidence of VAP or mortality rates.

Potentially serious adverse effects have been observed in patients receiving acidified feeding (gastrointestinal bleeding) or intermittent enteral feeding (increased gastric volume and lower volumes of feeding). Thus, to date, methods of enteral feeding aimed at reducing the incidence of VAP cannot be recommended for routine use.

Stress-Ulcer Prophylaxis

Gastric colonization by potentially pathogenic organisms increases with decreasing gastric acidity.[403] Thus, medications that decrease gastric acidity (antacids, H_2 blockers, proton pump inhibitors) may increase organism counts and increase the risk of VAP. In contrast, medications that do not affect gastric acidity (sucralfate) may not increase this risk. Several meta-analyses of more than 20 randomized trials have evaluated the risk for VAP associated with the methods used to prevent gastrointestinal bleeding in critically ill patients.[91,92,404] The largest randomized trial comparing ranitidine to sucralfate showed that ranitidine was superior in preventing gastrointestinal bleeding and did not increase the risk of VAP.[93] Therefore, despite the potential advantage of sucralfate (potentially less VAP with more gastrointestinal-bleeding) over H_2 blockers (potentially more VAP with less gastrointestinal bleeding) in preventing VAP, stress-ulcer prophylaxis with H_2 blockers appears to be safe in patients who are at high risk for bleeding as well as VAP.[405] Although proton pump inhibitors are now widely used for gastric bleeding prophylaxis in the ICU, based on their potentially higher efficacy, their use is associated with similar rates of nosocomial pneumonia as H_2 blockers.[404,406–410]

Selective Digestive Decontamination

Selective decontamination of the digestive tract (SDD) includes a short course of systemic antibiotic therapy, such as cefotaxime, trimethoprim, or a fluoroquinolone, and topical administration of nonabsorbable antibiotics (usually an aminoglycoside, polymyxin B, and amphotericin) to the mouth and stomach, in order to eradicate potentially pathogenic bacteria and yeast that may cause infections.[411] Since the original study published by Stoutenbeck et al in 1984,[412] which demonstrated a decrease of the overall infection rate in patients receiving the SDD regimen, more than forty randomized, controlled trials, and eight meta-analyses have been published. All eight meta-analyses reported a significant reduction in the risk of VAP, and four reported a significant reduction in mortality.[86,413–416] Recently, three prospective, randomized, controlled trials, all performed in ICUs with low rates of antibiotic resistance, have been published that were large enough to show a significant survival benefit in SDD-treated patients.[417–419] All three were in favor of treatment with SDD, the largest and most recent one by De Smet et al demonstrating a relative decrease in 28-day mortality rate (OR 0.83,95% CI, 0.72 to 0.97) and an absolute survival benefit of 3.5%.[419]

In spite of these benefits, widespread use of SDD in ICU patients remains controversial. The major concern with use of SDD is that it probably needs to be used in nearly all patients in a given ICU, and this widespread use has been shown in some studies to promote the emergence of resistant bacteria, particularly gram-positives such as MRSA.[420–424] This is likely to be even a greater problem in ICUs with a high baseline rate of resistance.[317,318,425] In contrast to what was expected, however, most studies that have evaluated this issue showed a lower incidence of colonization with (multi)resistant bacteria in SDD-treated patients than in control patients.[418,426] In a single-center observational study from Germany, 5-year use of SDD was not associated with an increase of MRSA or aminoglycoside and β-lactam resistance in gram-negative bacteria.[427] Putative explanations why colonization with resistant microorganisms is lower after treatment with SDD include the almost invariable sensibility of gram-negative aerobic bacteria for the commonly used combination of polymyxin E and tobramycin, the fact that treatment with polymyxin E rarely induces resistance, the very high local concentrations in the bowel of the used antibiotics, and the lower rate of use of systemic antibiotics in SDD-treated patients.[428]

Implementing a Structured Prevention Policy

The application of consistent evidence-based interventions to prevent VAP has been highly variable from one ICU to another and often suboptimal.[429–430] Moreover, no single preventive measure can succeed alone, emphasizing the need to use multifaceted and multidisciplinary programs to prevent VAP. Such programs are frequently referred to as "care bundles." A care bundle is a set of readily implementable interventions that are required to be undertaken for each patient on a regular basis.[431] The key goal is that every intervention must be implemented for every patient on every day of his or her stay in the ICU. Compliance is assessed for the bundle as a whole, so failure to complete even a single intervention means failure of the whole bundle at a particular assessment. The interventions need to be packaged in such a way that they are easy to assess for compliance, which usually means that no more than five interventions are included in each care bundle. The performance goal is to routinely achieve more than 95% compliance. Care bundles make it possible to introduce evidence-based preventive measures, including appropriate nurse staffing levels, hand hygiene with alcohol-based formulations, standardized weaning protocols and daily interruption of sedation, oral care with chlorhexidine, and keeping patients who receive enteral nutrition in a semirecumbent position.[252] All of these measures can be consistently applied to all patients in a coordinated way. The aim of care bundles is therefore only to facilitate and promote changes in patient care and encourage compliance with guidelines. Several studies using quasi-experimental designs confirmed the usefulness of this strategy for preventing VAP in the ICU.[134,355,432–444]

The lack of methodologic rigor of the reported studies, however, precludes any conclusive statements about “bundle care” effectiveness or cost-effectiveness. The exact set of key interventions that should be part of the “VAP-prevention bundle” is also not currently known as well as the optimal way for implementing it.[134,445–447] Successful VAP prevention requires an interdisciplinary team, educational interventions, system innovations, process indicator evaluation, and feedback to health care workers. As shown by a recent study, simply having a checklist available for reference without consideration of a robust implementation and adherence strategy is unlikely to maximize patient outcomes.[444] Whether this organization and data collection can be generalized to all ICUs remains to be determined, as well as the selection of the “optimal” bundle. In the meantime, clinical practice quality indicators must be developed in parallel with guidelines to check the adequacy between the two and to find solutions to improve guideline compliance.

In the United States, the Centers for Medicare and Medicaid Services has proposed stopping hospital reimbursements for care made necessary by preventable complications, including nosocomial infections, aiming for a zero-VAP rate.[448] Although this plan may have the desirable consequences of improving the quality of care, it also may penalize hospitals that admit high-risk patients and inadvertently encourage institutions to underreport VAP or to overuse antibiotics, thereby favoring dissemination of multidrug-resistant microorganisms. This possibility further underscores the need to carefully evaluate all new strategies potentially aimed at preventing VAP against what represents best clinical practices.

Conclusion

VAP is associated with mortality in excess of that caused by the underlying disease alone, particularly in case of infection caused by high-risk pathogens, such as *P. aeruginosa* and MRSA. The high level of bacterial resistance observed in patients who develop VAP limits the treatment options available to clinicians and encourages the use of antibiotic regimens combining several broad-spectrum drugs, even if the pretest probability of the disease is low, because initial inappropriate antimicrobial therapy has been linked to poor prognosis. Besides its economic impact, this practice of “spiralling empiricism” increasingly leads to the unnecessary administration of antibiotics in many ICU patients without true infection, paradoxically resulting in the emergence of infections caused by more antibiotic-resistant microorganisms that are, in turn, associated with increased rates of patient mortality and morbidity. Every possible effort should therefore be made to obtain, before new antibiotics are administered, reliable pulmonary specimens for direct microscope examination and cultures from each patient clinically suspected of having developed VAP. Because respiratory tract colonization of ICU patients is generally very complex, corresponding to a mix of self-colonization and cross-transmission, only a multifaceted and multidisciplinary preventive program can be effective.

References

1. Chastre J, Fagon JY. Ventilator-associated pneumonia. *Am J Respir Crit Care Med.* 2002;165:867–903. [PubMed: 11934711]

2. American Thoracic Society, Infectious Diseases Society of America. Guidelines for the management of adults with hospital-acquired, ventilator-associated, and healthcare-associated pneumonia. *Am J Respir Crit Care Med.* 2005;171:388–416.

3. National nosocomial infections surveillance system report, data summary from January 1990–May 1999, issued June 1999. *Am J Infect Control.* 1999;27:520–532.

4. Bergmans DC, Bonten MJ, Gaillard CA, et al. Indications for antibiotic use in ICU patients: a one-year prospective surveillance. *J Antimicrob Chemother.* 1997;39:527–535. [PubMed: 9145828]

5. Kirtland SH, Corley DE, Winterbauer RH, et al. The diagnosis of ventilator-associated pneumonia: a comparison of histologic, microbiologic, and clinical criteria. *Chest.* 1997;112:445–457. [PubMed: 9266883]

6. Tejerina E, Esteban A, Fernandez-Segoviano P, et al. Accuracy of clinical definitions of ventilator-associated pneumonia: comparison with autopsy findings. *J Crit Care.* 2010;25:62–68. [PubMed: 19592209]

7. Klompas M. Interobserver variability in ventilator-associated pneumonia surveillance. *Am J Infect Control.* 2010;38:237–239. [PubMed: 20171757]

8. Fagon JY, Chastre J, Hance AJ, et al. Evaluation of clinical judgment in the identification and treatment of nosocomial pneumonia in ventilated patients. *Chest.* 1993;103:547–553. [PubMed: 8432152]

9. Schurink CA, Van Nieuwenhoven CA, Jacobs JA, et al. Clinical pulmonary infection score for ventilator-associated pneumonia: accuracy and inter-observer variability. *Intensive Care Med.* 2004;30:217–224. [PubMed: 14566455]

10. Vincent JL, Rello J, Marshall J, et al. International study of the prevalence and outcomes of infection in intensive care units. *JAMA.* 2009;302:2323–2329. [PubMed: 19952319]

11. Cross AS, Roup B. Role of respiratory assistance devices in endemic nosocomial pneumonia. *Am J Med.* 1981;70:681–685. [PubMed: 6938128]

12. Horan TC, Culver DH, Gaynes RP, et al. Nosocomial infections in surgical patients in the United States, January 1986–June 1992. National Nosocomial Infections Surveillance System. *Infect Control Hosp Epidemiol.* 1993;14:73–80. [PubMed: 8440883]

13. Cook DJ, Walter SD, Cook RJ, et al. Incidence of and risk factors for ventilator-associated pneumonia in critically ill patients. *Ann Intern Med.* 1998;129:433–440. [PubMed: 9735080]

14. Chastre J, Trouillet JL, Vuagnat A, et al. Nosocomial pneumonia in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 1998;157:1165–1172. [PubMed: 9563735]

15. Delclaux C, Roupie E, Blot F, et al. Lower respiratory tract colonization and infection during severe acute respiratory distress syndrome: incidence and diagnosis. *Am J Respir Crit Care Med.* 1997;156:1092–1098. [PubMed: 9351607]

16. Markowicz P, Wolff M, Djedaini K, et al. Multicenter prospective study of ventilator-associated pneumonia during acute respiratory distress syndrome. Incidence, prognosis, and risk factors. ARDS study group. *Am J Respir Crit Care Med.* 2000;161:1942–1948. [PubMed: 10852771]

17. Meduri GU, Reddy RC, Stanley T, El-Zeky F. Pneumonia in acute respiratory distress syndrome. A prospective evaluation of bilateral bronchoscopic sampling. *Am J Respir Crit Care Med.* 1998;158:870–875. [PubMed: 9731019]

18. Melsen WG, Rovers MM, Bonten MJ. Ventilator-associated pneumonia and mortality: a systematic review of observational studies. *Crit Care Med.* 2009;37:2709–2718. [PubMed: 19885994]

19. Fagon JY, Chastre J, Hance AJ, et al. Nosocomial pneumonia in ventilated patients: a cohort study evaluating attributable mortality and hospital stay. *Am J Med.* 1993;94:281–288. [PubMed: 8452152]

20. Fagon JY, Chastre J, Vuagnat A, et al. Nosocomial pneumonia and mortality among patients in intensive care units. *JAMA.* 1996;275:866–869. [PubMed: 8596225]

21. Vincent JL, Bihari DJ, Suter PM, et al. The prevalence of nosocomial infection in intensive care units in Europe. Results of the European prevalence of infection in intensive care (EPIC) study. Epic International Advisory Committee. *JAMA.* 1995;274:639–644. [PubMed: 7637145]

22. Kollef MH. Ventilator-associated pneumonia. A multivariate analysis. *JAMA.* 1993;270:1965–1970. [PubMed: 8411554]

23. Bueno-Cavanillas A, Delgado-Rodriguez M, Lopez-Luque A, et al. Influence of nosocomial infection on mortality rate in an intensive care unit. *Crit Care Med.* 1994;22:55–60. [PubMed: 8124975]

24. Nguile-Makao M, Zahar JR, Francais A, et al. Attributable mortality of ventilator-associated pneumonia: respective impact of main characteristics at ICU admission and VAP onset using conditional logistic regression and multi-state models. *Intensive Care Med.* 2010;36:781–789. [PubMed: 20232046]

25. Rello J, Ausina V, Ricart M, et al. Impact of previous antimicrobial therapy on the etiology and outcome of ventilator-associated pneumonia. *Chest.* 1993;104:1230–1235. [PubMed: 8404198]

26. Sanchez-Nieto JM, Torres A, Garcia-Cordoba F, et al. Impact of invasive and noninvasive quantitative culture sampling on outcome of ventilator-associated pneumonia: a pilot study [published erratum appears in *Am J Respir Crit Care Med.* 1998;157(3 pt 1):1005]. *Am J Respir Crit Care Med.* 1998;157:371–376. [PubMed: 9476845]

27. Ruiz M, Torres A, Ewig S, et al. Noninvasive versus invasive microbial investigation in ventilator-associated pneumonia: evaluation of outcome. *Am J Respir Crit Care Med.* 2000;162:119–125. [PubMed: 10903230]

28. Alvarez-Lerma F. Modification of empiric antibiotic treatment in patients with pneumonia acquired in the intensive care unit. ICU-acquired pneumonia study group. *Intensive Care Med.* 1996;22:387–394. [PubMed: 8796388]

29. Kollef MH. Antimicrobial therapy of ventilator-associated pneumonia: how to select an appropriate drug regimen [editorial; comment]. *Chest.* 1999;115:8–11. [PubMed: 9925056]

30. Luna CM, Vujacich P, Niederman MS, et al. Impact of BAL data on the therapy and outcome of ventilator-associated pneumonia. *Chest.* 1997;111:676–685. [PubMed: 9118708]

31. Kollef MH, Sherman G, Ward S, Fraser VJ. Inadequate antimicrobial treatment of infections: a risk factor for hospital mortality among critically ill patients. *Chest.* 1999;115:462–474. [PubMed: 10027448]

32. Dupont H, Mentec H, Sollet JP, Bleichner G. Impact of appropriateness of initial antibiotic therapy on the outcome of ventilator-associated pneumonia. *Intensive Care Med.* 2001;27:355–362. [PubMed: 11396279]

33. Combes A, Luyt CE, Fagon JY, et al. Early predictors for infection recurrence and death in patients with ventilator-associated pneumonia. *Crit Care Med.* 2007;35:146–154. [PubMed: 17080004]

34. Schumacher M, Wangler M, Wolkewitz M, Beyersmann J. Attributable mortality due to nosocomial infections. A simple and useful application of multistate models. *Methods Inf Med.* 2007;46:595–600. [PubMed: 17938785]

35. Safdar N, Dezfulian C, Collard HR, Saint S. Clinical and economic consequences of ventilator-associated pneumonia: a systematic review. *Crit Care Med.* 2005;33:2184–2193. [PubMed: 16215368]

36. Eber MR, Laxminarayan R, Perencevich EN, Malani A. Clinical and economic outcomes attributable to health care-associated sepsis and pneumonia. *Arch Intern Med.* 2010;170:347–353. [PubMed: 20177037]

37. Heyland DK, Cook DJ, Griffith L, et al. The attributable morbidity and mortality of ventilator-associated pneumonia in the critically ill patient. The Canadian critical trials group. *Am J Respir Crit Care Med.* 1999;159:1249–1256. [PubMed: 10194173]

38. Baker AM, Meredith JW, Haponik EF. Pneumonia in intubated trauma patients. Microbiology and outcomes. *Am J Respir Crit Care Med.* 1996;153:343–349. [PubMed: 8542141]

39. Muscedere JG, Martin CM, Heyland DK. The impact of ventilator-associated pneumonia on the Canadian health care system. *J Crit Care.* 2008;23:5–10. [PubMed: 18359415]

40. Peleg AY, Hooper DC. Hospital-acquired infections due to gram-negative bacteria. *N Engl J Med.* 2010;362:1804–1813. [PubMed: 20463340]

41. Rello J, Ollendorf DA, Oster G, et al. Epidemiology and outcomes of ventilator-associated pneumonia in a large us database. *Chest.* 2002;122:2115–2121. [PubMed: 12475855]

42. Kollef MH, Morrow LE, Niederman MS, et al. Clinical characteristics and treatment patterns among patients with ventilator-associated pneumonia. *Chest.* 2006;129:1210–1218. [PubMed: 16685011]

43. Rello J, Diaz E, Rodriguez A. Etiology of ventilator-associated pneumonia. *Clin Chest Med.* 2005;26:87–95. [PubMed: 15802170]

44. Combes A, Figliolini C, Trouillet JL, et al. Incidence and outcome of polymicrobial ventilator-associated pneumonia. *Chest.* 2002;121:1618–1623. [PubMed: 12006452]

45. Trouillet JL, Chastre J, Vuagnat A, et al. Ventilator-associated pneumonia caused by potentially drug-resistant bacteria. *Am J Respir Crit Care Med.* 1998;157:531–539. [PubMed: 9476869]

46. Giantsou E, Liratzopoulos N, Efraimidou E, et al. Both early-onset and late-onset ventilator-associated pneumonia are caused mainly by potentially multiresistant bacteria. *Intensive Care Med.* 2005;31:1488–1494. [PubMed: 16151723]

47. Ibrahim EH, Ward S, Sherman G, Kollef MH. A comparative analysis of patients with early-onset vs late-onset nosocomial pneumonia in the ICU setting. *Chest.* 2000;117:1434–1442. [PubMed: 10807834]

48. Rello J, Sa-Borges M, Correa H, et al. Variations in etiology of ventilator-associated pneumonia across four treatment sites: implications for antimicrobial prescribing practices. *Am J Respir Crit Care Med.* 1999;160:608–613. [PubMed: 10430736]

49. Bruynseels P, Jorens PG, Demey HE, et al. Herpes simplex virus in the respiratory tract of critical care patients: a prospective study. *Lancet.* 2003;362:1536–1541. [PubMed: 14615108]

50. Luyt CE. Virus diseases in ICU patients: a long time underestimated; but be aware of overestimation. *Intensive Care Med.* 2006;32:968–970. [PubMed: 16791659]

51. Luyt CE, Combes A, Deback C, et al. Herpes simplex virus lung infection in patients undergoing prolonged mechanical ventilation. *Am J Respir Crit Care Med.* 2007;175:935–942. [PubMed: 17234903]

52. Chiche L, Forel JM, Papazian L. The role of viruses in nosocomial pneumonia. *Curr Opin Infect Dis.* 2011;24:152–156. [PubMed: 21252660]

53. Papazian L, Fraisse A, Garbe L, et al. Cytomegalovirus. An unexpected cause of ventilator-associated pneumonia. *Anesthesiology.* 1996;84:280–287. [PubMed: 8602657]

54. Heininger A, Haeberle H, Fischer I, et al. Cytomegalovirus reactivation and associated outcome of critically ill patients with severe sepsis. *Crit Care.* 2011;15:R77.

55. Heininger A, Jahn G, Engel C, et al. Human cytomegalovirus infections in nonimmunosuppressed critically ill patients. *Crit Care Med.* 2001;29:541–547. [PubMed: 11373417]

56. Delisle MS, Williamson DR, Perreault MM, et al. The clinical significance of candida colonization of respiratory tract secretions in critically ill patients. *J Crit Care.* 2008;23:11–17. [PubMed: 18359416]

57. el-Ebiary M, Torres A, Fabregas N, et al. Significance of the isolation of candida species from respiratory samples in critically ill, non-neutropenic patients. An immediate postmortem histologic study. *Am J Respir Crit Care Med.* 1997;156:583–590. [PubMed: 9279244]

58. Rello J, Esandi ME, Diaz E, et al. The role of candida sp. isolated from bronchoscopic samples in nonneutropenic patients. *Chest.* 1998;114:146–149. [PubMed: 9674461]

59. Williamson DR, Albert M, Perreault MM, et al. The relationship between candida species cultured from the respiratory tract and systemic inflammation in critically ill patients with ventilator-associated pneumonia. *Can J Anaesth.* 2011;58:275–284. [PubMed: 21287306]

60. Wood GC, Mueller EW, Croce MA, et al. *Candida* sp. isolated from bronchoalveolar lavage: clinical significance in critically ill trauma patients. *Intensive Care Med.* 2006;32:599–603. [PubMed: 16477410]

61. Ader F, Jawhara S, Nseir S, et al. Short term *Candida albicans* colonization reduces pseudomonas aeruginosa-related lung injury and bacterial burden in a murine model. *Crit Care.* 2011;15:R150.

62. Nseir S, Jozefowicz E, Cavestri B, et al. Impact of antifungal treatment on *Candida-Pseudomonas* interaction: a preliminary retrospective case-control study. *Intensive Care Med.* 2007;33:137–142. [PubMed: 17115135]

63. Roux D, Gaudry S, Dreyfuss D, et al. *Candida albicans* impairs macrophage function and facilitates pseudomonas aeruginosa pneumonia in rat. *Crit Care Med.* 2009;37:1062–1067. [PubMed: 19237918]

64. Azoulay E, Timsit JF, Tafflet M, et al. *Candida* colonization of the respiratory tract and subsequent pseudomonas ventilator-associated pneumonia. *Chest.* 2006;129:110–117. [PubMed: 16424420]

65. Craven DE, De Rosa FG, Thornton D. Nosocomial pneumonia: emerging concepts in diagnosis, management, and prophylaxis. *Curr Opin Crit Care.* 2002;8:421–429. [PubMed: 12357110]

66. Johanson WG, Pierce AK, Sanford JP, Thomas GD. Nosocomial respiratory infections with gram-negative bacilli. The significance of colonization of the respiratory tract. *Ann Intern Med.* 1972;77:701–706. [PubMed: 5081492]

67. Bonten MJ, Gaillard CA, de Leeuw PW, Stobberingh EE. Role of colonization of the upper intestinal tract in the pathogenesis of ventilator-associated pneumonia [see comments]. *Clin Infect Dis.* 1997;24:309–319. [PubMed: 9114178]

68. Bonten MJ, Bergmans DC, Speijer H, Stobberingh EE. Characteristics of polyclonal endemicity of pseudomonas aeruginosa colonization in intensive care units. Implications for infection control. *Am J Respir Crit Care Med.* 1999;160:1212–1219. [PubMed: 10508809]

69. Bonten MJ, Gaillard CA, van Tiel FH, et al. The stomach is not a source for colonization of the upper respiratory tract and pneumonia in ICU patients [see comments]. *Chest.* 1994;105:878–884. [PubMed: 8131556]

70. de Latorre FJ, Pont T, Ferrer A, et al. Pattern of tracheal colonization during mechanical ventilation. *Am J Respir Crit Care Med.* 1995;152:1028–1033.

71. Bonten MJ, Kullberg BJ, van Dalen R, et al. Selective digestive decontamination in patients in intensive care. *J Antimicrob Chemother.* 2000;46:351–362. [PubMed: 10980160]

72. Bonten MJ. Controversies on diagnosis and prevention of ventilator-associated pneumonia. *Diagn Microbiol Infect Dis.* 1999;34:199–204. [PubMed: 10403099]

73. Chevret S, Hemmer M, Carlet J, Langer M. Incidence and risk factors of pneumonia acquired in intensive care units. Results from a multicenter prospective study on 996 patients. European cooperative group on nosocomial pneumonia. *Intensive Care Med.* 1993;19:256–264. [PubMed: 8408934]

74. Celis R, Torres A, Gatell JM, et al. Nosocomial pneumonia. A multivariate analysis of risk and prognosis. *Chest.* 1988;93:318–324. [PubMed: 3338299]

75. Torres A, Aznar R, Gatell JM, et al. Incidence, risk, and prognosis factors of nosocomial pneumonia in mechanically ventilated patients. *Am Rev Respir Dis.* 1990;142:523–528. [PubMed: 2202245]

76. Craven DE, Kunches LM, Kilinsky V, et al. Risk factors for pneumonia and fatality in patients receiving continuous mechanical ventilation. *Am Rev Respir Dis.* 1986;133:792–796. [PubMed: 3706887]

77. Joshi N, Localio AR, Hamory BH. A predictive risk index for nosocomial pneumonia in the intensive care unit. *Am J Med.* 1992;93:135–142. [PubMed: 1497009]

78. Antonelli M, Moro ML, Capelli O, et al. Risk factors for early onset pneumonia in trauma patients. *Chest.* 1994;105:224–228. [PubMed: 8275735]

79. Cunnion KM, Weber DJ, Broadhead WE, et al. Risk factors for nosocomial pneumonia: comparing adult critical-care populations. *Am J Respir Crit Care Med.* 1996;153:158–162. [PubMed: 8542110]

80. Daschner F, Kappstein I, Engels I, et al. Stress ulcer prophylaxis and ventilation pneumonia: prevention by antibacterial cytoprotective agents? *Infect Control Hosp Epidemiol.* 1988;9:59–65. [PubMed: 3343501]

81. Garibaldi RA, Britt MR, Coleman ML, et al. Risk factors for postoperative pneumonia. *Am J Med.* 1981;70:677–680. [PubMed: 7211900]

82. Tillotson JR, Finland M. Secondary pulmonary infections following antibiotic therapy for primary bacterial pneumonia. *Antimicrob Agents Chemother.* 1968;8:326–330. [PubMed: 5735377]

83. Spencer RC. Predominant pathogens found in the European prevalence of infection in intensive care study. *Eur J Clin Microbiol Infect Dis.* 1996;15:281–285. [PubMed: 8781877]

84. Rello J, Ausina V, Ricart M, et al. Risk factors for infection by pseudomonas aeruginosa in patients with ventilator-associated pneumonia. *Intensive Care Med.* 1994;20:193–198. [PubMed: 8014285]

85. Jarvis WR, Edwards JR, Culver DH, et al. Nosocomial infection rates in adult and pediatric intensive care units in the United States. National nosocomial infections surveillance system. *Am J Med.* 1991;91:185S–191S.

86. D'Amico R, Pifferi S, Leonetti C, et al. Effectiveness of antibiotic prophylaxis in critically ill adult patients: systematic review of randomised controlled trials. *BMJ.* 1998;316:1275–1285. [PubMed: 9554897]

87. Sirvent JM, Torres A, El-Ebiary M, et al. Protective effect of intravenously administered cefuroxime against nosocomial pneumonia in patients with structural coma. *Am J Respir Crit Care Med.* 1997;155:1729–1734. [PubMed: 9154884]

88. George DL, Falk PS, Wunderink RG, et al. Epidemiology of ventilator-acquired pneumonia based on protected bronchoscopic sampling. *Am J Respir Crit Care Med.* 1998;158:1839–1847. [PubMed: 9847276]

89. Atherton ST, White DJ. Stomach as source of bacteria colonising respiratory tract during artificial ventilation. *Lancet.* 1978;2:968–969. [PubMed: 81991]

90. du Moulin GC, Paterson DG, Hedley-Whyte J, Lisbon A. Aspiration of gastric bacteria in antacid-treated patients: a frequent cause of postoperative colonisation of the airway. *Lancet.* 1982;1:242–245.

91. Cook DJ, Reeve BK, Guyatt GH, et al. Stress ulcer prophylaxis in critically ill patients. Resolving discordant meta-analyses. *JAMA.* 1996;275:308–314. [PubMed: 8544272]

92. Messori A, Trippoli S, Vaiani M, et al. Bleeding and pneumonia in intensive care patients given ranitidine and sucralfate for prevention of stress ulcer: meta-analysis of randomised controlled trials [in process citation]. *BMJ.* 2000;321:1103–1106. [PubMed: 11061729]

93. Cook D, Guyatt G, Marshall J, et al. A comparison of sucralfate and ranitidine for the prevention of upper gastrointestinal bleeding in patients requiring mechanical ventilation. Canadian critical care trials group [see comments]. *N Engl J Med.* 1998;338:791–797. [PubMed: 9504939]

94. Eom CS, Jeon CY, Lim JW, et al. Use of acid-suppressive drugs and risk of pneumonia: a systematic review and meta-analysis. *CMAJ.* 2011;183:310–319. [PubMed: 21173070]

95. Herzig SJ, Howell MD, Ngo LH, Marcantonio ER. Acid-suppressive medication use and the risk for hospital-acquired pneumonia. *JAMA.* 2009;301:2120–2128. [PubMed: 19470989]

96. Ben-Menachem T, Fogel R, Patel RV, et al. Prophylaxis for stress-related gastric hemorrhage in the medical intensive care unit. A randomized, controlled, single-blind study. *Ann Intern Med.* 1994;121:568–575. [PubMed: 8085688]

97. Eddleston JM, Pearson RC, Holland J, et al. Prospective endoscopic study of stress erosions and ulcers in critically ill adult patients treated with either sucralfate or placebo. *Crit Care Med.* 1994;22:1949–1954. [PubMed: 7988131]

98. Sottile FD, Marrie TJ, Prough DS, et al. Nosocomial pulmonary infection: possible etiologic significance of bacterial adhesion to endotracheal tubes. *Crit Care Med.* 1986;14:265–270. [PubMed: 3956213]

99. Spray SB, Zuidema GD, Cameron JL. Aspiration pneumonia; incidence of aspiration with endotracheal tubes. *Am J Surg.* 1976;131:701–703. [PubMed: 937649]

100. Rello J, Diaz E, Roque M, Valles J. Risk factors for developing pneumonia within 48 hours of intubation. *Am J Respir Crit Care Med.* 1999;159:1742–1746. [PubMed: 10351912]

101. Torres A, Gatell JM, Aznar E, et al. Re-intubation increases the risk of nosocomial pneumonia in patients needing mechanical ventilation. *Am J Respir Crit Care Med.* 1995;152:137–141. [PubMed: 7599812]

102. Rumbak MJ, Newton M, Truncale T, et al. A prospective, randomized, study comparing early percutaneous dilational tracheotomy to prolonged translaryngeal intubation (delayed tracheotomy) in critically ill medical patients. *Crit Care Med.* 2004;32:1689–1694.

[PubMed: 15286545]

103. Blot F, Similowski T, Trouillet JL, et al. Early tracheotomy versus prolonged endotracheal intubation in unselected severely ill ICU patients. *Intensive Care Med.* 2008;34:1779–1787. [PubMed: 18592210]

104. Bouderka MA, Fakhir B, Bouaggad A, et al. Early tracheostomy versus prolonged endotracheal intubation in severe head injury. *J Trauma.* 2004;57:251–254. [PubMed: 15345969]

105. Trouillet JL, Luyt CE, Guiguet M, et al. Early percutaneous tracheotomy versus prolonged intubation of mechanically ventilated patients after cardiac surgery: a randomized trial. *Ann Intern Med.* 2011;154:373–383. [PubMed: 21403073]

106. Terragni PP, Antonelli M, Fumagalli R, et al. Early vs late tracheotomy for prevention of pneumonia in mechanically ventilated adult ICU patients: a randomized controlled trial. *JAMA.* 2010;303:1483–1489. [PubMed: 20407057]

107. Moore FA, Moore EE, Jones TN, et al. Ten versus TPN following major abdominal trauma—reduced septic morbidity. *J Trauma.* 1989;29:916–922; discussion 922–913. [PubMed: 2501509]

108. Heyland DK, Cook DJ, Schoenfeld PS, et al. The effect of acidified enteral feeds on gastric colonization in critically ill patients: results of a multicenter randomized trial. Canadian critical care trials group. *Crit Care Med.* 1999;27:2399–2406. [PubMed: 10579255]

109. Montecalvo MA, Steger KA, Farber HW, et al. Nutritional outcome and pneumonia in critical care patients randomized to gastric versus jejunal tube feedings. The critical care research team. *Crit Care Med.* 1992;20:1377–1387. [PubMed: 1395657]

110. Hsu CW, Sun SF, Lin SL, et al. Duodenal versus gastric feeding in medical intensive care unit patients: a prospective, randomized, clinical study. *Crit Care Med.* 2009;37:1866–1872. [PubMed: 19384225]

111. Meer JA. Inadvertent dislodgement of nasoenteral feeding tubes: incidence and prevention. *JPEN J Parenter Enteral Nutr.* 1987;11:187–189. [PubMed: 3108542]

112. Torres A, Serra-Batlles J, Ros E, et al. Pulmonary aspiration of gastric contents in patients receiving mechanical ventilation: the effect of body position. *Ann Intern Med.* 1992;116:540–543. [PubMed: 1543307]

113. Drakulovic MB, Torres A, Bauer TT, et al. Supine body position as a risk factor for nosocomial pneumonia in mechanically ventilated patients: a randomised trial. *Lancet.* 1999;354:1851–1858. [PubMed: 10584721]

114. Orozco-Levi M, Torres A, Ferrer M, et al. Semirecumbent position protects from pulmonary aspiration but not completely from gastroesophageal reflux in mechanically ventilated patients. *Am J Respir Crit Care Med.* 1995;152:1387–1390. [PubMed: 7551400]

115. Peterson M, Schwab W, McCutcheon K, et al. Effects of elevating the head of bed on interface pressure in volunteers. *Crit Care Med.* 2008;36(11):3038–3042. [PubMed: 18824905]

116. Silvestri L, Gregori D, van Saene HK, et al. Semirecumbent position to prevent ventilator-associated pneumonia is not evidence based. *J Crit Care.* 2010;25:152–153. [PubMed: 19828285]

117. van Nieuwenhoven CA, Vandenbroucke-Grauls C, van Tiel FH, et al. Feasibility and effects of the semirecumbent position to prevent ventilator-associated pneumonia: a randomized study. *Crit Care Med.* 2006;34:396–402.

118. Goularte TA, Manning M, Craven DE. Bacterial colonization in humidifying cascade reservoirs after 24 and 48 hours of continuous mechanical ventilation. *Infect Control.* 1987;8:200–203. [PubMed: 3647941]

119. Craven DE, Goularte TA, Make BJ. Contaminated condensate in mechanical ventilator circuits. A risk factor for nosocomial pneumonia? *Am Rev Respir Dis.* 1984;129:625–628. [PubMed: 6585160]

120. Craven DE, Steger KA, Barber TW. Preventing nosocomial pneumonia: state of the art and perspectives for the 1990s. *Am J Med.* 1991;91:44S–53S.

121. Martin C, Perrin G, Gevaudan MJ, et al. Heat and moisture exchangers and vaporizing humidifiers in the intensive care unit. *Chest.* 1990;97:144–149. [PubMed: 2295234]

122. Roustan JP, Kienlen J, Aubas P, et al. Comparison of hydrophobic heat and moisture exchangers with heated humidifier during prolonged mechanical ventilation. *Intensive Care Med.* 1992;18:97–100. [PubMed: 1613206]

123. Dreyfuss D, Djedaini K, Gros I, et al. Mechanical ventilation with heated humidifiers or heat and moisture exchangers: effects on patient colonization and incidence of nosocomial pneumonia. *Am J Respir Crit Care Med.* 1995;151:986–992. [PubMed: 7697277]

124. Hurni JM, Feihl F, Lazor R, et al. Safety of combined heat and moisture exchanger filters in long-term mechanical ventilation. *Chest.* 1997;111:686–691. [PubMed: 9118709]

125. Kirton OC, DeHaven B, Morgan J, et al. A prospective, randomized comparison of an in-line heat moisture exchange filter and heated wire humidifiers: rates of ventilator-associated early-onset (community-acquired) or late-onset (hospital-acquired) pneumonia and incidence of endotracheal tube occlusion. *Chest.* 1997;112:1055–1059. [PubMed: 9377917]

126. Djedaini K, Billiard M, Mier L, et al. Changing heat and moisture exchangers every 48 hours rather than 24 hours does not affect their efficacy and the incidence of nosocomial pneumonia. *Am J Respir Crit Care Med.* 1995;152:1562–1569. [PubMed: 7582295]

127. Conti G, De Blasi RA, Rocco M, et al. Effects of the heat-moisture exchangers on dynamic hyperinflation of mechanically ventilated COPD patients. *Intensive Care Med.* 1990;16:441–443. [PubMed: 2269712]

128. Ploysongsang Y, Branson R, Rashkin MC, Hurst JM. Pressure flow characteristics of commonly used heat-moisture exchangers. *Am Rev Respir Dis.* 1988;138:675–678. [PubMed: 3202420]

129. Dreyfuss D, Djedaini K, Weber P, et al. Prospective study of nosocomial pneumonia and of patient and circuit colonization during mechanical ventilation with circuit changes every 48 hours versus no change. *Am Rev Respir Dis.* 1991;143:738–743. [PubMed: 2008985]

130. Kollef MH, Shapiro SD, Fraser VJ, et al. Mechanical ventilation with or without 7-day circuit changes. A randomized controlled trial. *Ann Intern Med.* 1995;123:168–174. [PubMed: 7598297]

131. Long MN, Wickstrom G, Grimes A, et al. Prospective, randomized study of ventilator-associated pneumonia in patients with one versus three ventilator circuit changes per week. *Infect Control Hosp Epidemiol.* 1996;17:14–19. [PubMed: 8789682]

132. Centers for Disease Control and Prevention. Guidelines for prevention of nosocomial pneumonia. *MMWR Morb Mortal Wkly Rep.* 1997;46:1–79.

133. Masterton R, Craven D, Rello J, et al. Hospital-acquired pneumonia guidelines in Europe: a review of their status and future development. *J Antimicrob Chemother.* 2007;60:206–213. [PubMed: 17545144]

134. Rello J, Lode H, Cornaglia G, Masterton R. A European care bundle for prevention of ventilator-associated pneumonia. *Intensive Care Med.* 2010;36:773–780. [PubMed: 20237759]

135. Holzapfel L, Chastang C, Demingeon G, et al. A randomized study assessing the systematic search for maxillary sinusitis in nasotracheally mechanically ventilated patients. Influence of nosocomial maxillary sinusitis on the occurrence of ventilator-associated pneumonia. *Am J Respir Crit Care Med.* 1999;159:695–701. [PubMed: 10051239]

136. Rouby JJ, Laurent P, Gosnach M, et al. Risk factors and clinical relevance of nosocomial maxillary sinusitis in the critically ill. *Am J Respir Crit Care Med.* 1994;150:776–783. [PubMed: 8087352]

137. Kollef MH, Von Harz B, Prentice D, et al. Patient transport from intensive care increases the risk of developing ventilator-associated pneumonia. *Chest.* 1997;112:765–773. [PubMed: 9315813]

138. Wunderink RG. Clinical criteria in the diagnosis of ventilator-associated pneumonia. *Chest.* 2000;117:191S–194S.

139. Wunderink RG. Radiologic diagnosis of ventilator-associated pneumonia. *Chest.* 2000;117:188S–190S.

140. Wunderink RG, Woldenberg LS, Zeiss J, et al. The radiologic diagnosis of autopsy-proven ventilator-associated pneumonia. *Chest.* 1992;101:458–463. [PubMed: 1735272]

141. Meduri GU, Mauldin GL, Wunderink RG, et al. Causes of fever and pulmonary densities in patients with clinical manifestations of ventilator-associated pneumonia. *Chest.* 1994;106:221–235. [PubMed: 8020275]

142. Winer-Muram HT, Steiner RM, Gurney JW, et al. Ventilator-associated pneumonia in patients with adult respiratory distress syndrome: CT evaluation. *Radiology.* 1998;208:193–199. [PubMed: 9646813]

143. Fagon JY. Diagnosis and treatment of ventilator-associated pneumonia: fiberoptic bronchoscopy with bronchoalveolar lavage is essential. *Semin Respir Crit Care Med.* 2006;27:34–44. [PubMed: 16508880]

144. Pugin J, Auckenthaler R, Mili N, et al. Diagnosis of ventilator-associated pneumonia by bacteriologic analysis of bronchoscopic and nonbronchoscopic “blind” bronchoalveolar lavage fluid. *Am Rev Respir Dis.* 1991;143:1121–1129. [PubMed: 2024824]

145. Schurink CA, Nieuwenhoven CA, Jacobs JA, et al. Clinical pulmonary infection score for ventilator-associated pneumonia: accuracy and inter-observer variability. *Intensive Care Med.* 2004;30:217–224. [PubMed: 14566455]

146. Luyt CE, Chastre J, Fagon JY. Value of the clinical pulmonary infection score for the identification and management of ventilator-associated pneumonia. *Intensive Care Med.* 2004;30:844–852. [PubMed: 15127196]

147. Torres A, el-Ebiary M, Padro L, et al. Validation of different techniques for the diagnosis of ventilator-associated pneumonia. Comparison with immediate postmortem pulmonary biopsy. *Am J Respir Crit Care Med.* 1994;149:324–331. [PubMed: 8306025]

148. Croce MA, Swanson JM, Magnotti LJ, et al. The futility of the clinical pulmonary infection score in trauma patients. *J Trauma.* 2006;60:523–527; discussion 527–528. [PubMed: 16531849]

149. Fabregas N, Ewig S, Torres A, et al. Clinical diagnosis of ventilator associated pneumonia revisited: comparative validation using immediate post-mortem lung biopsies. *Thorax.* 1999;54:867–873. [PubMed: 10491448]

150. Fartoukh M, Maitre B, Honore S, et al. Diagnosing pneumonia during mechanical ventilation: the clinical pulmonary infection score revisited. *Am J Respir Crit Care Med.* 2003;168:173–179. [PubMed: 12738607]

151. Jung B, Embriaco N, Roux F, et al. Microbiological data, but not procalcitonin improve the accuracy of the clinical pulmonary infection score. *Intensive Care Med.* 2010;36:790–798. [PubMed: 20217042]

152. Rosbolt MB, Sterling ES, Fahy BG. The utility of the clinical pulmonary infection score. *J Intensive Care Med.* 2009;24:26–34. [PubMed: 19019839]

153. Shan J, Chen HL, Zhu JH. Diagnostic accuracy of clinical pulmonary infection score for ventilator-associated pneumonia: a meta-analysis. *Respir Care.* 2011;56:1087–1094. [PubMed: 21310117]

154. Zilberberg MD, Shorr AF. Ventilator-associated pneumonia: the clinical pulmonary infection score as a surrogate for diagnostics and outcome. *Clin Infect Dis.* 2010;51 Suppl 1:S131–S135.

155. Niederman MS. Appropriate use of antimicrobial agents: challenges and strategies for improvement. *Crit Care Med.* 2003;31:608–616. [PubMed: 12576973]

156. Kolleff MH. Appropriate antibiotic therapy for ventilator-associated pneumonia and sepsis: a necessity, not an issue for debate. *Intensive Care Med.* 2003;29:147–149. [PubMed: 12675039]

157. Iregui M, Ward S, Sherman G, et al. Clinical importance of delays in the initiation of appropriate antibiotic treatment for ventilator-associated pneumonia. *Chest.* 2002;122:262–268. [PubMed: 12114368]

158. Kumar A, Roberts D, Wood KE, et al. Duration of hypotension before initiation of effective antimicrobial therapy is the critical determinant of survival in human septic shock. *Crit Care Med.* 2006;34:1589–1596. [PubMed: 16625125]

159. Bochud PY, Bonten M, Marchetti O, Calandra T. Antimicrobial therapy for patients with severe sepsis and septic shock: an evidence-based review. *Crit Care Med.* 2004;32:S495–S512.

160. Dellinger RP, Levy MM, Carlet JM, et al. Surviving sepsis campaign: international guidelines for management of severe sepsis and septic shock: 2008. *Crit Care Med.* 2008;36:296–327. [PubMed: 18158437]

161. Kumar A. Optimizing antimicrobial therapy in sepsis and septic shock. *Crit Care Clin.* 2009;25:733–751. [PubMed: 19892250]

162. Micek ST, Welch EC, Khan J, et al. Empiric combination antibiotic therapy is associated with improved outcome against sepsis due to gram-negative bacteria: a retrospective analysis. *Antimicrob Agents Chemother.* 2010;54:1742–1748. [PubMed: 20160050]

163. Park DR. Antimicrobial treatment of ventilator-associated pneumonia. *Respir Care.* 2005;50:932–952. [PubMed: 15972114]

164. Kollef MH, Fraser VJ. Antibiotic resistance in the intensive care unit. *Ann Intern Med.* 2001;134:298–314. [PubMed: 11182841]

165. Carmeli Y, Troillet N, Karchmer AW, Samore MH. Health and economic outcomes of antibiotic resistance in pseudomonas aeruginosa. *Arch Intern Med.* 1999;159:1127–1132. [PubMed: 10335691]

166. Goldmann DA, Weinstein RA, Wenzel RP, et al. Strategies to prevent and control the emergence and spread of antimicrobial-resistant microorganisms in hospitals. A challenge to hospital leadership. *JAMA.* 1996;275:234–240. [PubMed: 8604178]

167. Neu HC. The crisis in antibiotic resistance. *Science.* 1992;257:1064–1073. [PubMed: 1509257]

168. Neuhauser MM, Weinstein RA, Rydman R, et al. Antibiotic resistance among gram-negative bacilli in us intensive care units: implications for fluoroquinolone use. *JAMA.* 2003;289:885–888. [PubMed: 12588273]

169. Struelens MJ, Byl B, Vincent JL. Antibiotic policy: a tool for controlling resistance of hospital pathogens. *Clin Microbiol Infect.* 1999;5:S19–S24.

170. Rello J, Rue M, Jubert P, et al. Survival in patients with nosocomial pneumonia: impact of the severity of illness and the etiologic agent. *Crit Care Med.* 1997;25:1862–1867. [PubMed: 9366771]

171. Kollef MH. Inadequate antimicrobial treatment: an important determinant of outcome for hospitalized patients. *Clin Infect Dis.* 2000;31 Suppl 4:S131–S138.

172. Kollef MH. Treatment of ventilator-associated pneumonia: get it right from the start. *Crit Care Med.* 2003;31:969–970. [PubMed: 12627014]

173. Kollef MH, Ward S, Sherman G, et al. Inadequate treatment of nosocomial infections is associated with certain empiric antibiotic choices. *Crit Care Med.* 2000;28:3456–3464. [PubMed: 11057801]

174. Leone M, Bourgoin A, Giuly E, et al. Influence on outcome of ventilator-associated pneumonia in multiple trauma patients with head trauma treated with selected digestive decontamination. *Crit Care Med.* 2002;30:1741–1746. [PubMed: 12163786]

175. Ibrahim EH, Sherman G, Ward S, et al. The influence of inadequate antimicrobial treatment of bloodstream infections on patient outcomes in the ICU setting. *Chest.* 2000;118:146–155. [PubMed: 10893372]

176. Torres A, Ewig S. Diagnosing ventilator-associated pneumonia. *N Engl J Med.* 2004;350:433–435. [PubMed: 14749448]

177. Cook D, Mandell L. Endotracheal aspiration in the diagnosis of ventilator-associated pneumonia. *Chest.* 2000;117:195S–197S.

178. Jourdain B, Novara A, Joly-Guillou ML, et al. Role of quantitative cultures of endotracheal aspirates in the diagnosis of nosocomial pneumonia. *Am J Respir Crit Care Med.* 1995;152:241–246. [PubMed: 7599831]

179. Borderon E, Leprince A, Gueveler C, Borderon JC. The diagnostic value of quantitative bacteriology in tracheal aspirates compared to lung biopsy [author's transl; in French]. *Rev Fr Mal Respir.* 1981;9:229–239. [PubMed: 7302346]

180. Singh N, Rogers P, Atwood CW, et al. Short-course empiric antibiotic therapy for patients with pulmonary infiltrates in the intensive care unit. A proposed solution for indiscriminate antibiotic prescription. *Am J Respir Crit Care Med.* 2000;162:505–511. [PubMed: 10934078]

181. Bartlett JG, Finegold SM. Bacteriology of expectorated sputum with quantitative culture and wash technique compared to transtracheal aspirates. *Am Rev Respir Dis.* 1978;117:1019–1027. [PubMed: 352206]

182. Baselski V. Microbiologic diagnosis of ventilator-associated pneumonia. *Infect Dis Clin North Am.* 1993;7:331–357. [PubMed: 8345172]

183. Chastre J, Fagon JY, Bornet-Lecso M, et al. Evaluation of bronchoscopic techniques for the diagnosis of nosocomial pneumonia. *Am J Respir Crit Care Med.* 1995;152:231–240. [PubMed: 7599829]

184. Baselski VS, Wunderink RG. Bronchoscopic diagnosis of pneumonia. *Clin Microbiol Rev.* 1994;7:533–558. [PubMed: 7834604]

185. Meduri GU, Chastre J. The standardization of bronchoscopic techniques for ventilator- associated pneumonia. *Chest.* 1992;102:557S–564S.

186. Timsit JF, Misset B, Azoulay E, et al. Usefulness of airway visualization in the diagnosis of nosocomial pneumonia in ventilated patients. *Chest.* 1996;110:172–179. [PubMed: 8681623]

187. Rouby JJ. Histology and microbiology of ventilator-associated pneumonias. *Semin Respir Infect.* 1996;11:54–61. [PubMed: 8885062]

188. Croce MA, Fabian TC, Waddle-Smith L, et al. Utility of gram's stain and efficacy of quantitative cultures for posttraumatic pneumonia: a prospective study. *Ann Surg.* 1998;227:743–751; discussion 751–755. [PubMed: 9605666]

189. Chastre J, Fagon JY, Soler P, et al. Diagnosis of nosocomial bacterial pneumonia in intubated patients undergoing ventilation: comparison of the usefulness of bronchoalveolar lavage and the protected specimen brush [published erratum appears in *Am J Med.* 1989;86(2):258]. *Am J Med.* 1988;85:499–506. [PubMed: 3177397]

190. Allaouchiche B, Jaumain H, Chassard D, Bouletreau P. Gram stain of bronchoalveolar lavage fluid in the early diagnosis of ventilator-associated pneumonia. *Br J Anaesth.* 1999;83:845–849. [PubMed: 10700780]

191. Torres A, El-Ebiary M, Fabregas N, et al. Value of intracellular bacteria detection in the diagnosis of ventilator associated pneumonia. *Thorax.* 1996;51:378–384. [PubMed: 8733489]

192. Veber B, Souweine B, Gachot B, et al. Comparison of direct examination of three types of bronchoscopy specimens used to diagnose nosocomial pneumonia. *Crit Care Med.* 2000;28:962–968. [PubMed: 10809267]

193. Timsit JF, Cheval C, Gachot B, et al. Usefulness of a strategy based on bronchoscopy with direct examination of bronchoalveolar lavage fluid in the initial antibiotic therapy of suspected ventilator-associated pneumonia. *Intensive Care Med.* 2001;27:640–647. [PubMed: 11398689]

194. de Jaeger A, Litalien C, Lacroix J, et al. Protected specimen brush or bronchoalveolar lavage to diagnose bacterial nosocomial pneumonia in ventilated adults: a meta-analysis. *Crit Care Med.* 1999;27:2548–2560.

195. Torres A, El-Ebiary M. Bronchoscopic BAL in the diagnosis of ventilator-associated pneumonia. *Chest.* 2000;117:198S–202S.

196. Johanson WG Jr, Seidenfeld JJ, Gomez P, et al. Bacteriologic diagnosis of nosocomial pneumonia following prolonged mechanical ventilation. *Am Rev Respir Dis.* 1988;137:259–264. [PubMed: 3341619]

197. Dreyfuss D, Mier L, Le Bourdelles G, et al. Clinical significance of borderline quantitative protected brush specimen culture results. *Am Rev Respir Dis.* 1993;147:946–951. [PubMed: 8466131]

198. Marquette CH, Herengt F, Mathieu D, et al. Diagnosis of pneumonia in mechanically ventilated patients. Repeatability of the protected specimen brush. *Am Rev Respir Dis.* 1993;147:211–214. [PubMed: 8420420]

199. Timsit JF, Misset B, Francoual S, et al. Is protected specimen brush a reproducible method to diagnose ICU-acquired pneumonia? *Chest.* 1993;104:104–108. [PubMed: 8325050]

200. Gerbeaux P, Ledoray V, Boussuges A, et al. Diagnosis of nosocomial pneumonia in mechanically ventilated patients: repeatability of the bronchoalveolar lavage. *Am J Respir Crit Care Med.* 1998;157:76–80. [PubMed: 9445281]

201. McGowan JE. Antimicrobial resistance in hospital organisms and its relation to antibiotic use. *Rev Infect Dis.* 1983;5:1033–1048. [PubMed: 6318289]

202. Heyland DK, Cook DJ, Marshall J, et al. The clinical utility of invasive diagnostic techniques in the setting of ventilator-associated pneumonia. Canadian Critical Care Trials Group. *Chest.* 1999;115:1076–1084. [PubMed: 10208211]

203. Croce MA, Fabian TC, Shaw B, et al. Analysis of charges associated with diagnosis of nosocomial pneumonia: can routine bronchoscopy be justified? *J Trauma.* 1994;37:721–727. [PubMed: 7966468]

204. Liu YC, Huang WK, Huang TS, Kunin CM. Inappropriate use of antibiotics and the risk for delayed admission and masked diagnosis of infectious diseases: a lesson from Taiwan. *Arch Intern Med.* 2001;161:2366–2370. [PubMed: 11606153]

205. Baughman RP. Nonbronchoscopic evaluation of ventilator-associated pneumonia. *Semin Respir Infect.* 2003;18:95–102. [PubMed: 12840790]

206. Jorda R, Parras F, Ibanez J, et al. Diagnosis of nosocomial pneumonia in mechanically ventilated patients by the blind protected telescoping catheter. *Intensive Care Med.* 1993;19:377–382. [PubMed: 8270716]

207. Souweine B, Veber B, Bedos JP, et al. Diagnostic accuracy of protected specimen brush and bronchoalveolar lavage in nosocomial pneumonia: impact of previous antimicrobial treatments. *Crit Care Med.* 1998;26:236–244. [PubMed: 9468159]

208. Prats E, Dorca J, Pujol M, et al. Effects of antibiotics on protected specimen brush sampling in ventilator-associated pneumonia. *Eur Respir J.* 2002;19:944–951. [PubMed: 12030737]

209. Montravers P, Fagon JY, Chastre J, et al. Follow-up protected specimen brushes to assess treatment in nosocomial pneumonia. *Am Rev Respir Dis.* 1993;147:38–44. [PubMed: 8420428]

210. Assicot M, Gendrel D, Carsin H, et al. High serum procalcitonin concentrations in patients with sepsis and infection. *Lancet.* 1993;341:515–518. [PubMed: 8094770]

211. Christ-Crain M, Opal SM. Clinical review: the role of biomarkers in the diagnosis and management of community-acquired pneumonia. *Crit Care.* 2010;14:203. [PubMed: 20236471]

212. Schuetz P, Albrich W, Christ-Crain M, et al. Procalcitonin for guidance of antibiotic therapy. *Expert Rev Anti Infect Ther.* 2010;8:575–587. [PubMed: 20455686]

213. Cairns S, Thomas JG, Hooper SJ, et al. Molecular analysis of microbial communities in endotracheal tube biofilms. *PLoS One.* 2011;6:e14759.

214. Briel M, Schuetz P, Mueller B, et al. Procalcitonin-guided antibiotic use vs a standard approach for acute respiratory tract infections in primary care. *Arch Intern Med.* 2008;168:2000–2007. [PubMed: 18852401]

215. Christ-Crain M, Jaccard-Stolz D, Bingisser R, et al. Effect of procalcitonin-guided treatment on antibiotic use and outcome in lower respiratory tract infections: cluster-randomised, single-blinded intervention trial. *Lancet.* 2004;363:600–607. [PubMed: 14987884]

216. Christ-Crain M, Stolz D, Bingisser R, et al. Procalcitonin-guidance of antibiotic therapy in community-acquired pneumonia: a randomized trial. *Am J Respir Crit Care Med.* 2006;174:84–93. [PubMed: 16603606]

217. Schuetz P, Christ-Crain M, Thomann R, et al. Effect of procalcitonin-based guidelines vs standard guidelines on antibiotic use in lower respiratory tract infections: the ProHOSP randomized controlled trial. *JAMA.* 2009;302:1059–1066. [PubMed: 19738090]

218. Stolz D, Christ-Crain M, Bingisser R, et al. Antibiotic treatment of exacerbations of COPD: a randomized, controlled trial comparing procalcitonin-guidance with standard therapy. *Chest.* 2007;131:9–19. [PubMed: 17218551]

219. Cuquemelle E, Soulis F, Villers D, et al. Can procalcitonin help identify associated bacterial infection in patients with severe influenza pneumonia? A multicentre study. *Intensive Care Med.* 2011;37:796–800. [PubMed: 21369807]

220. Gibot S, Cravoisy A, Levy B, et al. Soluble triggering receptor expressed on myeloid cells and the diagnosis of pneumonia. *N Engl J Med.* 2004;350:451–458. [PubMed: 14749453]

221. Luyt CE, Combes A, Reynaud C, et al. Usefulness of procalcitonin for the diagnosis of ventilator-associated pneumonia. *Intensive Care Med.* 2008;34:1434–1440. [PubMed: 18421435]

222. Luyt CE, Combes A, Trouillet JL, Chastre J. Value of the serum procalcitonin level to guide antimicrobial therapy for patients with ventilator-associated pneumonia. *Semin Respir Crit Care Med.* 2011;32:181–187. [PubMed: 21506054]

223. Luyt CE, Guerin V, Combes A, et al. Procalcitonin kinetics as a prognostic marker of ventilator-associated pneumonia. *Am J Respir Crit Care Med.* 2005;171:48–53. [PubMed: 15447947]

224. Gibot S, Cravoisy A. Soluble form of the triggering receptor expressed on myeloid cells-1 as a marker of microbial infection. *Clin Med Res.* 2004;2:181–187. [PubMed: 15931355]

225. Determann RM, Millo JL, Gibot S, et al. Serial changes in soluble triggering receptor expressed on myeloid cells in the lung during development of ventilator-associated pneumonia. *Intensive Care Med.* 2005;31:1495–1500. [PubMed: 16195904]

226. Gibot S, Cravoisy A, Dupays R, et al. Combined measurement of procalcitonin and soluble trem-1 in the diagnosis of nosocomial sepsis. *Scand J Infect Dis.* 2007;39:604–608. [PubMed: 17577825]

227. Horonenko G, Hoyt JC, Robbins RA, et al. Soluble triggering receptor expressed on myeloid cell-1 is increased in patients with ventilator-associated pneumonia: a preliminary report. *Chest.* 2007;132:58–63. [PubMed: 17505044]

228. Flanagan PG, Jackson SK, Findlay G. Diagnosis of gram negative, ventilator associated pneumonia by assaying endotoxin in bronchial lavage fluid. *J Clin Pathol.* 2001;54:107–110. [PubMed: 11215277]

229. Kollef MH, Eisenberg PR, Ohlendorf MF, Wick MR. The accuracy of elevated concentrations of endotoxin in bronchoalveolar lavage fluid for the rapid diagnosis of gram-negative pneumonia. *Am J Respir Crit Care Med.* 1996;154:1020–1028. [PubMed: 8887601]

230. Nys M, Ledoux D, Canivet JL, et al. Correlation between endotoxin level and bacterial count in bronchoalveolar lavage fluid of ventilated patients. *Crit Care Med.* 2000;28:2825–2830. [PubMed: 10966257]

231. Pugin J, Auckenthaler R, Delaspre O, et al. Rapid diagnosis of gram negative pneumonia by assay of endotoxin in bronchoalveolar lavage fluid. *Thorax.* 1992;47:547–549. [PubMed: 1412100]

232. Baker AM, Bowton DL, Haponik EF. Decision making in nosocomial pneumonia. An analytic approach to the interpretation of quantitative bronchoscopic cultures. *Chest.* 1995;107:85–95. [PubMed: 7813319]

233. Sterling TR, Ho EJ, Brehm WT, Kirkpatrick MB. Diagnosis and treatment of ventilator-associated pneumonia—impact on survival. A decision analysis. *Chest.* 1996;110:1025–1034. [PubMed: 8874264]

234. Sole Violan J, Fernandez JA, Benitez AB, et al. Impact of quantitative invasive diagnostic techniques in the management and outcome of mechanically ventilated patients with suspected pneumonia. *Crit Care Med.* 2000;28:2737–2741.

235. Fagon JY, Chastre J, Wolff M, et al. Invasive and noninvasive strategies for management of suspected ventilator-associated pneumonia. A randomized trial. *Ann Intern Med.* 2000;132:621–630. [PubMed: 10766680]

236. Heyland D, Dodek P, Muscedere J, Day A. A randomized trial of diagnostic techniques for ventilator-associated pneumonia. *N Engl J Med.* 2006;355:2619–2630.

237. Bonten MJ, Bergmans DC, Stobberingh EE, et al. Implementation of bronchoscopic techniques in the diagnosis of ventilator-associated pneumonia to reduce antibiotic use. *Am J Respir Crit Care Med.* 1997;156:1820–1824. [PubMed: 9412561]

238. Joffe AR, Muscedere J, Marshall JC, et al. The safety of targeted antibiotic therapy for ventilator-associated pneumonia: a multicenter observational study. *J Crit Care.* 2008;23:82–90. [PubMed: 18359425]

239. Shorr AF, Sherner JH, Jackson WL, Kollef MH. Invasive approaches to the diagnosis of ventilator-associated pneumonia: a meta-analysis. *Crit Care Med.* 2005;33:46–53. [PubMed: 15644647]

240. Garnacho-Montero J, Sa-Borges M, Sole-Violan J, et al. Optimal management therapy for pseudomonas aeruginosa ventilator-associated pneumonia: an observational, multicenter study comparing monotherapy with combination antibiotic therapy. *Crit Care Med.* 2007;35:1888–1895. [PubMed: 17581492]

241. Kumar A. Early antimicrobial therapy in severe sepsis and septic shock. *Curr Infect Dis Rep.* 2010;12:336–344. [PubMed: 21308515]

242. Martinez JA, Cobos-Trigueros N, Soriano A, et al. Influence of empiric therapy with a beta-lactam alone or combined with an aminoglycoside on prognosis of bacteremia due to gram-negative microorganisms. *Antimicrob Agents Chemother.* 2010;54:3590–3596. [PubMed: 20585123]

243. Hospital-acquired pneumonia in adults: diagnosis, assessment of severity, initial antimicrobial therapy, and preventive strategies. A consensus statement, American Thoracic Society, November 1995. *Am J Respir Crit Care Med.* 1996;153:1711–1725.

244. Trouillet JL, Vuagnat A, Combes A, et al. *Pseudomonas aeruginosa* ventilator-associated pneumonia: comparison of episodes due to piperacillin-resistant versus piperacillin-susceptible organisms. *Clin Infect Dis.* 2002;34:1047–1054. [PubMed: 11914992]

245. Carmeli Y, Troillet N, Eliopoulos GM, Samore MH. Emergence of antibiotic-resistant pseudomonas aeruginosa: comparison of risks associated with different antipseudomonal agents. *Antimicrob Agents Chemother.* 1999;43:1379–1382. [PubMed: 10348756]

246. Bhat S, Fujitani S, Potoski BA, et al. Pseudomonas aeruginosa infections in the intensive care unit: can the adequacy of empirical beta-lactam antibiotic therapy be improved? *Int J Antimicrob Agents.* 2007;30:458–462. [PubMed: 17703923]

247. Michel F, Franceschini B, Berger P, et al. Early antibiotic treatment for BAL-confirmed ventilator-associated pneumonia: a role for routine endotracheal aspirate cultures. *Chest.* 2005;127:589–597. [PubMed: 15706001]

248. Depuydt P, Blot S, Benoit D, Decruyenaere J. Improving the adequacy of empirical beta-lactam therapy for *Pseudomonas aeruginosa* ventilator-associated pneumonia. *Int J Antimicrob Agents.* 2008.

249. Leone M, Garcin F, Bouvenot J, et al. Ventilator-associated pneumonia: breaking the vicious circle of antibiotic overuse. *Crit Care Med.* 2007;35:379–385. [PubMed: 17205011]

250. Bouza E, Perez A, Munoz P, et al. Ventilator-associated pneumonia after heart surgery: a prospective analysis and the value of surveillance. *Crit Care Med.* 2003;31:1964–1970. [PubMed: 12847390]

251. Hayon J, Figliolini C, Combes A, et al. Role of serial routine microbiologic culture results in the initial management of ventilator-associated pneumonia. *Am J Respir Crit Care Med.* 2002;165:41–46. [PubMed: 11779728]

252. Chastre J. Conference summary: ventilator-associated pneumonia. *Respir Care.* 2005;50:975–983. [PubMed: 15972117]

253. Muller B. Procalcitonin and ventilator-associated pneumonia: yet another breath of fresh air. *Am J Respir Crit Care Med.* 2005;171:2–3. [PubMed: 15615888]

254. Micek ST, Ward S, Fraser VJ, Kollef MH. A randomized controlled trial of an antibiotic discontinuation policy for clinically suspected ventilator-associated pneumonia. *Chest.* 2004;125:1791–1799. [PubMed: 15136392]

255. Niederman MS. The importance of de-escalating antimicrobial therapy in patients with ventilator-associated pneumonia. *Semin Respir Crit Care Med.* 2006;27:45–50. [PubMed: 16508881]

256. File TM Jr. Recommendations for treatment of hospital-acquired and ventilator-associated pneumonia: review of recent international guidelines. *Clin Infect Dis.* 2010;51 Suppl 1:S42–S47.

257. Kuti JL, Shore E, Palter M, Nicolau DP. Tackling empirical antibiotic therapy for ventilator-associated pneumonia in your ICU: guidance for implementing the guidelines. *Semin Respir Crit Care Med.* 2009;30:102–115. [PubMed: 19199192]

258. Micek ST, Heuring TJ, Hollands JM, et al. Optimizing antibiotic treatment for ventilator-associated pneumonia. *Pharmacotherapy.* 2006;26:204–213. [PubMed: 16466325]

259. Rello J, Vidaur L, Sandiumenge A, et al. De-escalation therapy in ventilator-associated pneumonia. *Crit Care Med.* 2004;32:2183–2190. [PubMed: 15640629]

260. Depuydt P, Blot S. Antibiotic therapy for ventilator-associated pneumonia: de-escalation in the real world. *Crit Care Med.* 2007;35:632–633. [PubMed: 17251705]

261. Ewig S. Nosocomial pneumonia: de-escalation is what matters. *Lancet Infect Dis.* 2011;11:155–157. [PubMed: 21256087]

262. Hibbard ML, Kopelman TR, O'Neill PJ, et al. Empiric, broad-spectrum antibiotic therapy with an aggressive de-escalation strategy does not induce gram-negative pathogen resistance in ventilator-associated pneumonia. *Surg Infect (Larchmt).* 2010;11:427–432. [PubMed: 20818984]

263. Giantsou E, Liratzopoulos N, Efraimidou E, et al. De-escalation therapy rates are significantly higher by bronchoalveolar lavage than by tracheal aspirate. *Intensive Care Med.* 2007;33:1533–1540. [PubMed: 17549458]

264. Lodise TP, Drusano GL. Pharmacokinetics and pharmacodynamics: optimal antimicrobial therapy in the intensive care unit. *Crit Care Clin.* 2011;27:1–18. [PubMed: 21144983]

265. Lodise TP Jr, Lomaestro B, Drusano GL. Piperacillin-tazobactam for pseudomonas aeruginosa infection: clinical implications of an extended-infusion dosing strategy. *Clin Infect Dis.* 2007;44:357–363. [PubMed: 17205441]

266. Patel N, Cardone K, Grabe DW, et al. Use of pharmacokinetic and pharmacodynamic principles to determine optimal administration of daptomycin in patients receiving standardized thrice-weekly hemodialysis. *Antimicrob Agents Chemother.* 2011;55:1677–1683. [PubMed: 21282429]

267. Kiem S, Schentag JJ. Relationship of minimal inhibitory concentration and bactericidal activity to efficacy of antibiotics for treatment of ventilator-associated pneumonia. *Semin Respir Crit Care Med.* 2006;27:51–67. [PubMed: 16508882]

268. Hidayat LK, Hsu DI, Quist R, et al. High-dose vancomycin therapy for methicillin-resistant *Staphylococcus aureus* infections: efficacy and toxicity. *Arch Intern Med.* 2006;166:2138–2144. [PubMed: 17060545]

269. Moise-Broder PA, Forrest A, Birmingham MC, Schentag JJ. Pharmacodynamics of vancomycin and other antimicrobials in patients with *Staphylococcus aureus* lower respiratory tract infections. *Clin Pharmacokinet.* 2004;43:925–942. [PubMed: 15509186]

270. Moise-Broder PA, Sakoulas G, Forrest A, Schentag JJ. Vancomycin in vitro bactericidal activity and its relationship to efficacy in clearance of methicillin-resistant *Staphylococcus aureus* bacteremia. *Antimicrob Agents Chemother.* 2007.

271. Forrest A, Nix DE, Ballow CH, et al. Pharmacodynamics of intravenous ciprofloxacin in seriously ill patients. *Antimicrob Agents Chemother.* 1993;37:1073–1081. [PubMed: 8517694]

272. Kashuba AD, Nafziger AN, Drusano GL, Bertino JS Jr. Optimizing aminoglycoside therapy for nosocomial pneumonia caused by gram-negative bacteria. *Antimicrob Agents Chemother.* 1999;43:623–629. [PubMed: 10049277]

273. Peloquin CA, Cumbo TJ, Nix DE, et al. Evaluation of intravenous ciprofloxacin in patients with nosocomial lower respiratory tract infections. Impact of plasma concentrations, organism, minimum inhibitory concentration, and clinical condition on bacterial eradication. *Arch Intern Med.* 1989;149:2269–2273. [PubMed: 2508586]

274. Schentag JJ, Birmingham MC, Paladino JA, et al. In nosocomial pneumonia, optimizing antibiotics other than aminoglycosides is a more important determinant of successful clinical outcome, and a better means of avoiding resistance. *Semin Respir Infect.* 1997;12:278–293. [PubMed: 9436955]

275. Schentag JJ, Strenkoski-Nix LC, Nix DE, Forrest A. Pharmacodynamic interactions of antibiotics alone and in combination. *Clin Infect Dis.* 1998;27:40–46. [PubMed: 9675447]

276. Schentag JJ. Antimicrobial action and pharmacokinetics/pharmacodynamics: the use of AUIC to improve efficacy and avoid resistance. *J Chemother.* 1999;11:426–439. [PubMed: 10678784]

277. Schentag JJ. Pharmacokinetic and pharmacodynamic surrogate markers: studies with fluoroquinolones in patients. *Am J Health Syst Pharm.* 1999;56:S21–S24.

278. Ambrose PG, Bhavnani SM, Ellis-Grosse EJ, Drusano GL. Pharmacokinetic-pharmacodynamic considerations in the design of hospital-acquired or ventilator-associated bacterial pneumonia studies: look before you leap! *Clin Infect Dis.* 2010;51 Suppl 1:S103–S110.

279. Chastre J, Wunderink R, Prokocimer P, et al. Efficacy and safety of intravenous infusion of doripenem versus imipenem in ventilator-associated pneumonia: a multicenter, randomized study. *Crit Care Med.* 2008;36:1089–1096. [PubMed: 18379232]

280. Delattre IK, Musuamba FT, Nyberg J, et al. Population pharmacokinetic modeling and optimal sampling strategy for bayesian estimation of amikacin exposure in critically ill septic patients. *Ther Drug Monit.* 2010;32:749–756. [PubMed: 20962708]

281. Taccone FS, de Backer D, Laterre PF, et al. Pharmacokinetics of a loading dose of amikacin in septic patients undergoing continuous renal replacement therapy. *Int J Antimicrob Agents.* 2011;37:531–535. [PubMed: 21489756]

282. Taccone FS, Laterre PF, Dugernier T, et al. Insufficient beta-lactam concentrations in the early phase of severe sepsis and septic shock. *Crit Care.* 2010;14:R126.

283. Ceftazidime combined with a short or long course of amikacin for empirical therapy of gram-negative bacteremia in cancer patients with granulocytopenia. The EORTC International Antimicrobial Therapy Cooperative Group. *N Engl J Med.* 1987;317:1692–1698.

284. Hilf M, Yu VL, Sharp J, et al. Antibiotic therapy for pseudomonas aeruginosa bacteremia: outcome correlations in a prospective study of 200 patients. *Am J Med.* 1989;87:540–546. [PubMed: 2816969]

285. Korvick JA, Bryan CS, Farber B, et al. Prospective observational study of klebsiella bacteremia in 230 patients: outcome for antibiotic combinations versus monotherapy. *Antimicrob Agents Chemother.* 1992;36:2639–2644. [PubMed: 1482131]

286. Cometta A, Baumgartner JD, Lew D, et al. Prospective randomized comparison of imipenem monotherapy with imipenem plus netilmicin for treatment of severe infections in nonneutropenic patients. *Antimicrob Agents Chemother.* 1994;38:1309–1313. [PubMed: 8092830]

287. Cometta A, Calandra T, Gaya H, et al. Monotherapy with meropenem versus combination therapy with ceftazidime plus amikacin as empiric therapy for fever in granulocytopenic patients with cancer. The International Antimicrobial Therapy Cooperative Group of the European Organization for Research and Treatment of Cancer and the Gruppo Italiano Malattie Ematologiche Maligne dell'Adulto Infection Program. *Antimicrob Agents Chemother.* 1996;40:1108–1115. [PubMed: 8723449]

288. Cometta A, Zinner S, de Bock R, et al. Piperacillin-tazobactam plus amikacin versus ceftazidime plus amikacin as empiric therapy for fever in granulocytopenic patients with cancer. The international antimicrobial therapy cooperative group of the European Organization for Research and Treatment of Cancer. *Antimicrob Agents Chemother.* 1995;39:445–452. [PubMed: 7726513]

289. Dupont H, Carbon C, Carlet J. Monotherapy with a broad-spectrum beta-lactam is as effective as its combination with an aminoglycoside in treatment of severe generalized peritonitis: a multicenter randomized controlled trial. The Severe Generalized Peritonitis Study Group. *Antimicrob Agents Chemother.* 2000;44:2028–2033. [PubMed: 10898671]

290. Bochud PY, Glauser MP, Calandra T. Antibiotics in sepsis. *Intensive Care Med.* 2001;27:S33–S48.

291. Leibovici L, Paul M, Poznanski O, et al. Monotherapy versus beta-lactam-aminoglycoside combination treatment for gram-negative bacteremia: a prospective, observational study. *Antimicrob Agents Chemother.* 1997;41:1127–1133. [PubMed: 9145881]

292. Paul M, Soares-Weiser K, Leibovici L. Beta lactam monotherapy versus beta lactam-aminoglycoside combination therapy for fever with neutropenia: systematic review and meta-analysis. *BMJ.* 2003;326:1111. [PubMed: 12763980]

293. Marcus R, Paul M, Elphick H, Leibovici L. Clinical implications of beta-lactam-aminoglycoside synergism: systematic review of randomised trials. *Int J Antimicrob Agents.* 2011;37:491–503. [PubMed: 21292449]

294. Chamot E, Boffi El Amari E, Rohner P, Van Delden C. Effectiveness of combination antimicrobial therapy for pseudomonas aeruginosa bacteremia. *Antimicrob Agents Chemother.* 2003;47:2756–2764. [PubMed: 12936970]

295. Dennesen PJ, van der Ven AJ, Kessels AG, et al. Resolution of infectious parameters after antimicrobial therapy in patients with ventilator-associated pneumonia. *Am J Respir Crit Care Med.* 2001;163:1371–1375. [PubMed: 11371403]

296. Luna CM, Blanzaco D, Niederman MS, et al. Resolution of ventilator-associated pneumonia: prospective evaluation of the clinical pulmonary infection score as an early clinical predictor of outcome. *Crit Care Med.* 2003;31:676–682. [PubMed: 12626968]

297. Combes A, Figliolini C, Trouillet JL, et al. Factors predicting ventilator-associated pneumonia recurrence. *Crit Care Med.* 2003;31:1102–1107. [PubMed: 12682479]

298. Chastre J, Wolff M, Fagon JY, et al. Comparison of 8 vs 15 days of antibiotic therapy for ventilator-associated pneumonia in adults: a randomized trial. *JAMA.* 2003;290:2588–2598. [PubMed: 14625336]

299. Ibrahim EH, Ward S, Sherman G, et al. Experience with a clinical guideline for the treatment of ventilator-associated pneumonia. *Crit Care Med.* 2001;29:1109–1115. [PubMed: 11395584]

300. Bloos F, Marshall JC, Dellinger RP, et al. Multinational, observational study of procalcitonin in ICU patients with pneumonia requiring mechanical ventilation: a multicenter observational study. *Crit Care.* 2011;15:R88.

301. Charles PE, Tinel C, Barbar S, et al. Procalcitonin kinetics within the first days of sepsis: relationship with the appropriateness of antibiotic therapy and the outcome. *Crit Care.* 2009;13:R38.

302. Briel M, Christ-Crain M, Young J, et al. Procalcitonin-guided antibiotic use versus a standard approach for acute respiratory tract infections in primary care: study protocol for a randomised controlled trial and baseline characteristics of participating general practitioners. *BMC Fam Pract.* 2005;6:34. [PubMed: 16107222]

303. Nobre V, Harbarth S, Graf JD, et al. Use of procalcitonin to shorten antibiotic treatment duration in septic patients: a randomized trial. *Am J Respir Crit Care Med.* 2008;177:498–505. [PubMed: 18096708]

304. Stolz D, Smyrnios N, Eggimann P, et al. Procalcitonin for reduced antibiotic exposure in ventilator-associated pneumonia: a randomised study. *Eur Respir J.* 2009;34:1364–1375. [PubMed: 19797133]

305. Bouadma L, Luyt CE, Tubach F, et al. Use of procalcitonin to reduce patients' exposure to antibiotics in intensive care units (PRORATA trial): a multicentre randomised controlled trial. *Lancet.* 2010;375:463–474. [PubMed: 20097417]

306. Hochreiter M, Kohler T, Schweiger AM, et al. Procalcitonin to guide duration of antibiotic therapy in intensive care patients: a randomized prospective controlled trial. *Crit Care.* 2009;13:R83.

307. Kopterides P, Siempos II, Tsangaris I, et al. Procalcitonin-guided algorithms of antibiotic therapy in the intensive care unit: a systematic review and meta-analysis of randomized controlled trials. *Crit Care Med.* 2010;38:2229–2241. [PubMed: 20729729]

308. Schroeder S, Hochreiter M, Koehler T, et al. Procalcitonin (pct)-guided algorithm reduces length of antibiotic treatment in surgical intensive care patients with severe sepsis: results of a prospective randomized study. *Langenbecks Arch Surg.* 2009;394:221–226. [PubMed: 19034493]

309. Ioannidou E, Siempos, II, Falagas ME. Administration of antimicrobials via the respiratory tract for the treatment of patients with nosocomial pneumonia: a meta-analysis. *J Antimicrob Chemother.* 2007;60:1216–1226. [PubMed: 17934205]

310. Goldstein I, Wallet F, Robert J, et al. Lung tissue concentrations of nebulized amikacin during mechanical ventilation in piglets with healthy lungs. *Am J Respir Crit Care Med.* 2002;165:171–175. [PubMed: 11790649]

311. Palmer LB, Smaldone GC, Chen JJ, et al. Aerosolized antibiotics and ventilator-associated tracheobronchitis in the intensive care unit. *Crit Care Med.* 2008;36:2008–2013. [PubMed: 18552684]

312. Kofteridis DP, Alexopoulou C, Valachis A, et al. Aerosolized plus intravenous colistin versus intravenous colistin alone for the treatment of ventilator-associated pneumonia: a matched case-control study. *Clin Infect Dis.* 2010;51:1238–1244. [PubMed: 20973727]

313. Rattanaumpawan P, Lorsutthitham J, Ungprasert P, et al. Randomized controlled trial of nebulized colistimethate sodium as adjunctive therapy of ventilator-associated pneumonia caused by gram-negative bacteria. *J Antimicrob Chemother.* 2010;65:2645–2649. [PubMed: 20876621]

314. Goldstein I, Wallet F, Nicolas-Robin A, et al. Lung deposition and efficiency of nebulized amikacin during *Escherichia coli* pneumonia in ventilated piglets. *Am J Respir Crit Care Med.* 2002;166:1375–1381. [PubMed: 12406838]

315. Luyt CE, Eldon MA, Stass H, et al. Pharmacokinetics and tolerability of amikacin administered as bay41-6551 aerosol in mechanically ventilated patients with gram-negative pneumonia and acute renal failure. *J Aerosol Med Pulm Drug Deliv.* 2011;24:183–190. [PubMed: 21361783]

316. Niederman MS, Chastre J, Corkery K, et al. Inhaled amikacin reduces iv antibiotic use in intubated mechanically ventilated patients [abstract]. *Am J Respir Crit Care Med.* 2007;175:A326.

317. Dodek P, Keenan S, Cook D, et al. Evidence-based clinical practice guideline for the prevention of ventilator-associated pneumonia. *Ann Intern Med.* 2004;141:305–313. [PubMed: 15313747]

318. Collard HR, Saint S, Matthay MA. Prevention of ventilator-associated pneumonia: an evidence-based systematic review. *Ann Intern Med.* 2003;138:494–501. [PubMed: 12639084]

319. Flaherty JP, Weinstein RA. Infection control and pneumonia prophylaxis strategies in the intensive care unit. *Semin Respir Infect.* 1990;5:191–203. [PubMed: 2255804]

320. du Moulin G. Minimizing the potential for nosocomial pneumonia: architectural, engineering, and environmental considerations for the intensive care unit. *Eur J Clin Microbiol Infect Dis.* 1989;8:69–74.

321. Needleman J, Buerhaus P, Pankratz VS, et al. Nurse staffing and inpatient hospital mortality. *N Engl J Med.* 2011;364:1037–1045. [PubMed: 21410372]

322. Needleman J, Buerhaus P, Mattke S, et al. Nurse-staffing levels and the quality of care in hospitals. *N Engl J Med.* 2002;346:1715–1722. [PubMed: 12037152]

323. Aiken LH, Clarke SP, Cheung RB, et al. Educational levels of hospital nurses and surgical patient mortality. *JAMA.* 2003;290:1617–1623. [PubMed: 14506121]

324. Pittet D, Mourouga P, Perneger TV. Compliance with handwashing in a teaching hospital. Infection control program. *Ann Intern Med.* 1999;130:126–130. [PubMed: 10068358]

325. Pittet D. Improving compliance with hand hygiene in hospitals. *Infect Control Hosp Epidemiol.* 2000;21:381–386. [PubMed: 10879568]

326. Girou E, Chai SH, Oppein F, et al. Misuse of gloves: the foundation for poor compliance with hand hygiene and potential for microbial transmission? *J Hosp Infect.* 2004;57:162–169. [PubMed: 15183248]

327. Girou E, Loyeau S, Legrand P, et al. Efficacy of handrubbing with alcohol based solution versus standard handwashing with antiseptic soap: randomised clinical trial. *BMJ.* 2002;325:362. [PubMed: 12183307]

328. Haley RW, Culver DH, White JW, et al. The efficacy of infection surveillance and control programs in preventing nosocomial infections in U.S. hospitals. *Am J Epidemiol.* 1985;121:182–205. [PubMed: 4014115]

329. Gruson D, Hilbert G, Vargas F, et al. Rotation and restricted use of antibiotics in a medical intensive care unit. Impact on the incidence of ventilator-associated pneumonia caused by antibiotic-resistant gram-negative bacteria. *Am J Respir Crit Care Med.* 2000;162:837–843. [PubMed: 10988092]

330. Bochicchio GV, Napolitano L, Joshi M, et al. Blood product transfusion and ventilator-associated pneumonia in trauma patients. *Surg Infect (Larchmt).* 2008;9:415–422. [PubMed: 18759678]

331. Earley AS, Gracias VH, Haut E, et al. Anemia management program reduces transfusion volumes, incidence of ventilator-associated pneumonia, and cost in trauma patients. *J Trauma.* 2006;61:1–5; discussion 5–7. [PubMed: 16832243]

332. Hortal J, Giannella M, Perez MJ, et al. Incidence and risk factors for ventilator-associated pneumonia after major heart surgery. *Intensive Care Med.* 2009;35:1518–1525. [PubMed: 19557389]

333. Shorr AF, Duh MS, Kelly KM, Kollef MH. Red blood cell transfusion and ventilator-associated pneumonia: a potential link? *Crit Care Med.* 2004;32:666–674. [PubMed: 15090945]

334. Vandromme MJ, McGwin G Jr, Marques MB, et al. Transfusion and pneumonia in the trauma intensive care unit: an examination of the temporal relationship. *J Trauma.* 2009;67:97–101. [PubMed: 19590316]

335. Yepes D, Gil B, Hernandez O, et al. Ventilator associated pneumonia and transfusion, is there really an association? *BMC Pulm Med.* 2006;6:18. [PubMed: 16869962]

336. Rello J, Sonora R, Jubert P, et al. Pneumonia in intubated patients: role of respiratory airway care. *Am J Respir Crit Care Med.* 1996;154:111–115. [PubMed: 8680665]

337. Needleman IG, Hirsch NP, Leemans M, et al. Randomized controlled trial of toothbrushing to reduce ventilator-associated pneumonia pathogens and dental plaque in a critical care unit. *J Clin Periodontol.* 2011;38:246–252. [PubMed: 21223352]

338. Nava S, Ambrosino N, Clini E, et al. Noninvasive mechanical ventilation in the weaning of patients with respiratory failure due to chronic obstructive pulmonary disease. A randomized, controlled trial. *Ann Intern Med.* 1998;128:721–728. [PubMed: 9556465]

339. Kress JP, Pohlman AS, O'Connor MF, Hall JB. Daily interruption of sedative infusions in critically ill patients undergoing mechanical ventilation. *N Engl J Med.* 2000;342:1471–1477. [PubMed: 10816184]

340. Krishnan JA, Moore D, Robeson C, et al. A prospective, controlled trial of a protocol-based strategy to discontinue mechanical ventilation. *Am J Respir Crit Care Med.* 2004;169:673–678. [PubMed: 14726421]

341. Brook AD, Ahrens TS, Schaiff R, et al. Effect of a nursing-implemented sedation protocol on the duration of mechanical ventilation. *Crit Care Med.* 1999;27:2609–2615. [PubMed: 10628598]

342. Barrientos-Vega R, Mar Sanchez-Soria M, Morales-Garcia C, et al. Prolonged sedation of critically ill patients with midazolam or propofol: impact on weaning and costs. *Crit Care Med.* 1997;25:33–40. [PubMed: 8989173]

343. Kollef MH, Shapiro SD, Silver P, et al. A randomized, controlled trial of protocol-directed versus physician-directed weaning from mechanical ventilation. *Crit Care Med.* 1997;25:567–574. [PubMed: 9142019]

344. Girou E, Schortgen F, Delclaux C, et al. Association of noninvasive ventilation with nosocomial infections and survival in critically ill patients. *JAMA.* 2000;284:2361–2367. [PubMed: 11066187]

345. Kagramanov V, Lyman A. Noninvasive ventilation and nosocomial infection. *JAMA.* 2001;285:881. [PubMed: 11180722]

346. Carlucci A, Richard JC, Wysocki M, et al. Noninvasive versus conventional mechanical ventilation. An epidemiologic survey. *Am J Respir Crit Care Med.* 2001;163:874–880. [PubMed: 11282759]

347. Nourdine K, Combes P, Carton MJ, et al. Does noninvasive ventilation reduce the ICU nosocomial infection risk? A prospective clinical survey. *Intensive Care Med.* 1999;25:567–573. [PubMed: 10416907]

348. Brochard L, Mancebo J, Wysocki M, et al. Noninvasive ventilation for acute exacerbations of chronic obstructive pulmonary disease. *N Engl J Med.* 1995;333:817–822. [PubMed: 7651472]

349. Confalonieri M, Parigi P, Scartabellati A, et al. Noninvasive mechanical ventilation improves the immediate and long-term outcome of COPD patients with acute respiratory failure. *Eur Respir J.* 1996;9:422–430. [PubMed: 8729999]

350. Wood KA, Lewis L, Von Harz B, Kollef MH. The use of noninvasive positive pressure ventilation in the emergency department: results of a randomized clinical trial. *Chest.* 1998;113:1339–1346. [PubMed: 9596317]

351. Antonelli M, Conti G. Noninvasive positive pressure ventilation as treatment for acute respiratory failure in critically ill patients. *Crit Care.* 2000;4:15–22. [PubMed: 11094492]

352. Antonelli M, Conti G, Rocco M, et al. A comparison of noninvasive positive-pressure ventilation and conventional mechanical ventilation in patients with acute respiratory failure. *N Engl J Med.* 1998;339:429–435. [PubMed: 9700176]

353. Alexiou VG, Ierodiakonou V, Dimopoulos G, Falagas ME. Impact of patient position on the incidence of ventilator-associated pneumonia: a meta-analysis of randomized controlled trials. *J Crit Care.* 2009;24:515–522. [PubMed: 19327314]

354. Scales DC, Dainty K, Hales B, et al. A multifaceted intervention for quality improvement in a network of intensive care units: a cluster randomized trial. *JAMA.* 2011;305:363–372. [PubMed: 21248161]

355. Bouadma L, Deslandes E, Lolom I, et al. Long-term impact of a multifaceted prevention program on ventilator-associated pneumonia in a medical intensive care unit. *Clin Infect Dis.* 2010;51:1115–1122. [PubMed: 20936973]

356. Tablan OC, Anderson LJ, Besser R, et al. Guidelines for preventing health-care–associated pneumonia, 2003: recommendations of CDC and the healthcare infection control practices advisory committee. *MMWR Recomm Rep.* 2004;53:1–36. [PubMed: 15048056]

357. Koenig SM, Truwit JD. Ventilator-associated pneumonia: diagnosis, treatment, and prevention. *Clin Microbiol Rev.* 2006;19:637–657. [PubMed: 17041138]

358. Mandell LA, Wunderink RG, Anzueto A, et al. Infectious diseases society of America/American Thoracic Society consensus guidelines on the management of community-acquired pneumonia in adults. *Clin Infect Dis.* 2007;44 Suppl 2:S27–S72.

359. Masterton RG, Galloway A, French G, et al. Guidelines for the management of hospital-acquired pneumonia in the UK: report of the working party on hospital-acquired pneumonia of the British Society for Antimicrobial Chemotherapy. *J Antimicrob Chemother.* 2008.

360. Klompas M. Prevention of ventilator-associated pneumonia. *Expert Rev Anti Infect Ther.* 2010;8:791–800. [PubMed: 20586564]

361. Vincent JL, de Souza Barros D, Cianferoni S. Diagnosis, management and prevention of ventilator-associated pneumonia: an update. *Drugs.* 2010;70:1927–1944. [PubMed: 20883051]

362. Choi SC, Nelson LD. Kinetic therapy in critically ill patients: combined results based on meta-analysis. *J Crit Care.* 1992;7:57–62.

363. Traver GA, Tyler ML, Hudson LD, et al. Continuous oscillation: outcome in critically ill patients. *J Crit Care.* 1995;10:97–103. [PubMed: 7496451]

364. Kollef M, Pittet D, Sanchez Garcia M, et al. A randomized double-blind trial of iseganan in prevention of ventilator-associated pneumonia. *Am J Respir Crit Care Med.* 2006;173:91–97. [PubMed: 16192451]

365. Fourrier F, Dubois D, Pronnier P, et al. Effect of gingival and dental plaque antiseptic decontamination on nosocomial infections acquired in the intensive care unit: a double-blind placebo-controlled multicenter study. *Crit Care Med.* 2005;33:1728–1735. [PubMed: 16096449]

366. Bellissimo-Rodrigues F, Bellissimo-Rodrigues WT, Viana JM, et al. Effectiveness of oral rinse with chlorhexidine in preventing nosocomial respiratory tract infections among intensive care unit patients. *Infect Control Hosp Epidemiol.* 2009;30:952–958. [PubMed: 19743899]

367. DeRiso AJ 2nd, Ladowski JS, Dillon TA, et al. Chlorhexidine gluconate 0.12% oral rinse reduces the incidence of total nosocomial respiratory infection and nonprophylactic systemic antibiotic use in patients undergoing heart surgery. *Chest.* 1996;109:1556–1561. [PubMed: 8769511]

368. Genuit T, Bochicchio G, Napolitano LM, et al. Prophylactic chlorhexidine oral rinse decreases ventilator-associated pneumonia in surgical ICU patients. *Surg Infect (Larchmt).* 2001;2:5–18. [PubMed: 12594876]

369. Koeman M, van der Ven AJ, Hak E, et al. Oral decontamination with chlorhexidine reduces the incidence of ventilator-associated pneumonia. *Am J Respir Crit Care Med.* 2006;173:1348–1355. [PubMed: 16603609]

370. Kollef MH. Prevention of hospital-associated pneumonia and ventilator-associated pneumonia. *Crit Care Med.* 2004;32:1396–1405. [PubMed: 15187525]

371. Panchabhai TS, Dangayach NS, Krishnan A, et al. Oropharyngeal cleansing with 0.2% chlorhexidine for prevention of nosocomial pneumonia in critically ill patients: an open-label randomized trial with 0.01% potassium permanganate as control. *Chest.* 2009;135:1150–1156. [PubMed: 19420193]

372. Scannapieco FA, Yu J, Raghavendran K, et al. A randomized trial of chlorhexidine gluconate on oral bacterial pathogens in mechanically ventilated patients. *Crit Care.* 2009;13:R117.

373. Segers P, Speekenbrink RG, Ubbink DT, et al. Prevention of nosocomial infection in cardiac surgery by decontamination of the nasopharynx and oropharynx with chlorhexidine gluconate: a randomized controlled trial. *JAMA.* 2006;296:2460–2466. [PubMed: 17119142]

374. Silvestri L, van Saene HK, Milanese M, et al. Prevention of ventilator-associated pneumonia by use of oral chlorhexidine. *Infect Control Hosp Epidemiol.* 2009;30:101–102. [PubMed: 19067601]

375. Silvestri L, van Saene HK, Zandstra DF, et al. SDD, SOD, or oropharyngeal chlorhexidine to prevent pneumonia and to reduce mortality in ventilated patients: which manoeuvre is evidence-based? *Intensive Care Med.* 2010;36:1436–1437. [PubMed: 20237766]

376. Tantipong H, Morkchareonpong C, Jaiyindee S, Thamlikitkul V. Randomized controlled trial and meta-analysis of oral decontamination with 2% chlorhexidine solution for the prevention of ventilator-associated pneumonia. *Infect Control Hosp Epidemiol.* 2008;29:131–136. [PubMed: 18179368]

377. Chlebicki MP, Safdar N. Topical chlorhexidine for prevention of ventilator-associated pneumonia: a meta-analysis. *Crit Care Med.* 2007;35:595–602. [PubMed: 17205028]

378. Mahul P, Auboyer C, Jospe R, et al. Prevention of nosocomial pneumonia in intubated patients: respective role of mechanical subglottic secretions drainage and stress ulcer prophylaxis. *Intensive Care Med.* 1992;18:20–25. [PubMed: 1578042]

379. Valles J, Artigas A, Rello J, et al. Continuous aspiration of subglottic secretions in preventing ventilator-associated pneumonia. *Ann Intern Med.* 1995;122:179–186. [PubMed: 7810935]

380. Kollef MH, Skubas NJ, Sundt TM. A randomized clinical trial of continuous aspiration of subglottic secretions in cardiac surgery patients. *Chest.* 1999;116:1339–1346. [PubMed: 10559097]

381. Bo H, He L, Qu J. Influence of the subglottic secretion drainage on the morbidity of ventilator associated pneumonia in mechanically ventilated patients [in Chinese]. *Zhonghua Jie He He Hu Xi Za Zhi.* 2000;23:472–474. [PubMed: 11778260]

382. Smulders K, van Der Hoeven H, Weers-Pothoff I, Vandenbroucke-Grauls C. A randomized clinical trial of intermittent subglottic secretion drainage in patients receiving mechanical ventilation. *Chest.* 2002;121:858–862. [PubMed: 11888973]

383. Dezfulian C, Shojania K, Collard HR, et al. Subglottic secretion drainage for preventing ventilator-associated pneumonia: a meta-analysis. *Am J Med.* 2005;118:11–18. [PubMed: 15639202]

384. Lorente L, Lecuona M, Jimenez A, et al. Influence of an endotracheal tube with polyurethane cuff and subglottic secretion drainage on pneumonia. *Am J Respir Crit Care Med.* 2007;176:1079–1083. [PubMed: 17872488]

385. Lacherade JC, De Jonghe B, Guezennec P, et al. Intermittent subglottic secretion drainage and ventilator-associated pneumonia: a multicenter trial. *Am J Respir Crit Care Med.* 2010;182:910–917. [PubMed: 20522796]

386. Muscedere J, Rewa O, McKechnie K, et al. Subglottic secretion drainage for the prevention of ventilator-associated pneumonia: a systematic review and meta-analysis. *Crit Care Med.* 2011.

387. Berra L, De Marchi L, Panigada M, et al. Evaluation of continuous aspiration of subglottic secretion in an in vivo study. *Crit Care Med.* 2004;32:2071–2078. [PubMed: 15483416]

388. Girou E, Buu-Hoi A, Stephan F, et al. Airway colonisation in long-term mechanically ventilated patients. Effect of semi-recumbent position and continuous subglottic suctioning. *Intensive Care Med.* 2004;30:225–233. [PubMed: 14647884]

389. Berra L, Curto F, Li Bassi G, et al. Antimicrobial-coated endotracheal tubes: an experimental study. *Intensive Care Med.* 2008;34:1020–1029. [PubMed: 18418572]

390. Berra L, Kolobow T, Laquerriere P, et al. Internally coated endotracheal tubes with silver sulfadiazine in polyurethane to prevent bacterial colonization: a clinical trial. *Intensive Care Med.* 2008;34:1030–1037. [PubMed: 18418571]

391. Olson ME, Harmon BG, Kollef MH. Silver-coated endotracheal tubes associated with reduced bacterial burden in the lungs of mechanically ventilated dogs. *Chest.* 2002;121:863–870. [PubMed: 11888974]

392. Pacheco-Fowler V, Gaonkar T, Wyer PC, Modak S. Antiseptic impregnated endotracheal tubes for the prevention of bacterial colonization. *J Hosp Infect.* 2004;57:170–174. [PubMed: 15183249]

393. Rello J, Afessa B, Anzueto A, et al. Activity of a silver-coated endotracheal tube in preclinical models of ventilator-associated pneumonia and a study after extubation. *Crit Care Med.* 2010;38:1135–1140. [PubMed: 20081533]

394. Rello J, Kollef M, Diaz E, et al. Reduced burden of bacterial airway colonization with a novel silver-coated endotracheal tube in a randomized multiple-center feasibility study. *Crit Care Med.* 2006;34:2766–2772. [PubMed: 16957639]

395. Kollef MH, Afessa B, Anzueto A, et al. Silver-coated endotracheal tubes and incidence of ventilator-associated pneumonia: the nascent randomized trial. *JAMA.* 2008;300:805–813. [PubMed: 18714060]

396. Jongerden IP, Buiting AG, Leverstein-van Hall MA, et al. Effect of open and closed endotracheal suctioning on cross-transmission with gram-negative bacteria: a prospective crossover study. *Crit Care Med.* 2011;39:1313–1321. [PubMed: 21358397]

397. Stamm AM. Ventilator-associated pneumonia and frequency of circuit changes. *Am J Infect Control.* 1998;26:71–73. [PubMed: 9503116]

398. Davis K Jr, Evans SL, Campbell RS, et al. Prolonged use of heat and moisture exchangers does not affect device efficiency or frequency rate of nosocomial pneumonia [see comments]. *Crit Care Med.* 2000;28:1412–1418. [PubMed: 10834688]

399. Deppe SA, Kelly JW, Thoi LL, et al. Incidence of colonization, nosocomial pneumonia, and mortality in critically ill patients using a trach care closed-suction system versus an open-suction system: prospective, randomized study. *Crit Care Med.* 1990;18:1389–1393. [PubMed: 2245613]

400. Combes P, Fauvage B, Oleyer C. Nosocomial pneumonia in mechanically ventilated patients, a prospective randomised evaluation of the Steri-Cath closed suctioning system. *Intensive Care Med.* 2000;26:878–882. [PubMed: 10990101]

401. Yavagal DR, Karnad DR, Oak JL. Metoclopramide for preventing pneumonia in critically ill patients receiving enteral tube feeding: a randomized controlled trial. *Crit Care Med.* 2000;28:1408–1411. [PubMed: 10834687]

402. Kearns PJ, Chin D, Mueller L, et al. The incidence of ventilator-associated pneumonia and success in nutrient delivery with gastric versus small intestinal feeding: a randomized clinical trial. *Crit Care Med.* 2000;28:1742–1746. [PubMed: 10890612]

403. Donowitz LG, Page MC, Mileur BL, Guenthner SH. Alteration of normal gastric flora in critical care patients receiving antacid and cimetidine therapy. *Infect Control.* 1986;7:23–26. [PubMed: 3633241]

404. Marik PE, Vasu T, Hirani A, Pachinburavan M. Stress ulcer prophylaxis in the new millennium: a systematic review and meta-analysis. *Crit Care Med.* 2010;38:2222–2228. [PubMed: 20711074]

405. Kahn JM, Doctor JN, Rubenfeld GD. Stress ulcer prophylaxis in mechanically ventilated patients: integrating evidence and judgment using a decision analysis. *Intensive Care Med.* 2006;32:1151–1158. [PubMed: 16788804]

406. Huggins RM, Scates AC, Latour JK. Intravenous proton-pump inhibitors versus H_2-antagonists for treatment of GI bleeding. *Ann Pharmacother.* 2003;37:433–437. [PubMed: 12639176]

407. Kantorova I, Svoboda P, Scheer P, et al. Stress ulcer prophylaxis in critically ill patients: a randomized controlled trial. *Hepatogastroenterology.* 2004;51:757–761. [PubMed: 15143910]

408. Leontiadis GI, Sreedharan A, Dorward S, et al. Systematic reviews of the clinical effectiveness and cost-effectiveness of proton pump inhibitors in acute upper gastrointestinal bleeding. *Health Technol Assess.* 2007;11:iii-iv, 1–164.

409. Tofil NM, Benner KW, Fuller MP, Winkler MK. Histamine 2 receptor antagonists vs intravenous proton pump inhibitors in a pediatric intensive care unit: a comparison of gastric ph. *J Crit Care.* 2008;23:416–421. [PubMed: 18725049]

410. Lin PC, Chang CH, Hsu PI, et al. The efficacy and safety of proton pump inhibitors vs histamine-2 receptor antagonists for stress ulcer bleeding prophylaxis among critical care patients: a meta-analysis. *Crit Care Med.* 2010;38:1197–1205. [PubMed: 20173630]

411. van Saene HK, Stoutenbeek CC, Stoller JK. Selective decontamination of the digestive tract in the intensive care unit: current status and future prospects. *Crit Care Med.* 1992;20:691–703.

412. Stoutenbeek CP, van Saene HK, Miranda DR, Zandstra DF. The effect of selective decontamination of the digestive tract on colonisation and infection rate in multiple trauma patients. *Intensive Care Med.* 1984;10:185–192. [PubMed: 6470306]

413. Heyland DK, Cook DJ, Jaeschke R, et al. Selective decontamination of the digestive tract. An overview. *Chest.* 1994;105:1221–1229. [PubMed: 8018162]

414. Hurley JC. Prophylaxis with enteral antibiotics in ventilated patients: selective decontamination or selective cross-infection? *Antimicrob Agents Chemother.* 1995;39:941–947. [PubMed: 7786000]

415. Nathens AB, Marshall JC. Selective decontamination of the digestive tract in surgical patients: a systematic review of the evidence. *Arch Surg.* 1999;134:170–176. [PubMed: 10025458]

416. Oostdijk EA, de Smet AM, Kesecioglu J, Bonten MJ. The role of intestinal colonization with gram-negative bacteria as a source for intensive care unit-acquired bacteremia. *Crit Care Med.* 2011;39:961–966. [PubMed: 21283008]

417. Krueger WA, Lenhart FP, Neeser G, et al. Influence of combined intravenous and topical antibiotic prophylaxis on the incidence of infections, organ dysfunctions, and mortality in critically ill surgical patients: a prospective, stratified, randomized, double-blind, placebo-controlled clinical trial. *Am J Respir Crit Care Med.* 2002;166:1029–1037. [PubMed: 12379544]

418. de Jonge E, Schultz MJ, Spanjaard L, et al. Effects of selective decontamination of digestive tract on mortality and acquisition of resistant bacteria in intensive care: a randomised controlled trial. *Lancet.* 2003;362:1011–1016.

419. de Smet AM, Kluytmans JA, Cooper BS, et al. Decontamination of the digestive tract and oropharynx in ICU patients. *N Engl J Med.* 2009;360:20–31.

420. Dahms R, Carlson M, Lohr B, Beilman G. Selective digestive tract decontamination and vancomycin-resistant enterococcus isolation in the surgical intensive care unit. *Shock.* 2000;14:343–346. [PubMed: 11028554]

421. Ebner W, Kropec-Hubner A, Daschner FD. Bacterial resistance and overgrowth due to selective decontamination of the digestive tract. *Eur J Clin Microbiol Infect Dis.* 2000;19:243–247. [PubMed: 10834811]

422. Humphreys H, Winter R, Pick A. The effect of selective decontamination of the digestive tract on gastrointestinal enterococcal colonization in itu patients. *Intensive Care Med.* 1992;18:459–463. [PubMed: 1289369]

423. Kaufhold A, Behrendt W, Krauss T, van Saene H. Selective decontamination of the digestive tract and methicillin-resistant *Staphylococcus aureus. Lancet.* 1992;339:1411–1412. [PubMed: 1350817]

424. Oostdijk EA, de Smet AM, Blok HE, et al. Ecological effects of selective decontamination on resistant gram-negative bacterial colonization. *Am J Respir Crit Care Med.* 2010;181:452–457. [PubMed: 19965807]

425. Bonten MJ. Prevention of infection in the intensive care unit. *Curr Opin Crit Care.* 2004;10:364–368. [PubMed: 15385752]

426. de Smet AM, Kluytmans JA, Blok HE, et al. Selective digestive tract decontamination and selective oropharyngeal decontamination and antibiotic resistance in patients in intensive-care units: an open-label, clustered group-randomised, crossover study. *Lancet Infect Dis.* 2011;11:372–380.

427. Heininger A, Meyer E, Schwab F, et al. Effects of long-term routine use of selective digestive decontamination on antimicrobial resistance. *Intensive Care Med.* 2006;32:1569–1576. [PubMed: 16896852]

428. van Essen EH, de Jonge E. Selective decontamination of the digestive tract (SDD): is the game worth the candle? *Semin Respir Crit Care Med.* 2011;32:236–242.

429. Rello J, Lorente C, Bodi M, et al. Why do physicians not follow evidence-based guidelines for preventing ventilator-associated pneumonia?: a survey based on the opinions of an international panel of intensivists. *Chest.* 2002;122:656–661. [PubMed: 12171847]

430. Ricart M, Lorente C, Diaz E, et al. Nursing adherence with evidence-based guidelines for preventing ventilator-associated pneumonia. *Crit Care Med.* 2003;31:2693–2696. [PubMed: 14605543]

431. Pronovost P, Needham D, Berenholtz S, et al. An intervention to decrease catheter-related bloodstream infections in the ICU. *N Engl J Med.* 2006;355:2725–2732. [PubMed: 17192537]

432. Apisarnthanarak A, Pinitchai U, Thongphubeth K, et al. Effectiveness of an educational program to reduce ventilator-associated pneumonia in a tertiary care center in Thailand: a 4-year study. *Clin Infect Dis.* 2007;45:704–711. [PubMed: 17712753]

433. Babcock HM, Zack JE, Garrison T, et al. An educational intervention to reduce ventilator-associated pneumonia in an integrated health system: a comparison of effects. *Chest.* 2004;125:2224–2231. [PubMed: 15189945]

434. Bouadma L, Mourvillier B, Deiler V, et al. Changes in knowledge, beliefs, and perceptions throughout a multifaceted behavioral program aimed at preventing ventilator-associated pneumonia. *Intensive Care Med.* 2010;36:1341–1347. [PubMed: 20431867]

435. Bouadma L, Mourvillier B, Deiler V, et al. A multifaceted program to prevent ventilator-associated pneumonia: impact on compliance with preventive measures. *Crit Care Med.* 2010;38:789–796. [PubMed: 20068461]

436. Salahuddin N, Zafar A, Sukhyani L, et al. Reducing ventilator-associated pneumonia rates through a staff education programme. *J Hosp Infect.* 2004;57:223–227. [PubMed: 15236851]

437. Zack JE, Garrison T, Trovillion E, et al. Effect of an education program aimed at reducing the occurrence of ventilator-associated pneumonia. *Crit Care Med.* 2002;30:2407–2412. [PubMed: 12441746]

438. Al-Tawfiq JA, Abed MS. Decreasing ventilator-associated pneumonia in adult intensive care units using the institute for healthcare improvement bundle. *Am J Infect Control.* 2010;38:552–556. [PubMed: 20400203]

439. Blamoun J, Alfakir M, Rella ME, et al. Efficacy of an expanded ventilator bundle for the reduction of ventilator-associated pneumonia in the medical intensive care unit. *Am J Infect Control.* 2009;37:172–175. [PubMed: 18945515]

440. Cocanour CS, Peninger M, Domonoske BD, et al. Decreasing ventilator-associated pneumonia in a trauma ICU. *J Trauma.* 2006;61:122–129; discussion 129–130. [PubMed: 16832259]

441. Hawe CS, Ellis KS, Cairns CJ, Longmate A. Reduction of ventilator-associated pneumonia: active versus passive guideline implementation. *Intensive Care Med.* 2009;35:1180–1186. [PubMed: 19308354]

442. Resar R, Pronovost P, Haraden C, et al. Using a bundle approach to improve ventilator care processes and reduce ventilator-associated pneumonia. *Jt Comm J Qual Patient Saf.* 2005;31:243–248. [PubMed: 15960014]

443. Tolentino-DelosReyes AF, Ruppert SD, Shiao SY. Evidence-based practice: use of the ventilator bundle to prevent ventilator-associated pneumonia. *Am J Crit Care.* 2007;16:20–27. [PubMed: 17192523]

444. Weiss CH, Moazed F, McEvoy CA, et al. Prompting physicians to address a daily checklist and process of care and clinical outcomes: a single-site study. *Am J Respir Crit Care Med.* 2011;184:680–686. [PubMed: 21616996]

445. Kollef M. Smart approaches for reducing nosocomial infections in the ICU. *Chest.* 2008;134:447–456. [PubMed: 18682466]

446. Wip C, Napolitano L. Bundles to prevent ventilator-associated pneumonia: how valuable are they? *Curr Opin Infect Dis.* 2009;22:159–166. [PubMed: 19276975]

447. Zilberberg MD, Shorr AF, Kollef MH. Implementing quality improvements in the intensive care unit: ventilator bundle as an example. *Crit Care Med.* 2009;37:305–309. [PubMed: 19050626]

448. Edwards JR, Peterson KD, Andrus ML, et al. National Healthcare Safety Network (NHSN) report, data summary for 2006, issued June 2007. *Am J Infect Control.* 2007;35:290–301. [PubMed: 17577475]